# 46 Primary and Secondary Hypertension

Aldo J. Peixoto | Vivek Bhalla | Debbie L. Cohen | George L. Bakris[†]

## CHAPTER OUTLINE



## KEY POINTS

- The prevalence of hypertension is rising, treatment is suboptimal, and control is insufficient. It is imperative that clinicians focus on screening patients and promptly initiating lifestyle changes and, if appropriate, antihypertensive therapy to mitigate the adverse cardiovascular effects of hypertension.
- Out-of-office blood pressure (BP) monitoring is necessary for the diagnosis of hypertension. Home BP or 24-hour ambulatory BP monitoring are equally recommended. If unavailable, multiple office measurements on different occasions are necessary. The threshold for diagnosis of hypertension is BP $\geq$130/80 mm Hg.
- The initial evaluation of the hypertensive patient should include ruling out common causes of secondary hypertension, especially parenchymal kidney disease, aldosterone excess, and renovascular disease. Clinicians should use standard clinical and laboratory tests to rule out these disorders.
- Primary aldosteronism is increasingly recognized as a common cause of secondary hypertension and should be formally sought through screening measurement of plasma renin activity (with or without plasma aldosterone) in patients with risk factors for primary aldosteronism and in all patients with resistant hypertension.
- Target BP for patients with high CV risk is <130/80 mm Hg. Treatment should include medications from one of the four core classes (ACE inhibitors, angiotensin receptor blockers, thiazide diuretics, and calcium channel blockers). The choice often depends on associated comorbidities. Combination therapy is often necessary and leads to more prompt achievement of BP control.

Hypertension is a major contributor to premature morbidity and mortality worldwide. A pooled analysis of the published global trends showed that more than 1 billion aged 30 to 79 years have hypertension.[1] This figure indicates a doubling of hypertension prevalence between 1990 and 2019. This includes nations in all continents and of different socioeconomic profiles. As emerging countries have improved sanitation and other basic public health measures, cardiovascular (CV) disease has or soon will become the most common cause of death, and hypertension will be its most common reversible risk factor. Among individuals aged 50 and older worldwide, ischemic heart disease and stroke are the two leading causes of disability, accounting for 23.4% of disability-adjusted life-years for ages 50 to 74 years and 34.1% for those aged 75 years or older.[2] Hypertension is the most important modifiable risk factor for stroke and heart failure and a significant factor for ischemic heart disease. Therefore hypertension directly impacts these trends and demands interventions for its better detection and treatment.

† Deceased.

## HYPERTENSION DEFINITIONS

Traditionally, high blood pressure (BP) has been defined as a persistent BP elevation in the office at or above 140/90 mm Hg. The "threshold BP value" to secure a diagnosis of hypertension comes from large epidemiologic studies demonstrating risk of death at levels above 140/90 mm Hg.[3] In the United States, the 2017 ACC/AHA guidelines shifted the definition of hypertension down to levels above 130/80 mm Hg (Table 46.1).[4] Their justification was the strong relations among BP, CV events, and death in observational studies and data from clinical studies showing improved CV outcomes and survival with systolic BP levels below 130 mm Hg. However, despite acknowledging the observational data linking adverse outcomes to BP at levels below 140/90 mm Hg, most other organizations continue to use the 140/90 mm Hg definition, including the 2023 European Society of Hypertension guidelines.[5] It should be noted that the relation between BP and CV mortality is log-linear, and a threshold-driven, binary definition does not fully capture the nuance between BP and outcomes with precision; the risk of cardiovascular mortality increases above BP levels of 115/75 mm Hg, doubling for every 20/10 mm Hg increase above this level.[3]

## EPIDEMIOLOGY

Hypertension is widely treated because of its increased risk for long-term CV and renal morbidity and mortality and the efficacy of treatment in reducing events. The risks attributable to elevated BP levels have been documented in numerous epidemiologic studies, beginning in 1948 with the Framingham Heart Study and extending to the present. Meta-analyses of pooled data confirm the robust, continuous relation between BP level and cerebrovascular and coronary heart disease in Western and Eastern populations. Elevated BP has been linked directly in epidemiologic studies to incident left ventricular hypertrophy (LVH), heart failure, peripheral vascular disease, carotid atherosclerosis, end-stage kidney disease, and "subclinical CV disease." Hypertension often coexists with other risk factors including diabetes mellitus, metabolic syndrome with insulin resistance, hyperlipidemia, obesity, an unhealthy diet, and physical inactivity. The presence of more than one risk factor increases the risk of CV events. The overall prevalence of hypertension in the United States has continued to increase. According to the latest National Health and Nutrition Examination Survey (NHANES) data, when defining BP >140/90 mm Hg, the prevalence increased in the overall population from 31.5% in 2009–2012 to 32.9% in 2017–2020. When considering the AHA/ACC definition of BP >130/80 mm Hg, prevalence is much higher and increased from 45.8% to 46.5% over the same period.[6]

**Table 46.1 Categorization of Blood Pressure for Hypertension and Its Related Diagnoses in the United States[a]**

| BP Category[b] | SBP | | DBP |
|---|---|---|---|
| Normal | <120 mm Hg | and | <80 mm Hg |
| Elevated | 120–129 mm Hg | and | <80 mm Hg |
| **Hypertension** | | | |
| • Stage 1 | 130–139 mm Hg | or | 80–89 mm Hg |
| • Stage 2 | 140–159 mm Hg | or | 90–99 mm Hg |

[a]If the systolic and diastolic blood pressure levels fall into two different diagnostic categories, the higher category is used (e.g., 162/92 mm Hg is stage 3 hypertension; 134/72 mm Hg is stage 1).

[b]Definitions are from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults: Executive Summary: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:1269–1324. Different classification schemes are used outside the United States.

## AGE AND HYPERTENSION

Older age is a major risk factor for developing hypertension and a strong confounder of its independent risk for CV and renal events. Above the age of 40 years, younger individuals have higher risk for death for any given BP increase[3] (Fig. 46.1). The prevalence of hypertension increases with age and according to the latest NHANES data from 2017–2020; when BP is defined as SBP ≥140/90 mm Hg; the prevalence of hypertension in adults aged 65 to 74 years was 64% and in adults aged 75 years or older was 74.5%[6] (Table 46.2) A meta-analysis of more than 1 million adults demonstrated that risk increases in all age groups with SBP >115 mm Hg or DBP >75 mm Hg with a doubling of the risk of death from heart disease or stroke for every 20 mm Hg higher SBP and 10 mm Hg higher DBP.[3] SBP and pulse pressure are more potent predictors of risk in patients older than the age of 50 to 60 years,[7] whereas DBP is a better predictor of mortality in individuals younger than 50 years of age.[8] When SBP is <130 mm Hg, regardless of age, DBP does not associate with CV risk.[9,10] The absolute risk of CV disease is dependent on older age and other CV risk factors in addition to the level of BP.[11] Antihypertensive drug therapy, however, reduces the risk for CV events across the full age spectrum and has its greatest *absolute* benefit in older persons including those older than the age of 80 years.[12-14]

Hypertension in children and adolescents also contributes to premature atherosclerosis and early development of CV disease as adults. The diagnosis of hypertension in children and adolescents is becoming more prevalent due to the epidemic of obesity in young Americans.[15] Current U.S. guidelines[16] suggest diagnosing hypertension in children younger than 13 years based on new normative pediatric blood BP tables based on normal-weight children as opposed to children with overweight/obesity (those with a BMI ≥85th percentile). In adolescents ≥13 years of age, diagnosis of hypertension has been simplified and aligns with the AHA/ACC adult BP guidelines.[17] There is also a more limited recommendation to perform screening BP measurements only at annual preventive care visits; however, BP should be checked at every health care encounter for persons with obesity, those taking medications known to increase BP, with kidney disease, a history of aortic arch

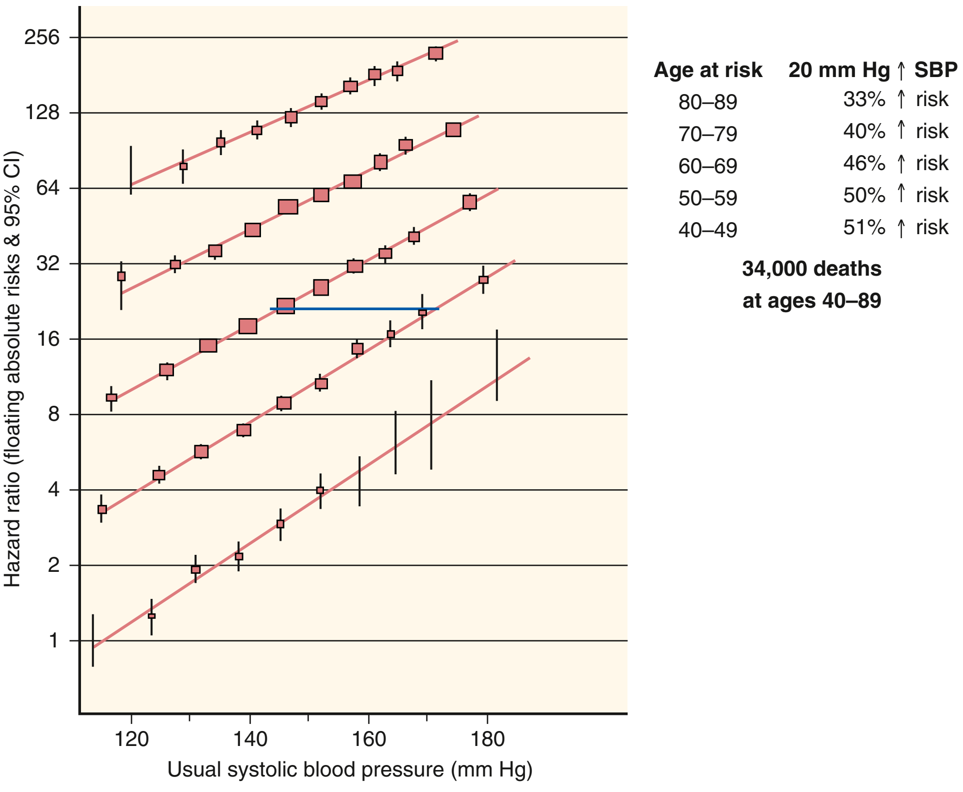


**Fig. 46.1 Coronary heart disease mortality according to systolic blood pressure and age.** (Reprinted with permission from Elsevier, from Lewington S, Clarke R, Qizilbash N, et al. Age-specific relevance of usual blood pressure to vascular mortality: a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet.* 2002;360:1903–1913.)

**Table 46.2 Trends in Hypertension Prevalence in the United States According to Age, Sex, and Race-Ethnicity**

| | Calendar Period | | | |
|---|---|---|---|---|
| **Population Evaluated** | **2009-2012** | **2013-2016** | **2017-2020** | ***P* Trend** |
| Overall | 31.5 (30.3-32.8) | 32.0 (30.6-33.3) | 32.9 (31.0-34.7) | 0.218 |
| **Age group, year** | | | | |
| 18-44 | 9.2 (8.3-10.2) | 10.1 (8.9-11.4) | 9.8 (8.3-11.2) | 0.570 |
| 45-64 | 40.0 (37.4-42.7) | 40.8 (38.5-43.1) | 43.6 (40.2-47.0) | 0.100 |
| 65-74 | 63.9 (60.2-67.7) | 64.2 (60.1-68.3) | 64.1 (59.8-68.4) | 0.956 |
| ≥75 | 76.8 (74.5-79.1) | 73.5 (70.2-76.9) | 74.5 (70.8-78.3) | 0.318 |
| **Sex** | | | | |
| Women | 30.8 (29.4-32.1) | 30.9 (29.4-32.5) | 31.5 (29.6-33.4) | 0.493 |
| Men | 32.2 (30.3-34.0) | 32.8 (31.0-34.6) | 34.1 (31.8-36.4) | 0.192 |
| **Race-ethnicity** | | | | |
| Non-Hispanic White | 30.3 (28.7-31.9) | 30.6 (28.9-32.4) | 30.7 (28.1-33.2) | 0.870 |
| Non-Hispanic Black | 44.2 (42.4-46.1) | 44.1 (42.0-6.2) | 46.6 (44.2-49.0) | 0.163 |
| Non-Hispanic Asian | 27.0 (23.4-30.7)[a] | 28.9 (26.5-31.2) | 33.5 (30.7-36.2) | 0.003 |
| Hispanic | 29.4 (27.2-31.5) | 30.0 (27.8-32.2) | 33.2 (29.9-36.6) | 0.029 |

[a]Data for non-Hispanic Asian adults from 2009 to 2012 represent 2011 to 2012 as the public use NHANES data set does not provide information for non-Hispanic Asian participants from 2009 to 2010.

Hypertension was defined according to the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure as SBP ≥140 mm Hg or DBP ≥90 mm Hg or self-reported antihypertensive medication use. Numbers in the table are estimated percentage (95% CI) of the U.S adult population or subgroup with hypertension.

*DBP,* Diastolic blood pressure; NHANES, National Health and Nutrition Examination Survey; *SBP,* systolic blood pressure.

From Muntner P, Miles MA, Jaeger BC, et al. Blood pressure control among US adults, 2009 to 2012 through 2017 to 2020. *Hypertension*.

obstruction or coarctation, or diabetes. Trained health care professionals in the office setting should make a diagnosis of HTN if a child or adolescent has auscultatory confirmed BP readings ≥95th percentile at three different visits, and guidelines encourage an expanded role for ambulatory BP monitoring (ABPM) in the diagnosis and management of pediatric hypertension.[16]

## SEX AND HYPERTENSION

Hypertension is a major problem for both men and women, but men tend to develop it at an earlier age (Table 46.2), which is also true of the adverse clinical consequences of hypertension. Among individuals older than 70 years, women are more likely to have hypertension[18,19] and to have a CV event rate comparable with men. Age and body mass index have been much stronger predictors of incident hypertension than sex in epidemiologic studies. Drug treatment of hypertension yields roughly the same benefits for women and men.[20,21]

## RACE/ETHNICITY AND HYPERTENSION

The NHANES survey from 2017–2020 demonstrated, similar to previous surveys, a higher prevalence of hypertension in non-Hispanic Blacks (46.6%) than in non-Hispanic Whites (30.7%)[6] (Table 46.2). The prevalence was unchanged in non-Hispanic Whites but increased as compared with NHANES 2013-2016 in non-Hispanic Blacks (from 44.1%–46.6%) and to a greater extent in non-Hispanic Asians (from 28.9%–33.5%) and Hispanics (from 30 to 33.2%).[6] Although the prevalence of hypertension has increased in non-Hispanic Asians and Hispanics, non-Hispanic Blacks still have the highest overall prevalence of hypertension in the United States. The prevalence of hypertension is geographically heterogeneous, with the highest prevalence in Blacks and Whites in the southeastern United States ("the Stroke Belt"). Perhaps because of the persistent historical difference in BP control rates, the adverse long-term consequences of hypertension are still more common in Blacks than Whites, but disparities have decreased in the past decade.[18] In 2014, the age-adjusted death rate from heart disease was 24% higher in Blacks, as was stroke (by 41%) and hypertension or hypertensive renal disease (by 111%).[22] In 2021, the prevalence of ESKD among Black individuals was more than four times that of White individuals. The prevalence among Hispanic and Native American individuals was more than twice as high as among White individuals.[23] Although diuretics or calcium channel blockers (CCBs) may have an advantage as initial therapy for reducing CV events in Blacks, an angiotensin-converting enzyme (ACE) inhibitor (ACEI) was better than either a CCB or a β-blocker in preventing the decline of kidney function in African Americans with hypertensive nephrosclerosis.[18,21,24] Most current hypertension guidelines therefore recommend controlling BP with multidrug regimens in all racial/ethnic groups.

# BLOOD PRESSURE CONTROL RATES

Despite major progress in identifying the risks associated with elevated BP and demonstrating that reducing BP to within a certain range reduces risk for death from CV disease and stroke, as well as kidney disease progression, improved methods of achieving and sustaining BP control are needed. We have more than 125 different medications, most of which are available in generic formulations, from eight different antihypertensive drug classes to help lower BP, and more than 20 single-pill combination agents. Despite the array of medication choices, BP control remains suboptimal in many parts of the world.[6,25]

BP control rates (to <140/90 mm Hg) have improved substantially in the United States since 1974 but have declined from 52.8% in 2009 to 2012 to 48.2% in 2017 to 2020.[6] During the same time period, the age-specific proportion of U.S. adults with controlled BP declined among those ≥75 years of age (from 46.0%–36.8%), and BP control also declined among women (from 56.3%–47.9%) and non-Hispanic Black adults (from 48.9%–37.4%). Among U.S. adults with hypertension taking antihypertensive medication, the age-adjusted proportion with controlled BP was 69.9% in 2009–2012 and 67.7% in 2017 to 2020, respectively.[18]

The age-adjusted proportion of U.S. adults with hypertension who were aware they had hypertension declined from 82.4% in 2009 to 2012 to 79.1% in 2017 to 2020. Declines in hypertension awareness were more common among U.S. adults ≥75 years of age, women, and non-Hispanic Black adults. Among persons aware that they had hypertension, the age-adjusted proportion taking antihypertensive medication was 91.9% from 2009 to 2012, 89.2% in 2013 to 2016, and 90.6% in 2017 to 2020. The age-adjusted proportion of non-Hispanic Black adults taking antihypertensive medication significantly declined from 90.9% in 2009 to 2012 to 87.1% in 2017 to 2020.[6]

The prevalence of uncontrolled hypertension is higher for undiagnosed, untreated, or older individuals and for systolic (rather than diastolic) BP. Some health care delivery organizations have reported BP control rates in excess of 60% to 80%.[26] These improvements in BP control have been attributed to systems improvements with increased home BP monitoring, use of electronic medical records and remote transmission of home BP readings via wireless or Bluetooth technology through the electronic medical record, and more widespread use of single-pill combination therapy.[26,27]

Hypertension, once identified, should be controlled promptly. The wisdom of controlling BP over a relatively short time course after its discovery, rather than taking months to do so, was most clearly demonstrated in the Valsartan Antihypertensive Long-term Use Evaluation (VALUE) trial.[28] Although the randomized comparison was between high-risk patients with hypertension who received either valsartan or amlodipine initially, prevention of CV events was clearly better among individuals who achieved their goal BP during the *first 6 months* of treatment, regardless of initial randomized therapy. Similar long-term benefits of "early" control of BP have been seen for stroke or CV events in the Systolic Hypertension in Europe trial[29] and for death in the Systolic Hypertension in the Elderly Program (SHEP).[30]

# PATHOPHYSIOLOGY

The physiology that generates BP involves the integration of cardiac output (CO) and systemic vascular resistance

(SVR) (BP = CO × SVR), with each of these having its own determinants ([CO = heart rate × stroke volume]; [SVR = 80 × (mean arterial pressure − central venous pressure)/CO]). This view is simplified, but it provides a framework to define the relevant factors in BP regulation. Changes in CO typically produce short-lived BP changes (hours to days), as adaptive mechanisms adjust SVR to normalize BP. However, changes in SVR are able to produce sustained increases in BP. The following sections summarize relevant mechanisms leading to BP regulation.

## SALT SENSITIVITY AND PRESSURE NATRIURESIS

Through observations by multiple groups and experiments performed by physiologists including Guyton, the role of dietary sodium intake plays a major role in the pathogenesis of hypertension. Dahl's hypothesis articulated in 1963 suggested that higher dietary sodium intake within a population was associated with a higher prevalence of hypertension.[31] Several groups have also made this association even when comparing groups with differences in dietary sodium that have few other confounders that could modulate BP.[32,33] In populations with low dietary sodium intake, BP does not increase with age, but in modern society with ubiquitously high dietary sodium, hypertension is increasingly common and prevalence rises sharply with aging.

Humans have the ability to match sodium excretion to relatively large swings (40–100×) in sodium intake such that BP is only modestly altered by a few mm Hg. Initially, acute rises in dietary sodium increase BP in healthy individuals, but both renal and extrarenal compensatory processes result in a modest rise in BP.[34,35] One example of chronic adaptation to hypertension is the renin-angiotensin-aldosterone system, described as follows.[36]

Guyton summarized the work of several investigators as to the mechanism of how an increase in sodium intake leads to hypertension. Although an increase in volume is the initiating event, the rise in CO is transient, as the excess sodium is excreted. However, that short-term rise in CO leads to a rise in BP that is sustained through a rise in peripheral vascular resistance via autoregulation. Thus a volume-driven event leads to hypertension, but after a few days of physiologic adaptation, the signature of that hypertension is an increase in total peripheral resistance. The higher BP will remain until the sodium intake returns to normal. Thus Guyton added a critical nuance to Dahl's hypothesis. In healthy controls, sodium intake does not increase BP, but in those with a "volume factor" (i.e., a rise in sodium intake, in the presence of a "kidney factor" to diminish sodium excretion), higher sodium intake does increase BP (Fig. 46.2).[37-39]

Salt sensitivity is defined by the change in BP with a change in sodium intake. The characteristic of salt sensitivity is not binary (i.e., salt-sensitive vs. salt-resistant) but as a continuum. An extension of Guyton's work is that salt sensitivity implies a "kidney factor" to raise BP. Each of these factors can be considered to shift the long-term kidney function curve to the right. Several mechanisms are known to increase salt sensitivity including CKD, genetic or acquired gain-of-function changes in the machinery for sodium reabsorption in different nephron segments, and many other factors (i.e., intrinsic to the kidney or external

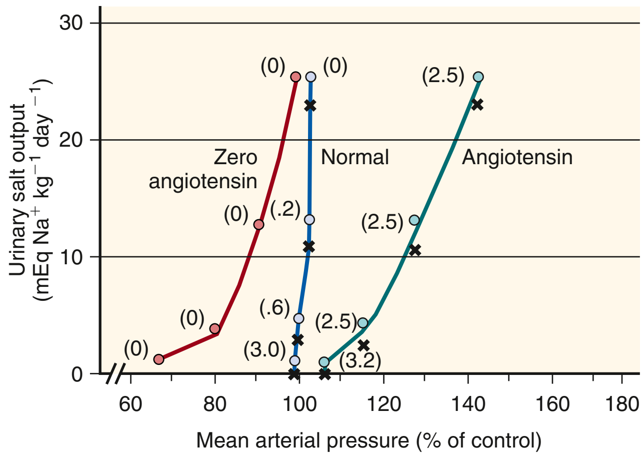


**Fig. 46.2 Pressure-natriuresis curves in dogs at different levels of sodium intake (reflected in urinary salt output, y-axis).** In the normal state, massive fluctuations in sodium intake produce minimal changes in blood pressure (BP). In states of high angiotensin II (Ang II) levels (Ang II infusion in this model), BP increases significantly, even with modest increases in sodium intake. Conversely, in states where Ang II is absent or low (administration of captopril in this model), the increased sodium results in increased BP only at low BP levels; once BP becomes normal, the relationship is similar to that of normal kidneys. Values in parentheses are the relative estimated concentrations of Ang II (1 being the reference). (From Guyton AC, Coleman TG, Young DB, et al. Salt balance and long-term blood pressure control. *Annu Rev Med.* 1980;31:15–27.)

factors acting on the kidney, such as neural, vascular, or hormonal) that deserve their own categories as clinically relevant, pathophysiologic mechanisms of hypertension. Some of these factors by historical convention fall under the heading of secondary causes of hypertension (e.g., primary aldosteronism and renovascular hypertension), while others are considered causes or exacerbating factors for primary/idiopathic/essential hypertension (e.g., obesity).

The interplay between renal sodium retention and hypertension involves changes in sodium handling throughout the nephron. A theory with substantial experimental support proposes that increased renal vasoconstriction due to a variety of possible mechanisms (e.g., increased levels of angiotensin II, catecholamines, or uric acid, and progressive aging) induces a preglomerular (afferent) arteriolopathy that results in impaired sodium filtration.[40,41] In addition, renal vasoconstriction results in tubular ischemia, another mediator of sodium avidity. Several studies have also shown that Blacks have higher salt sensitivity due to several factors,[42] in part due to higher effective response to aldosterone.[43,44]

A key factor in regulation of BP is the phenomenon of pressure natriuresis. Pressure natriuresis is defined as the increase in renal sodium excretion because of mild increases in BP, typically due to extracellular fluid volume expansion, allowing BP to remain in the normal range.[45-47] This concept is essential to the understanding of the sustainability of hypertension. If one understands the "set-point BP" as the BP at the point when extracellular volume and pressure natriuresis are in equilibrium, it necessarily follows that a change in BP can only be sustained if pressure natriuresis is abnormal. Pressure natriuresis occurs over hours to days and is modulated by both biophysical and humoral factors.

In the normal state, increased sodium intake causes an increase in extracellular volume and BP. Because of the steep relationship between volume and pressure, small increases in BP produce natriuresis that restores sodium balance and returns BP to normal (Fig. 46.2). Expansion of extracellular fluid volume and increased BP result in a rise in blood flow through the vasa recta, which stimulates the production of paracrine factors such as nitric oxide (NO) and ATP, which can inhibit tubular sodium reabsorption at multiple sites of the nephron.[48] Nitric oxide also blunts the myogenic response of arteriolar autoregulation, thus allowing increased blood flow that is necessary to increase renal blood flow and interstitial pressure (see later).[47,48]

The pressure-natriuresis process is also mediated by biophysical factors. Increased renal interstitial hydrostatic pressure is an important factor. Sodium-loading results in increased pressure in the vasa recta, which has noticeably poor autoregulation, while pressure in cortical peritubular capillaries remains normal.[49,50] Vasa recta blood flow approximates 10% of total renal blood flow. This increase in interstitial pressure inhibits sodium transport largely by increasing 20-hydroxyeicosatetraenoic acid, an inhibitor of sodium-potassium adenosine triphosphatase ($Na^+$-$K^+$-ATPase), whose inhibition causes decreased activity of the $Na^+$-$H^+$-exchanger isoform 3 (NHE3).[49] In addition, increased interstitial pressure limits proximal tubular paracellular pathways, thus maximizing natriuresis.

Because abnormalities in pressure-sodium relationships are essential to maintaining chronic elevations in BP, they represent a fundamental step in the pathogenesis of any type of hypertension in both the primary and maintenance phases of most secondary causes, such as renal and renovascular hypertension, hyperaldosteronism, glucocorticoid excess, coarctation of the aorta, and pheochromocytoma.

## GENETICS OF HYPERTENSION

Hypertension clusters in families; an individual with a family history of hypertension has a fourfold greater chance of developing hypertension.[51] It is estimated that the heritability of hypertension ranges from 27% to 68%.[52] Genome-wide association studies (GWAS) in several multinational cohorts have identified a large number of single-nucleotide polymorphisms (SNPs) associated with hypertension.[52,53] However, these individual SNPs (>900 identified by 2023)[52] are responsible for only minor BP effects (0.5–1 mm Hg), and the overall impact of these identified SNPs on the overall BP variance is only approximately 1% to 2%.[51] The shortcomings of GWAS and other large population approaches are multiple.[54] The SNP platforms used for testing are not hypothesis driven; they simply include common genetic variants for exploratory analyses that might provide clues for molecular pathways leading to better understanding of disease or new targets for therapy. To advance progress toward personalized medicine in hypertension, a GWAS based on the large U.K. Biobank Study cohort also performed functional and transcript expression analyses of candidate genes in target tissues in vitro (vascular smooth muscle cells, aortic fibroblasts, and endothelial cells) with the goal of identifying potential therapeutic targets.[53] The study also developed and validated an unbiased genetic risk score that included clinical and genotype information including data on a total of 107 independent risk loci to generate estimates of risk of hypertension and risk of specific hypertension-related outcomes (stroke, coronary disease, and any cardiovascular outcome). These analyses showed a sex-adjusted systolic BP difference of 9.3 mm Hg between the lowest and highest risk quintile (higher in the high-risk group, which also had 2.32-fold greater odds of hypertension and a 1.35-fold increase in the odds of adverse cardiovascular events.[53] These results indicate the potential value for genetic risk–based clinical scoring.

BP measurement is another important concern, as it is not uniform in these large, population-based GWASs. In addition, large numbers of patients are receiving treatment at the time of testing, thus limiting the strength of any associations. Finally, hypertensive phenotypes are not well defined, so patients with very different phenotypes (e.g., isolated diastolic hypertension and isolated systolic hypertension of the young, isolated systolic hypertension of older adults) are all lumped together.[55] We now understand that each of these phenotypes likely has different underlying pathophysiologic mechanisms and that even within each group there is substantial variability in hemodynamic profile.[56,57]

With the improvement in techniques that allow expeditious, cheaper whole-exome or whole-genome analyses and the expansion of precision medicine, it is possible that greater mechanistic insights into the genetics of hypertension will become available. Unfortunately, however, the heavy influence of lifestyle and environmental factors in hypertension makes it unlikely that the simple analysis of genome sequence will yield groundbreaking discoveries..[58] Instead, the systematic evaluation of epigenetic modification associated with detailed functional phenotyping should allow us to advance understanding of the complex interplay of lifestyle and environmental factors in patients with hypertension, particularly among patients with primary or "essential" hypertension.[58,59]

Whereas the attempts at using GWAS to understand essential hypertension have not been particularly fruitful, the study of monogenic causes of hypertension, although rare, has provided substantial insight into the pathogenesis of hypertension. Whether as yet uncharacterized variants in genes known to cause hypertension play a key role in essential hypertension will require further whole exome and whole-genome studies. Coupled with family-based studies, the use of diverse racial/ethnic cohorts to enable inherited or de novo rare variant detection aids the discovery of genetic causes of most chronic diseases including hypertension.[60,61] For example, the prevalence of Liddle syndrome or a Liddle phenotype with low renin, low aldosterone is more common in Blacks.[62] Of the monogenic forms of hypertension with well-described molecular mechanisms in adrenal cortex or the distal nephron, many share one thing in common: a defect in renal sodium handling. This commonality is consistent with the Guytonian hypothesis and highlights the importance of the kidney in regulating BP by way of sodium balance.

In Liddle syndrome, mutations in subunits of the epithelial sodium channel (ENaC) (*SCNN1A, SCNN1B, SCNN1G*) lead to increased ENaC expression and decreased removal from the luminal membrane, both of which contribute to persistent channel activation leading to sodium avidity, volume expansion, hypertension, hypokalemia, and metabolic alkalosis with suppressed plasma renin activity and aldosterone.[63]

In Gordon syndrome (pseudohypoaldosteronism type 2), various mutations have been described, leading to activation of the thiazide-sensitive sodium-chloride cotransporter (NCC, *SLC12A3*).[63] These mutations were initially mapped to the with-no-lysine (WNK) kinases 1 and 4, which stimulate NCC phosphorylation and activity. Mutations in two E3 ubiquitin ligase complex proteins (kelch-like 3 and cullin 3) were discovered later.[64] Collectively, these mutations are responsible for most cases of the syndrome. The shared mechanism is related to reduced degradation of kelch-like 3 and cullin 3 targets, WNK1, and WNK4 and therefore higher phosphorylation of NCC leading to the clinical phenotype of hypertension, hyperkalemia, and metabolic acidosis.

Mutations in mineralocorticoid receptor activation can also produce hypertensive syndromes, such as hypertension exacerbated by pregnancy (Geller syndrome), in which there is constitutive activity of the receptor in addition to marked sensitivity to progesterone, leading to hypertension during pregnancy in addition to chronic, severe hypertension with hypokalemia.[65] Likewise, increases in aldosterone production are due to a chimeric gene crossover of the promoter for 11β-hydroxylase and the aldosterone synthase gene *CYP11B2*. Adrenocorticotropic hormone (ACTH) stimulates this promoter and hence aldosterone synthase expression independent of angiotensin II or plasma potassium. Such patients have hyperaldosteronism that is blunted by glucocorticoid-induced ACTH suppression, thus the term *glucocorticoid-remediable hyperaldosteronism*.[66]

Other patients may have "apparent mineralocorticoid excess" due to mutations in the 11β-hydroxysteroid dehydrogenase type 2 gene. This enzyme is responsible for the conversion of cortisol to the inactive cortisone in target epithelia, including renal tubular epithelial cells of the distal nephron. Extracellular cortisol is typically ~1000× fold more abundant than aldosterone. Thus mutations in this gene permit cortisol to almost constitutively activate the mineralocorticoid receptor leading to a state of apparent mineralocorticoid excess (salt-sensitive hypertension, hypokalemia, metabolic alkalosis) with suppressed plasma renin activity and aldosterone.

Similarly, patients with congenital adrenal hyperplasia due to 11β-hydroxylase or 17α-hydroxylase deficiency have an excess production of 21-hydroxylated steroids such as deoxycorticosterone and corticosterone, which are potent activators of the mineralocorticoid receptor, thus also producing the syndrome of apparent mineralocorticoid excess in addition to the well-known sexual developmental abnormalities of the syndromes.[67]

In addition to hypertension-causing mutations that stimulate renal sodium reabsorption, human genetics has revealed inactivating mutations in sodium transporters, sodium channels, or their regulatory machinery that lower BP. These include but are not limited to Bartter syndrome (mutations in *SLC12A1, KCNJ1, CLCNKB, BSND, CASR, MAGED2*), Gitelman syndrome *(SLC12A3),* and pseudohypoaldosteronism type 1 *(SCNN1A, SCNN1B, SCNN1G),* which impair *NKCC2, NCC,* or *ENaC,* respectively. Taken together, these findings provide strong evidence for the role of renal sodium handling in the genesis of hypertension.

As discussed further later, endothelial and vascular smooth muscle cells also help to determine BP. The elucidation of the molecular genetics of the rare syndrome of autosomal hypertension with brachydactyly points to a vascular etiology that does not involve sodium reabsorption within the kidney.[68] This is a monogenic form of salt-resistant hypertension transmitted via gain-of-function mutations in the phosphodiesterase 3A gene *(PDE3A),* which results in increased protein kinase A–mediated PDE3A phosphorylation and increased cAMP-hydrolytic activity. These cellular actions result in enhanced cell proliferation in vascular smooth muscle and dysregulated parathyroid hormone-related peptide physiology, accounting for the hypertensive and skeletal phenotypes.[68]

## NONOSMOTIC SODIUM STORAGE

The paradigm of sodium balance described earlier assumes that sodium and its accompanying anion are osmotically active and therefore retained isosmotically with water. However, this paradigm cannot explain the observation that acute sodium loading in humans and animals results in positive sodium balance without the expected water (weight) gain. Consequently, sodium may accumulate without water, most prominently in the skin,[69] where negatively charged glycosaminoglycans bind sodium.[70] This system of interstitial sodium buffering adds to the classical Guytonian approach wherein nonosmotic accumulation occurs acutely and is presumably followed by increased removal from skin (via an enhanced lymphatic network) for ultimate renal excretion.

The mechanisms explaining isosmotic sodium storage are under intense investigation.[71] Mice and rats receiving a high-salt diet develop hypertonicity of the skin interstitium, which triggers a series of mechanisms to keep interstitial volume constant.[72] The hypertonic sodium content activates the tonicity-responsive enhancer–binding protein (TonEBP) present in mononuclear cells infiltrating the skin. Consequently, these skin macrophages act as "local osmosensors" and secrete vascular endothelial growth factor type C, resulting in increased density and hyperplasia of the skin lymphocapillary network and increased endothelial nitric oxide synthase (eNOS). If these responses are blocked, skin sodium accumulates and salt-sensitive hypertension develops.[72,73] These studies were primarily performed in preclinical models, but studies in patients treated with vascular endothelial growth factor receptor inhibition showed consistent results.[74] Moreover, clinical conditions associated with sodium-sensitive hypertension also display higher amounts of skin sodium.[75,76] These findings link the mononuclear phagocyte system to extracellular fluid volume control.

The implications of these findings were addressed in a study evaluating sodium balance in cosmonauts undergoing prolonged training (up to 205 days) in a facility simulating life in space.[77] By carefully monitoring water and electrolyte intake and excretion, as well as factors regulating sodium balance, individuals exhibited a large variability in sodium excretion on a day-to-day basis despite relatively stable diets. In the long term, approximately 95% (70% to 103%) of ingested sodium was recovered, but daily sodium excretion during stable sodium intake varied considerably and was independent of BP and sodium intake. Instead, urine sodium excretion varied as a function of circaseptan fluctuations (6–9 days in this case) in levels of aldosterone and cortisol/cortisone. Moreover, total body sodium stores had even longer infradian fluctuations (averaging several weeks).

The factors regulating these intriguing changes are still unknown. These observations have clinical implications for the use of urine sodium excretion to assess sodium intake because they suggest wide day-to-day variations that cannot[77] be captured in a single 24-hour urine collection. Moreover, these findings help to inform potential causes (e.g., immune-mediated) for salt-sensitive hypertension.

## RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

The RAAS has wide-ranging effects on BP regulation. Fig. 46.3 summarizes the most relevant elements of the RAAS and its role in the pathogenesis of hypertension and its complications. The different elements of the RAAS have key roles in mediating sodium retention, pressure natriuresis, salt sensitivity, vasoconstriction, endothelium dysfunction, and vascular injury. RAAS inhibitors are highly effective in treating hypertension; these classes are recommended among several first-line antihypertensive agents.[17] Taken together, the RAAS has an important role in the pathogenesis of hypertension. However, there are some still unanswered issues. For example, in a large GWAS of 2.5 million genotyped or imputed SNPs in 69,395 individuals of European ancestry from 29 studies,[8] the meta-analysis failed to show any significant RAAS-related SNPs associated with this analysis. Another meta-analysis evaluating the relations among polymorphisms in key RAAS genes (*ACE, AGT,* and *CYP11B2*) and salt sensitivity also found no significant associations.[78] Despite these genetic inconsistencies, there is a wealth of experimental evidence linking the RAAS to hypertension.

Angiotensinogen is the precursor of all components of the RAAS. Angiotensinogen is made primarily in the liver, although can be produced in other tissues (e.g., adipocytes), and its regulation is distinct from that of renin and aldosterone. Angiotensinogen is the rate-limiting factor in the production of angiotensin II, and higher copy number of the angiotensinogen gene, *AGT,* and higher plasma levels of angiotensinogen are associated with higher BP.[79,80] Although research into the effects of global RAAS inhibition (via inhibition of angiotensinogen) has been delayed due to the lack of available pharmacologic inhibitors, its significance beyond downstream therapeutics has become clinically relevant with the development of RNA interference therapeutics that target translation of angiotensinogen mRNA.[81] Intriguing effects such as nonredundant benefits of angiotensinogen knockdown on BP despite the angiotensin receptor blocker irbesartan may elucidate additional aspects of systemic versus tissue-specific blockade of RAAS.[81]

Renin and prorenin are synthesized and stored in the juxtaglomerular cell apparatus and released in response to decreased renal afferent perfusion pressure, decreased sodium delivery to the macula densa, activation of renal nerves (via $\beta_1$-adrenergic receptor stimulation), and a variety of metabolic products including prostaglandin

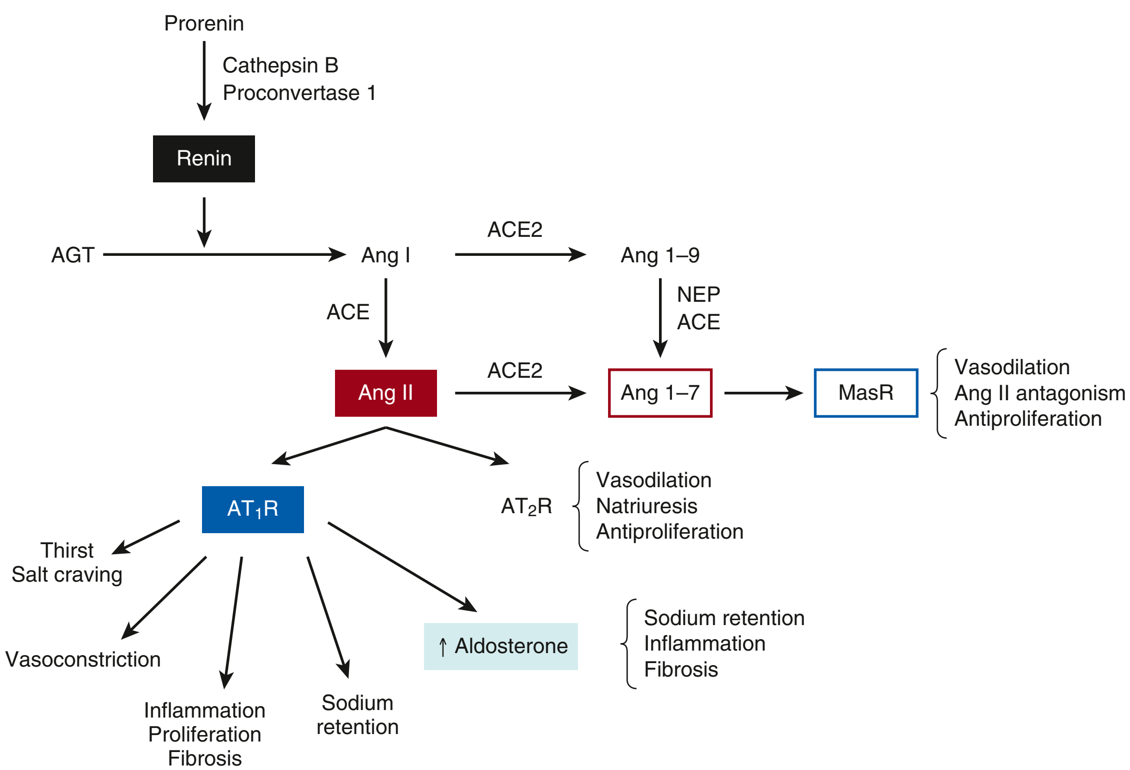


**Fig. 46.3 Key elements of the renin-angiotensin-aldosterone system.** *ACE,* Angiotensin-converting enzyme; *ACE2,* Angiotensin-converting enzyme type 2; *Ang,* angiotensin; *AGT,* angiotensinogen; *AT1R,* Ang II type 1 receptor; *AT2R,* Ang II type 2 receptor; *MasR,* Mas receptor; *NEP,* neutral endopeptidase.

$E_2$ and several others. Renin's main function is to cleave angiotensinogen into angiotensin I. Prorenin, previously viewed as an inactive substrate for renin production, stimulates the (pro)renin receptor (PRR). This receptor leads to more efficient cleavage of angiotensinogen and activates downstream intracellular signaling through the mitogen-activated protein (MAP) kinases extracellular signal–regulated kinases 1 and 2 (ERK1/2) pathways that have been associated with profibrotic effects in some, but not all, experimental models.[82,83] It is still uncertain if the PRR is involved in the genesis or complications of hypertension in a manner independent of the effects of angiotensin II; however, deletion of PRR from principal cells of the collecting duct lowers BP and sodium reabsorption.[82,84,85]

Angiotensin II, formed by the cleavage of angiotensin I by ACE, is at the center of the pathogenetic role of RAAS in hypertension. Primarily through its actions mediated by the angiotensin II type 1 receptor ($AT_1R$), angiotensin II is a potent vasoconstrictor of vascular smooth muscle, causing systemic vasoconstriction, as well as increased renovascular resistance and decreased medullary flow, the latter a mediator of salt sensitivity. Angiotensin II leads to increased sodium reabsorption in the proximal tubule by increasing the activity of NHE3, the sodium-bicarbonate exchanger, and $Na^+$-$K^+$-ATPase and by inducing aldosterone synthesis and release from the adrenal zona glomerulosa. In addition, angiotensin II is associated with endothelial cell dysfunction and produces extensive profibrotic and proinflammatory changes, largely mediated by increased oxidative stress, resulting in renal, cardiac, and vascular injury, thus providing a strong link between angiotensin II and target-organ injury in hypertension. Conversely, stimulation of the angiotensin II type 2 receptor ($AT_2R$) is associated with opposite effects, resulting in vasodilation, natriuresis, and antiproliferative effects.

Tissue expression of different elements of the RAAS is important, including the protection of animals from the development of hypertension after targeted elimination of renal ACE activity.[86] Because angiotensin II is necessary for normal renal development, experimental manipulation of the intrarenal RAAS demands the presence of some level of activity in the system. In experiments where the ACE gene was knocked out systemically and then reintroduced in myelomonocytic cells, there was no demonstrable renal ACE activity, but systemic levels generated from the myelomonocytic cells were enough to allow for normal renal development.[86] Animals with absent renal ACE activity have markedly blunted hypertensive responses to the infusion of angiotensin II or L-NAME (low renin, angiotensin II-independent model of hypertension). At face value, this indicates that the absence of renal ACE protects against hypertension irrespective of plasma angiotensin II concentrations, an unexpected finding given the traditional view of the RAAS, in which an infusion of angiotensin II bypasses ACE and produces sustained hypertension regardless of ACE activity[87] but well in line with growing evidence for the importance of ACE-induced intrarenal tubular generation of angiotensin II in the mediation of hypertension. However, because ACE was not expressed in other organs, either, it is also possible that the hypotensive effects of the absence of ACE activity were the result of absence of an effect in other locations, such as the vasculature.[88,89] This question remains unanswered while developments continue to occur in the understanding of the function of tissue RAAS (especially intrarenal), a system that has different regulators from the systemic RAAS and that, distinct from the endocrine RAAS, has feedforward mechanisms that generate further augmentation of local responses upon activation by the circulating RAAS.[88]

The relative importance of the renal and vascular effects of angiotensin II was evaluated in classic cross-transplantation studies using both wild-type mice and mice lacking the $AT_1R$.[90,91] By cross-transplanting kidneys of wild-type mice into $AT_1R$ knockout mice and vice versa, investigators were able to generate animals that were selective renal $AT_1R$ knockouts or selective systemic (nonrenal) $AT_1R$ knockouts. In physiologic conditions, renal, systemic, and total knockout mice had lower BP than wild-type mice, indicating a role of both renal and extrarenal $AT_1R$ in BP regulation.[91] Gain- and loss-of-function studies of proximal tubular AT1R are also consistent with a role for angiotensin II signaling in the kidney. The systemic absence of $AT_1R$ was associated with approximately 50% lower plasma aldosterone, but the lower BP observed in this group was independent of this lower aldosterone production, as BP remained low despite aldosterone infusions to supraphysiologic levels following adrenalectomy in the systemic knockout mice. In addition, the BP reduction in kidney knockout mice occurred despite normal aldosterone excretion, again confirming the independence of renal angiotensin II effects from aldosterone.

In the hypertensive environment, it is the presence of renal $AT_1R$ that mediates both hypertension and organ injury[91] (Fig. 46.4). When animals were infused with angiotensin II for 4 weeks, animals lacking renal $AT_1R$ did not develop sustained hypertension, whereas wild-type and systemic knockout mice had a significant increase in BP. Additionally, only animals with elevated BP developed cardiac hypertrophy and fibrosis. This indicates that cardiac injury is largely dependent on hypertension and not on the presence of $AT_1R$ in the heart, as the (hypertensive) systemic knockout mice developed significant cardiac abnormalities despite the absence of $AT_1R$ in the heart.[90] In summary, these experiments indicate that both systemic and renal actions of angiotensin II are relevant to physiologic BP regulation, but in hypertension, the detrimental effects of angiotensin II are mediated via its renal effects.

Aldosterone, the adrenocortical hormone synthesized in the zona glomerulosa, plays a critical role in hypertension through its well-known effects on sodium reabsorption that are largely mediated by genomic effects through the mineralocorticoid receptor leading to increased expression of ENaC. An extensive body of literature has identified other genomic and nongenomic effects of aldosterone with relevance to hypertension. Extensive nonepithelial effects include vascular smooth muscle cell proliferation, vascular extracellular matrix deposition, vascular remodeling, and fibrosis, and increased oxidative stress leading to endothelial dysfunction and vasoconstriction.[92]

Several other elements of the RAAS, including ACE2 and angiotensin (1-7), have been identified as having potential roles in BP regulation and angiotensin II–associated target-organ injury. ACE2 is expressed largely in heart, kidney,

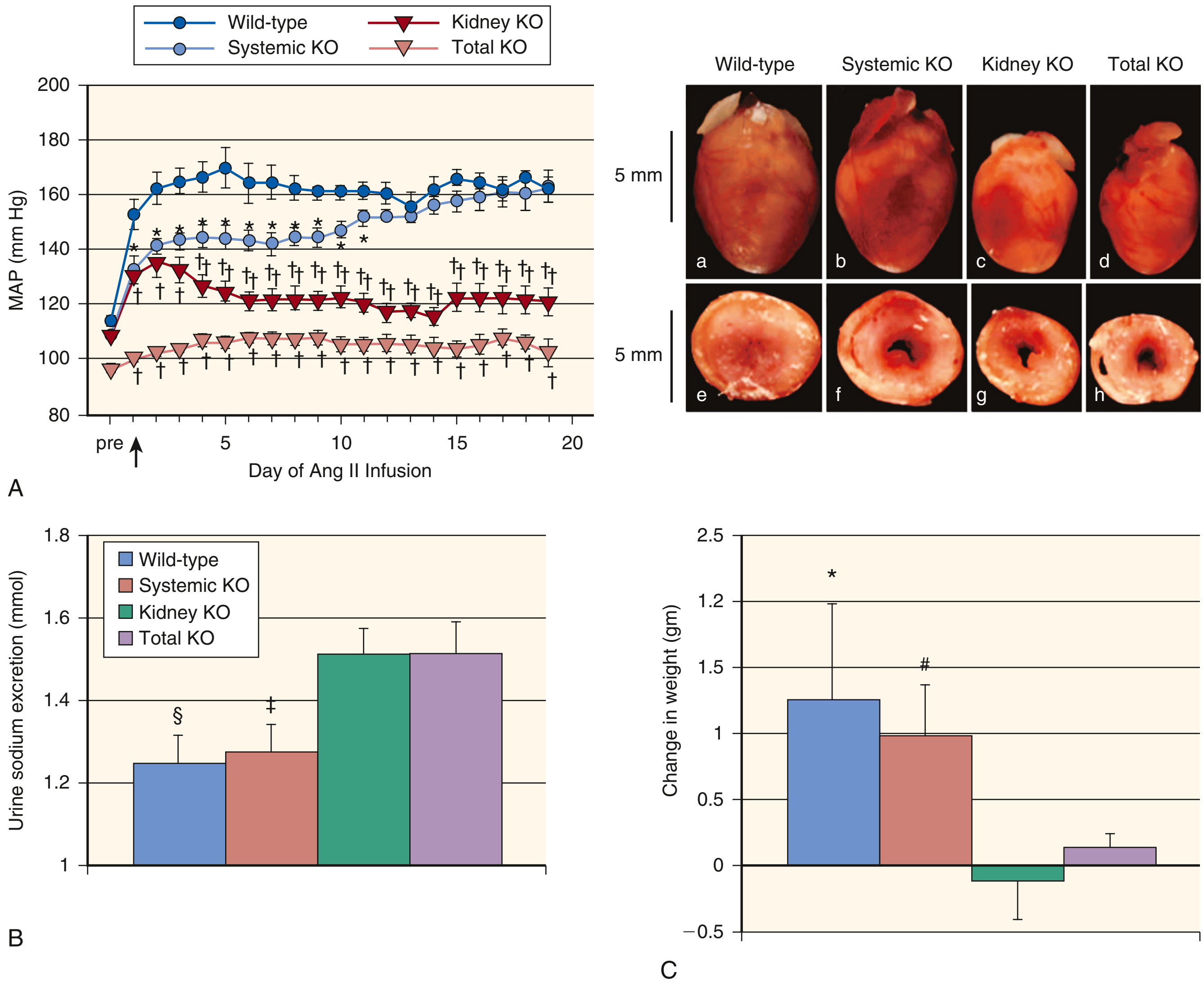


**Fig. 46.4 Effects of angiotensin II *(Ang II)* infusion on blood pressure (A), urinary sodium excretion (B), body weight (C), and cardiac hypertrophy (photos) according to renal and extrarenal presence of Ang II type 1 receptor.** See text for details. *KO,* Knockout; *MAP,* mean arterial pressure. (From Crowley SD, Gurley SB, Herrera MJ, et al. Ang II causes hypertension and cardiac hypertrophy through its receptors in the kidney. *Proc Natl Acad Sci U S A.* 2006;103:17985–17990. © 2006 National Academy of Sciences, U.S.A.)

and endothelium; it has partial homology to ACE and is unaffected directly by ACEIs.[93] ACE2 has a variety of substrates, but its most important action is the conversion of angiotensin II to angiotensin-(1-7). Angiotensin-(1-7) is formed primarily through the hydrolysis of angiotensin II by ACE2, and its actions are opposite to those of angiotensin II, including vasodilatory and antiproliferative properties that are mediated by the Mas receptor, a G-protein coupled receptor that, upon activation, forms complexes with the $AT_1R$, antagonizing the effects of angiotensin II. The vasodilatory effects are mediated by increased cyclic guanosine monophosphate, decreased norepinephrine release, and amplification of bradykinin effects. Studies have identified ACE2 and angiotensin-(1-7) as protective factors in the development of atherosclerosis and cardiac and kidney injury,[93,94] and administration of recombinant ACE2 or its activator, xanthenone, has resulted in improved endothelial function, decreased BP, and diminished renal, cardiac, and perivascular fibrosis in hypertensive animals.[95-97] However, a phase 1 study of recombinant ACE2 in healthy humans did not show BP-lowering effects despite appropriate modulation of the RAAS, including a sustained increase in angiotensin (1-7) levels.[98] The SARS-CoV-2 virus, the etiology of the COVID-19 pandemic, led to studies targeting the ACE2-Ang-(1-7) pathway. Using a synthetic angiotensin (1-7) (TXA-127) to augment the Ang-(1-7) pathway or an angiotensin II type 1 receptor–biased ligand (TRV-027) to block the angiotensin II pathway did not improve oxygen-free days or in-hospital mortality.[99] Whether investigation of these compounds or other therapeutics accelerated by the pandemic have the potential value in the landscape of hypertension treatment remains unknown.[100]

## SYMPATHETIC NERVOUS SYSTEM

The sympathetic nervous system (SNS) is activated consistently in patients with hypertension compared with normotensive individuals, particularly in the obese (Fig.

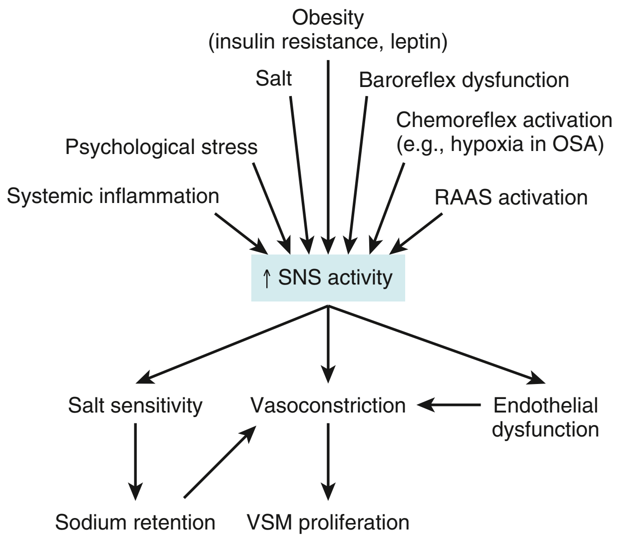


**Fig. 46.5 Causes and consequences of sympathetic nervous system activation in the pathogenesis of hypertension.** (*OSA,* Obstructive sleep apnea; *RAAS,* renin-angiotensin-aldosterone system; *SNS,* sympathetic nervous system; *VSM,* vascular smooth muscle.)

46.5). Many patients with hypertension are in a state of autonomic imbalance that encompasses increased sympathetic and decreased parasympathetic activity.[101,102] SNS hyperactivity is relevant to both the generation and maintenance of hypertension and is observed in human hypertension from early stages. For example, studies in humans have identified markers of sympathetic overactivity in normotensive individuals with a family history of hypertension.[101] Among patients with hypertension, increasing severity of hypertension is associated with increasing levels of sympathetic activity measured by microneurography.[103,104] In human hypertension, plasma catecholamine levels, microneurographic recordings, and systemic catecholamine spillover studies have shown consistent elevation of these markers in obesity, metabolic syndrome, and hypertension complicated by heart failure or kidney disease.[101] In addition, SNS hyperactivity is observed in most hypertensive subgroups, though it appears more pronounced in men than in women, and in younger than in older patients.

Several experimental models have outlined the importance of the SNS in generating hypertension. Different models of obesity-related hypertension indicate that the SNS is activated early in the development of increased adiposity,[102] and the key factor in the maintenance of sustained obesity-related hypertension is increased renal sympathetic nerve activity and its attendant sodium avidity.[102]

SNS-mediated induction of salt sensitivity is a key element to sustaining high BP in other models of hypertension as well. For instance, rats receiving daily infusions of phenylephrine for 8 weeks developed hypertension during the infusions, but BP normalized under a low-salt diet after discontinuation of phenylephrine.[105] However, once exposed to a high-salt diet, the animals again became hypertensive. The degree of BP elevation on a high-salt diet was directly related to the degree of renal tubulointerstitial fibrosis and decrement of GFR. These findings can be interpreted within the paradigm that catecholamine-induced hypertension causes renal interstitial injury that associates with a salt-sensitive phenotype even after sympathetic overactivity is no longer present.[105] In addition, enhanced SNS activity results in $\alpha_1$-receptor–mediated endothelial dysfunction, vasoconstriction, vascular smooth muscle proliferation, and arterial stiffness, all of which contribute to the development of hypertension.

Renalase is a flavoprotein highly expressed in kidney and heart that metabolizes catecholamines and catecholamine-like substances to aminochrome.[106] Tissue and plasma renalase is decreased in experimental models with renal mass reduction, and renalase knockout mice have increased BP and elevated circulating catecholamine levels. This phenotype is reversed by administration of recombinant renalase. Also of relevance to catecholamine metabolism is catestatin, a product of the proteolysis of the neuroendocrine peptide chromogranin A.[107] Catestatin acts at nicotinic cholinergic receptors in adrenal chromaffin cells as an inhibitor of catecholamine release. Chromogranin A knockout mice are hypertensive and have elevated catecholamine levels, both of which are normalized by administration of catestatin. Moreover, serum catestatin levels are decreased in patients with hypertension and their normotensive offspring, raising the possibility of a regulatory role in the development of hypertension. The role of renalase and catestatin in the modulation of SNS-mediated hypertension, as well as their possible value in the treatment of hypertension in humans remain uncertain.

Because increased SNS activity is associated with vascular smooth muscle proliferation, LVH, large artery stiffness, myocardial ischemia, and arrhythmogenesis, there is also a mechanistic role for SNS in the complications of hypertension. In support of this concept, there are several cohort studies reporting an association between physiologic or biochemical markers of SNS activation and adverse outcomes in heart failure, stroke, and end-stage kidney disease.[101,108] However, there are no such studies among patients with hypertension, and the indirect evaluation of the impact of treatment-induced heart rate reduction in hypertension has yielded paradoxical results.

In a meta-analysis of hypertension trials, heart rate reduction during treatment with β-blockers was associated with increased risk for death and CV events in patients with hypertension.[109] In contrast, in a large ($n$ = 10,000) patient outcome trial, a post hoc analysis of heart rate at baseline demonstrated that those with a resting heart rate above 80 beats per minute even with a BP below 140/90 mm Hg had a higher mortality rate.[110] Therefore while apparent that SNS activation is deleterious to patients with CV disease, and presumably with hypertension, a cause for the overactivity should be sought and an attempt made to inhibit that mechanism.

Although the role of the SNS has been known for decades, targeted treatment of this mechanism has not been a common treatment for essential hypertension. β-adrenergic receptor blockers can target β1ARs and inhibit renin release by the SNS. α-1 adrenergic receptor blockers (e.g., doxazosin, terazosin, and prazosin) are second-line antihypertensive agents utilized in states of refractory hypertension, concomitant benign prostatic hypertrophy in men, and α-2 adrenergic receptor agonists (e.g., clonidine, guanfacine, and α methyl dopa) may be useful in labile hypertension or individuals with baroreceptor dysfunction. However, procedures that ablate afferent and/or efferent renal nerves (i.e., renal denervation) may reduce medication burden and have shown modest success in lowering BP in randomized, blinded, sham-controlled clinical trials with a wide, interindividual range of efficacy. These approaches are now approved for treatment of uncontrolled hypertension in the United States. Whether these nonpharmacologic, irreversible approaches become a more common form of treatment around the world is yet unknown.[111-113]

## OBESITY

Approximately 60% to 70% of essential hypertension can be attributed to obesity.[114] Moreover, in large-scale epidemiologic studies such as the INTERSALT study, body mass index, a proxy for obesity, is directly associated with the degree of hypertension.[115] Moreover, in longitudinal observational analyses, each 10 kg of weight loss is associated with a –3.0/–2.2 mm Hg reduction in BP.[115] Obesity-related hypertension is characterized primarily by impaired sodium excretion and endothelial dysfunction, both of which are dependent on SNS overactivity, activation of the RAAS, and increased oxidative stress.[101,116] Fat tissue in obesity is hypertrophied and marked by increased macrophage infiltration.[117] As it is now well described, adipose tissue is not inert and secretes a wide range of cytokines and chemokines whose profile is abnormal in obesity, marked by increased levels of leptin, resistin, interleukin-6, and tumor necrosis factor-α secretion, elevated free fatty acid release, and reduced serum adiponectin concentrations. Decreased serum adiponectin concentrations contribute to insulin resistance, decreased induction of eNOS, and possibly increased sympathetic activity.

Resistin, an adipokine that promotes insulin resistance, impairs nitric oxide (NO) synthesis (eNOS inhibition), and enhances endothelin-1 (ET-1) production, shifts the vasodilation/vasoconstriction balance toward vasoconstriction. Hyperleptinemia directly stimulates the SNS through complex mechanisms that involve central leptin receptors, as well as activation of the pro-opiomelanocortin system (via the melanocortin 4 receptor).[116]

Lastly, visceral adipocyte mass is directly correlated with aldosterone secretion by the zona glomerulosa, a process mediated by angiotensinogen production by adipocytes, as well as increased secretion of Wnt signaling molecules that modulate steroidogenesis.[118-120] In addition (despite a lack of clarity regarding mechanism[s]), obese individuals tend to have lower natriuretic peptides than lean individuals,[116] and this relative deficiency is amplified in patients with obesity and hypertension.[121] All of these factors compound the tendency toward sodium retention and shifting the pressure-natriuresis curve to the right in obese individuals. Activation of these same systems leads to a proinflammatory state related to increased reactive oxygen species, factors directly associated with endothelial dysfunction and vascular proliferation. Obesity is also associated with higher rates of obstructive sleep apnea, which can contribute to elevated BP.[122] Therefore multiple mechanisms contribute to the development and maintenance of hypertension in obese individuals.

## NATRIURETIC PEPTIDES

Natriuretic peptides (atrial [ANP], brain [BNP], and urodilatin) also contribute to hypertension, as well as to salt sensitivity and heart failure.[123] These peptides have important natriuretic and vasodilatory properties that allow maintenance of sodium balance and BP during sodium loading. Upon administration of a sodium load, atrial and ventricular stretch lead to release of atrial natriuretic peptide (ANP) and BNP, respectively, which result in immediate BP lowering due to systemic vasodilation and decreased plasma volume, the latter caused by fluid shifts from the intravascular to the interstitial compartment.[124]

All natriuretic peptides increase GFR, which in volume-expanded states is mediated by an increase in efferent arteriolar tone. They also inhibit renal sodium reabsorption through both direct and indirect effects. Direct effects include decreased activity of $Na^+$-ATPase and ENaC in the distal nephron.[123] The inhibitory effects of natriuretic peptides on renin and aldosterone release mediate indirect effects on BP as well. Unfortunately, understanding the contribution of natriuretic peptides to the development of hypertension in humans is complicated by the elevation of their levels in association with increased BP (due to increased afterload) and hypertensive heart disease.

Some studies have tested whether polymorphisms in ANP or BNP genes resulting in higher levels of these peptides would associate with lower BP; results of these studies have been inconsistent and show small effect sizes.[125-127] There are no published studies evaluating sequential changes in natriuretic peptides and risk for incident hypertension.

There are several unanswered issues about the relation between natriuretic peptides and BP. For example, in a large GWAS of 2.5 million genotyped or imputed SNPs in 69,395 individuals of European ancestry briefly described earlier,[8] the majority of SNPs were directly or indirectly related to abnormalities in natriuretic peptides.

Attention has been given to corin, the serine protease that is largely expressed in the heart and converts pro-ANP and pro-BNP to their active forms. Genetic variants and defects of the enzyme that activates corin have been found in persons with hypertension, heart failure, and preeclampsia. Corin interacts with natriuretic peptides, and hence impaired intracellular trafficking, cell surface expression, and zymogen activation of corin will result in fluid retention.[16,128] Experiments suggest that states of corin deficiency are associated with sodium overload, heart failure, and salt-sensitive hypertension,[128,129] and several clinical studies have observed an association between lower serum corin concentrations and increased risk of adverse cardiovascular outcomes in patients with coronary disease, heart failure, and stroke.[16]

## ENDOTHELIUM

The endothelium is a major regulator of vascular tone and thus plays a key role in BP regulation. Endothelial cells produce a host of vasoactive substances, of which nitric oxide (NO) is the most relevant to BP regulation. Nitric oxide is continuously released by endothelial cells in response to flow-induced shear stress, leading to vascular smooth muscle relaxation through activation of guanylate cyclase and generation of intracellular cyclic guanosine monophosphate (cGMP).[130] Interruption of the production of cGMP via inhibition of the constitutively expressed endothelial NO synthase (eNOS) causes BP elevation and development of hypertension in both animals and humans. Using brachial artery flow-mediated vasodilation and measurement of urinary excretion of NO metabolites as methods to evaluate NO activity in humans, several studies have demonstrated decreased whole-body production of NO in patients with hypertension compared with normotensive controls.

Several elements are responsible for endothelial dysfunction in hypertension. Normotensive offspring of patients with hypertension have impaired endothelium-dependent vasodilation despite normal endothelium-independent responses, thus suggesting a genetic component to the development of endothelial dysfunction.[131] Besides direct pressure-induced injury in the setting of chronically elevated BP, a mechanism of major importance is increased oxidative stress. Reactive oxygen species are generated from enhanced activity of several enzyme systems; reduced nicotinamide adenine dinucleotide phosphate-oxidase (NADPH-oxidase), xanthine oxidase, and cyclo-oxygenase in particular; and decreased activity of the detoxifying enzyme superoxide dismutase.[130,132] Excess availability of superoxide anions leads to their binding to NO, leading to decreased NO bioavailability, in addition to generating the oxidant, proinflammatory peroxynitrite. It is the decreased NO bioavailability that links oxidative stress to endothelial dysfunction and hypertension.[131] Angiotensin II is a major enhancer of NADPH-oxidase activity and plays a central role in the generation of oxidative stress in hypertension, although several other factors are also involved including cyclic vascular stretch, ET-1, uric acid, systemic inflammation, norepinephrine, free fatty acids, and tobacco smoking.[133]

ET-1 is the endothelial cell product that counteracts NO to maintain balance between vasodilation and vasoconstriction. ET-1 expression is increased by shear stress, catecholamines, angiotensin II, hypoxia, and several proinflammatory cytokines such as tumor necrosis factor-α, interleukins 1 and 2, and transforming growth factor-β.[130] ET-1 is a potent vasoconstrictor through stimulation of ET-A receptors in vascular smooth muscle.[134] In hypertension, increased ET-1 levels are not consistently observed. However, there is a trend of increased sensitivity to the vasoconstrictor effects of ET-1. ET-1 therefore is considered a relevant mediator of BP elevation, as ET-A and ET-B receptor antagonists attenuate or abolish hypertension in several experimental models of hypertension (angiotensin II–mediated models, deoxycorticosterone acetate–salt hypertension, and Dahl salt-sensitive rats) and are effective in lowering BP in humans.[135] Endothelin receptor antagonists are utilized in the treatment of pulmonary hypertension and IgA nephropathy. Moreover, a dual endothelin receptor antagonist has shown efficacy to lower BP in patients with treatment-resistant hypertension.[136] Therefore an alternative and more direct treatment of the endothelium may soon provide an additional pathway to target hypertension in patients.

Endothelial cells also secrete a variety of other vasoregulatory substances. These include the vasodilating prostaglandin prostacyclin and several vasodilating endothelium-derived hyperpolarizing factors, the identity of which remains uncertain. In addition to ET-1, there are also endothelium-derived contracting factors, such as locally generated angiotensin II and vasoconstricting prostanoids such as thromboxane $A_2$ and prostaglandin $A_2$. The balance of these factors, along with NO and ET-1, determine the final impact of the endothelium on vascular tone.

Other non–endothelium-derived factors may be of relevance to the genesis of hypertension via endothelium dysfunction. Much attention is given to uric acid, which can induce endothelial dysfunction and produce salt-sensitive hypertension through mechanisms that involve renal microvascular injury.[137,138] These changes can be abrogated by therapies that lower serum urate in animals and may be of value in lowering BP and limiting kidney injury in humans with hyperuricemia. Current evidence from randomized trials, however, only supports a modest BP-lowering effect of allopurinol in hyperuricemic hypertensive adolescents,[139] with no observable effect of allopurinol or probenecid on flow-mediated vasodilatation in obese nonhypertensive adults with mild hyperuricemia.[140] These findings suggest an age-dependent effect or an effect of duration of the exposure to hyperuricemia, even if mild. Also of relevance is the possible role of high dietary fructose consumption in intracellular adenosine triphosphate depletion, increased oxidative stress, increased uric acid production, and endothelial dysfunction.[141,142]

Nitric oxide is not the only gasotransmitter relevant to vascular biology. Hydrogen sulfide ($H_2S$) is another compound that has received attention due to potential treatment relevance.[143] $H_2S$ is a vasodilating gas produced from sulfated amino acids by action of one of two key enzymes, cystathionine γ-lyase (CSE) and cystathionine β-synthase. CSE homozygous knockout mice have ~80% decreased $H_2S$ expression in heart and aorta and ~60% reduction in serum and both homozygous and heterozygous animals demonstrate impaired endothelial function and develop age-dependent hypertension.[144] The mechanisms underlying the BP effects of $H_2S$ are multiple, including enhanced NO-mediated vasodilation, activation of potassium-ATP channels, activation of protein kinase G1α, inhibition of phosphodiesterase type 5, and inhibition of SNS activity.[143] Modulation of serum $H_2S$ concentrations with the administration of gaseous $H_2S$ or $H_2S$ donors (sodium hydrosulfide or sodium thiosulfate) results in lower BP and decreased cardiovascular and renal injury in several experimental models.[143] If delivery systems permit and successful oral use of these agents are developed, it is possible that $H_2S$ may become a therapeutic target in hypertension and vascular disease.

Taken together, the net result observed in patients with hypertension is one of endothelial dysfunction. In cross-sectional analyses, the lesser the degree of forearm flow-mediated vasodilation, the greater the prevalence of hypertension.[132,145] Prospective cohort studies have used flow-mediated vasodilation as a measure of endothelial dysfunction (regardless of specific mechanism) to evaluate its relation with hypertension and test whether endothelial dysfunction is cause or consequence of hypertension, or both.[146] These studies have shown conflicting results, but the larger of them were unable to demonstrate an association between endothelial dysfunction and incident hypertension among 3500 patients followed for a median of 4.8 years.[146]

Furthermore, endothelial dysfunction carries a genetic predisposition that is independent of BP and may be improved by agents that have little or no impact on BP (e.g., some antioxidants).[147] Therefore as it currently stands, the evidence is stronger for endothelial dysfunction as a consequence, not a cause, of hypertension.[145,147]

## ARTERIAL STIFFNESS IN HYPERTENSION

Arterial stiffness is clearly associated with hypertension, particularly the syndromes of isolated systolic hypertension and hypertension with widened pulse pressure. Arterial stiffness develops as a result of structural changes in large arteries, particularly elastic arteries.[148] These include loss of elastic fibers and substitution with less distensible collagen fibers. Factors strongly associated with arterial stiffening include aging, hypertension, diabetes mellitus, CKD, smoking, and high sodium intake.[149]

The most commonly used measure to assess arterial stiffness in humans is carotid-femoral pulse wave velocity (cf-PWV). The traditional view linking arterial stiffness (measured as increased cf-PWV) to hypertension is that faster PWV produces faster reflection of the incident pulse wave, which results in an earlier reflected wave that returned to the central circulation before the end of systole, resulting in increased systolic BP.[150] While these mechanisms still hold true, newer data have highlighted the relevance of two other factors: increased amplitude of the forward wave and increased characteristic impedance of the proximal aorta.[150] When these specific factors are taken into account, the relative contribution of wave reflection to the observed age-dependent change in pulse pressure is only 4% to 11%.

Arterial stiffness was previously thought to be a consequence of hypertension. Cyclical pulsatile load is associated with fracture of elastin fibers and wall stiffening, and increased distending pressure demands recruitment of the less distensible collagen fibers, thus making vessels stiffer.[148] Evidence from several studies, however, indicates that arterial stiffness may precede and predispose to hypertension.[150] For example, in the Framingham Heart Study, markers of arterial stiffness (cf-PWV and amplitude of the forward pressure wave) were associated with a 30% to 60% increased risk for incident hypertension (per standard deviation of each variable) during 7 years of follow-up in a cohort with baseline mean age of 60 years.[151] Conversely, baseline BP did not associate with future changes in arterial stiffness. Other studies corroborate these findings, but other studies also suggest a bidirectional relation such that arterial stiffness is also a consequence of chronic hypertension.[150] Conversely, a cohort study of younger adults (baseline age 36 years) indicated that higher BP was associated with more severe large artery stiffness, not the opposite.[152] These differences in results between younger and older adult populations may indicate that earlier in life, hypertension is mediated by factors that are largely independent of large vessel stiffness, whereas later in life, arterial stiffness is more likely to contribute to the development of hypertension.

Arterial stiffening is relevant to target-organ damage in hypertension. Increased PWV is associated with mortality and an increased risk of CV events,[153] as well as with a variety of subclinical CV injury markers including coronary calcification, cerebral white matter lesions, ankle-brachial index, and albuminuria. The relation between PWV and cardiovascular complications is easily grasped: Increased impedance to left ventricular ejection results in LVH, diastolic dysfunction, and subendocardial myocardial ischemia.

The relations among PWV, brain complications, and kidney complications are more complex. It is apparent that the mechanism of damage of these organs, which are characterized by vasculature with high flow and low impedance, is mediated by increased transmission of increased pulsatile pressure to the brain and kidney parenchymata. The reason for this is related to the abnormal process of "impedance matching." For individuals with normal vessels, the elastic arteries are much less stiff than muscular arteries, thus creating an impedance mismatch. This mismatch provokes wave reflection, thus protecting the tissue located distally to the reflection point from injury caused by traveling pulse wave. In states of increased arterial stiffness, the stiffening of elastic arteries approximates the stiffness of muscular arteries, thus eliminating the protective impedance mismatch. Once impedances become matched, there is less reflection and potential for more severe tissue injury, as supported by a growing body of clinical and experimental literature.[150]

## ROLE OF THE IMMUNE SYSTEM IN HYPERTENSION

Immune responses, both innate and adaptive, participate in several of the mechanisms discussed earlier, including the generation of reactive oxygen species, mediation of the afferent arteriolopathy thought important to maintain salt sensitivity, activation by storage of skin sodium, and participation in the inflammatory changes noted in the kidneys, vessels, and brain in hypertension.[154-156] Innate responses, especially those mediated by macrophages, have been linked to hypertension induced by angiotensin II, aldosterone, and NO antagonism.

Macrophages play opposing but important roles in the regulation of BP. As noted in the section on "Nonosmotic Sodium Storage" earlier, macrophages in the skin interstitial sodium levels, and depletion of macrophages and consequent TonEBP/VEGFC signaling, causes hypertension.[72,73] Conversely, macrophages in the kidney and vasculature contribute to hypertension. Reductions in macrophage infiltration of the kidney or the periadventitial space of the aorta and medium-sized vessels lead to improvements in BP and salt sensitivity in several experimental models.[155,157]

Adaptive responses via T cells have been linked to the genesis and complications of hypertension. T cells express $AT_1R$ and mediate angiotensin II-dependent hypertension, as demonstrated by the observations that adoptive transfer of T cells restored the hypertensive phenotype in response to angiotensin II infusion that was absent in mice without lymphocytes.[157] Abnormalities in both proinflammatory T cells and regulatory T cells alike are implicated in complications of hypertension, as they appear to regulate vascular and renal inflammation that underlies target-organ injury. Suppression of these inflammatory responses can improve BP control.[154,155,158,159] T helper 17 cells that produce interleukin-17 are critical for sustained angiotensin II-dependent hypertension in mice. Serum interleukin-17 concentrations are higher in patients with hypertension, and interleukin 17 has been shown to independently cause hypertension and kidney disease.[160] The pleiotropic actions of interleukin-17 are reviewed elsewhere in the text but include a number of mechanisms addressed here including increased renal sodium retention, endothelial dysfunction, and arterial stiffness.

How are these T cells activated? High sodium, whether by high dietary sodium or more specifically, by increased interstitial sodium, is known to activate/polarize immune cells including T helper 17 cells to a proinflammatory and high-producing interleukin-17 state.[161,162] Dendritic cells are sensors for high dietary sodium by activation of serum-and-glucocorticoid kinase 1 (SGK1) that, in turn, stimulates ENaC expression on the surface of dendritic cells to promote sodium entry and isoketal (isolevuglandin) formation and isoketal adducts on intracellular proteins in mice. These adducts promote activation of T cells and cytokines including the aforementioned interleukin-17.[163] These dendritic cells are also activated by hypertension, establishing a positive feedback loop. Isoketal production and proinflammatory cytokines are also elevated in patients with hypertension versus controls.[164]

B lymphocytes may also play a causative role in hypertension, as suggested by reports of several autoantibodies including agonistic antibodies against adrenergic receptors, vascular calcium channels, and $AT_1R$, and antibodies against endothelial cells causing endothelial dysfunction, or heat shock proteins (hsp70) causing salt-sensitive hypertension. Further research will determine if manipulation of immune targets is of value in the prevention and treatment of human hypertension.

## OTHER METABOLIC PEPTIDES AND HYPERTENSION

Several vasodilating substances act as compensatory vasodilators to balance the heavily provasoconstrictive milieu in hypertension. Some of these vasodilators act primarily through an increase in NO release from endothelial cells, such as calcitonin gene–related peptide, adrenomedullin, and substance P. The glucose-regulating, gut hormone glucagon-like peptide-1 (GLP-1) has vasodilating properties, and its administration to Dahl salt-sensitive rats improves endothelial function, induces natriuresis, and lowers BP.[165] Additionally, a meta-analysis of GLP-1 receptor agonists to treat obesity in humans demonstrated a modest but significant BP reduction (–3.4/–1.1 mm Hg) compared with placebo.[166]

## GUT MICROBIOME

The gut microbiome may be related to several humoral and hemodynamic aspects of hypertension, including neuroimmunologic aspects of BP regulation including regulation of the RAAS and activation of the SNS.[167] Bacteria that produce short-chain fatty acids (SCFAs) (especially acetate, propionate, and butyrate) are seen as favorable to produce an environment of lower BP.[167] A proof-of-concept study demonstrated that an unfavorable gut microbiome profile (i.e., one that leads to less available SCFAs) is seen in spontaneously hypertensive rats as compared with normotensive Wistar Kyoto rates and is observed in animals exposed to angiotensin II infusions.[168] In addition, analyses of stool samples of 10 normotensive and 7 hypertensive patients showed a similar pattern.[168] Germ-free mice compared with controls with a presumably conventional microbiome exhibit lower BP in response to angiotensin II and lower vascular fibrosis.[169] Although the role of microbiome dysbiosis in hypertension has been shown in multiple species, the precise molecular mechanisms are not clear. There are multiple SCFAs; each can act through multiple G-protein coupled olfactory receptors; each of these receptors has multiple SCFA ligands; and these receptors are expressed in multiple tissues including endothelial cells, immune cells, and juxtaglomerular cells.[170,171] There may also be regulatory networks in the microbiome and the host that alter the levels of each of these components. For example, in a study of 70 patients, the intestinal production of SCFAs and the expression level of cognate G-protein coupled receptors on immune cells differed between normotensive patients and those with hypertension, raising the possibility that shifts in microbial gene pathways may alter responses to BP-regulatory metabolites.[172]

High dietary sodium can also alter the bacterial components of the microbiome (dysbiosis) in mice. Moreover, fecal transplant of microbiome from high sodium-fed mice induces isoketal protein adducts, proinflammatory T cells, interleukin-17, and elevated BP.[173] The mechanism of how dietary sodium altered the microbiome in this study is unknown.

The use of probiotics, typically as yogurt or other dairy products, is the most commonly used approach to alter the gut microbiome. A meta-analysis of nine randomized clinical trials investigating the use of probiotics to lower BP showed an average BP reduction of 3.6/2.4 mm Hg.[174] Only one study included hypertensive patients (average BP reduction 9.7/4.4 mm Hg), but an analysis of BP lowering according to baseline BP above or below 130/85 mm Hg showed no differences in systolic BP and a small difference in diastolic BP reduction (2.7 vs. 0.9 mm Hg).

These early results indicate that modification of the gut microbiome may alter hemodynamics and that the use of probiotics, presumably through such microbiome changes, may have salutary effects on BP, though the magnitude and clinical relevance of this effect remain to be determined. Further research to target specific ligands and/or receptors may be useful to better understand the role of the microbiome in hypertension.

# CLINICAL EVALUATION

The evaluation of patients with hypertension focuses on six key components: 1. the confirmation that the patient is indeed hypertensive through careful measurements of BP; 2. an assessment of clinical features that might suggest specific remediable causes of hypertension; 3. the identification of comorbid conditions that confer additional CV risk, or that may inform treatment decisions; 4. the discussion of patient-related lifestyle factors and preferences that will affect management; 5. the systematic evaluation of hypertensive target-organ damage; and 6. shared decision making about the treatment plan. To accomplish this, the clinician often needs multiple visits, a targeted clinical examination, and selected laboratory and imaging tests.

## HISTORY AND PHYSICAL EXAMINATION

The medical history and physical examination are essential to uncovering possible secondary causes of hypertension, identifying signs and symptoms suggestive of hypertensive target-organ damage, and diagnosing comorbid conditions that may affect treatment decisions. While the focus is traditionally on the CV, central nervous system, and kidneys, a complete review of systems is indicated when the patient is first evaluated, as some patients will present with hypertension because of sleep apnea (snoring, witnessed apneas/gasping), hyperthyroidism or hypothyroidism (each with their litany of possible symptoms), hyperparathyroidism (symptoms of hypercalcemia), Cushing syndrome (symptoms of cortisol excess), pheochromocytoma or paraganglioma (symptoms of catecholamine excess), or acromegaly with its distinctive physical findings. These conditions are discussed in detail later in this chapter.

High BP is typically asymptomatic, but some symptoms are common among patients with very high BP levels, such as headaches, epistaxis, dyspnea, chest pain, and faintness, all of which were present in more than 10% of patients presenting with diastolic BP levels above 120 mm Hg.[175] Other common symptoms are nocturia and unsteady gait, whereas treated patients often complain of fatigue in addition to symptoms of overtreatment and those related to specific side effects of medications.

When searching for target-organ damage, one looks for signs and symptoms to suggest a previous stroke or transient ischemic attack, previous or ongoing coronary ischemia, heart failure, peripheral arterial disease, or a history of kidney disease or current symptoms such as hematuria or flank pain.

Obtaining a detailed family history as it pertains to hypertension is important. Focus should be on the development of hypertension at a young age or clustering of endocrine (pheochromocytoma, multiple endocrine neoplasia, primary aldosteronism) or kidney problems (autosomal dominant polycystic kidney disease or other inherited forms of kidney disease). The young patient with hypertension and a family history of hypertension poses a particular challenge and should be evaluated in detail. Table 46.3 provides a guide to possible causes to be entertained.[176]

Knowledge of several conditions with potential relevance to treatment is important. For example, issues related to CV risk management such as diabetes mellitus, hypercholesterolemia, and tobacco use need to be evaluated. Patients with established CV disease will need some treatments for both their hypertension and their underlying disorder (e.g., β-blockers for angina pectoris), so knowledge of specific CV diagnoses is essential. Lastly, some non-CV conditions may have an impact on treatment options. For example, patients with reactive airway disease (asthma) probably should not receive β-blockers, patients with prostatic hyperplasia may benefit from a regimen that includes an α-blocker, and patients with attention-deficit/hyperactivity disorder or anxiety may benefit from a central sympatholytic (e.g., guanfacine), whereas those with major depression should probably not be treated with this drug class.

It is also important to recognize that it is during the history that the clinician has the opportunity to explore issues related to lifestyle, cultural beliefs, and patient preferences that will be essential in designing an effective treatment plan. It is important to define dietary and physical activity patterns and, when problems are identified, to determine if the patient is willing and/or able to modify them. Cultural beliefs related to the treatment of hypertension, health literacy, and mistrust in physicians and the pharmaceutical industry are several of the items that can affect the relationship with the patient and that should be openly raised. It is only then that patients will be able to participate in shared decision making about their treatment, an essential tenet of patient-centered care.

The physical examination is designed to complement items discussed in the history. One should pay attention to syndromic features of cortisol excess (moon face, central obesity, frontal balding, cervical and supraclavicular fat deposits, skin thinning, abdominal striae), hyperthyroidism (tachycardia, anxiety, lid lag/proptosis, hypertelorism, pretibial myxedema), hypothyroidism (bradycardia, coarse facial features, macroglossia, myxedema, hyporeflexia), acromegaly (frontal bossing, widened nose, enlarged jaw, dental separation, acral enlargement), neurofibromatosis (neurofibromas, café au lait spots, as neurofibromatosis is associated with pheochromocytoma and renal artery stenosis), or tuberous sclerosis (hypopigmented ash leaf patches, facial angiofibromas, as tuberous sclerosis is associated with renal hypertension, usually related to angiomyolipomas). Other even rarer associations exist but fall beyond the scope of this chapter.

In younger patients or in patients with unexplained, difficult-to-treat hypertension, it is worth exploring the possibility of coarctation of the aorta by measurement of BP in both arms and in one thigh. If present, there will be a significantly lower BP in the thigh (typically by >30 mm Hg). Sometimes, in case of a lesion proximal to the left subclavian, there may be a significant interarm difference, lower on the left. In addition, there is significant decrease in intensity of the femoral pulses and a palpable radial-femoral pulse delay.

Funduscopic examination is valuable in many patients, particularly those with severe hypertension (BP ≥180/110 mm Hg). The retinal changes are associated with severity of both acute and chronic BP elevation. Acute changes can happen quite abruptly (hours to days) and range from arteriolar spasm in most patients with uncontrolled BP to retinal infarcts (exudates) and microvascular rupture (flame hemorrhages), to papilledema once the protection afforded by vasoconstriction is overcome. Chronic changes take much longer to develop and include vascular tortuosity (arteriovenous nicking) due to perivascular fibrosis, followed by progressive arteriolar wall thickening that prevents visualization of the blood column, thus leading to the appearance of copper wiring, then silver wiring. Several studies have demonstrated a relationship between severity of hypertensive retinopathy and risk for LVH and stroke. Smartphone-based technology allows the use of a condensing lens coupled with the smartphone's own camera for video and photography of the retina (undilated pupil) in place of a conventional ophthalmoscope.[177,178] These devices can be used even by inexperienced operators, and the images can be shared for more precise diagnosis.

**Table 46.3 Clinical Clues to Guide the Investigation in Young Hypertensive Patients With a Potential Hereditary Cause**

| Possible Causes of Familial Hypertension | | Clinical Clues |
|---|---|---|
| **Catecholamine-Producing Tumors** | | |
| Pheochromocytoma/ Paraganglioma | Familial cases are responsible for up to 40% of cases. | Paroxysmal palpitations, headaches, diaphoresis, pale flushing. Syndromic features of any of the associated disorders (see PPGL section for details). |
| Neuroblastomas (adrenal) | 1%-2% of neuroblastomas are familial. | Symptoms of the abdominal tumor (pain, mass) or catecholamine release (same as PPGL). |
| **Parenchymal Kidney Disease** | | |
| Glomerulonephritis | Alport disease (X-linked, AR or AD), familial IgA nephropathy (AD with incomplete penetrance). | Proteinuria, hematuria, low eGFR. |
| Polycystic kidney disease | ADPKD type 1 or 2, ARPKD. | Multiple kidney cysts (as few as 3 in patients under the age of 30). |
| **Adrenocortical Disease** | | |
| Glucocorticoid-remediable aldosteronism (familial hyperaldosteronism type 1) | AD chimeric fusion of the 11-β-hydroxylase and aldosterone synthase genes. | Cerebral hemorrhages at young age, cerebral aneurysms. Mild hypokalemia. High plasma aldo, low renin. |
| Familial hyperaldosteronism type 2 | Mutation in *CLCN2* gene (chloride channel protein 2 [ClC-2]). | Severe hypertension in early adulthood. High plasma aldo, low renin. No response to glucocorticoid treatment. |
| Familial hyperaldosteronism type 3 | Mutation in *KCJN5* gene (G-protein–activated inward rectifier potassium channel 4 [GIRK4]). | Severe hypertension in childhood with extensive target-organ damage. High plasma aldo, low renin. Marked bilateral adrenal enlargement. |
| Familial hyperaldosteronism type 4 | Mutation in *CACNA1H* gene (voltage-dependent T-type calcium channel subunit α-1H [Cav3.2]). | Early onset, accompanied by developmental delay or attention-deficit disorder. |
| Congenital adrenal hyperplasia | AR mutations in 11-β hydroxylase or 17-hydroxylase. | Hirsutism, virilization. Hypokalemia and metabolic alkalosis. Low plasma aldo and renin. |
| **Monogenic Primary Renal Tubular Defects** | | |
| Gordon syndrome | AD mutations of *KLHL3, CUL3, WNK1, WNK4*. AR mutations of *KLHL3*. | Hyperkalemia and metabolic acidosis with normal renal function. |
| Liddle syndrome | AD mutations of the epithelial sodium channel. | Hypokalemia and metabolic alkalosis. Low plasma aldo and renin. |
| Apparent mineralocorticoid excess | AD mutation in 11-β-hydroxysteroid dehydrogenase type 2. | Hypokalemia and metabolic alkalosis. Low plasma aldo and renin. |
| Geller syndrome | AD mutation in the mineralocorticoid receptor. | Hypokalemia and metabolic alkalosis. Low plasma aldo and renin. Increased BP during pregnancy or exposure to spironolactone. |
| **Unknown Mechanisms** | | |
| Hypertension-brachydactyly syndrome | AD mutation in the phosphodiesterase 3 *(PDE3)* gene. | Short fingers (small phalanges) and short stature. Brainstem compression from vascular tortuosity in the posterior fossa. |

*AD,* Autosomal dominant; *Aldo,* aldosterone; *AR*, autosomal recessive; *PKD*, polycystic kidney disease; *PPGL,* pheochromocytoma/paraganglioma.
Modified from Peixoto AJ. Attending rounds: a young patient with a family history of hypertension. *Clin J Am Soc Nephrol*. 2014;9:2164–2172.

The CV examination focuses on the identification of volume overload (jugular venous distension, lung crackles, peripheral edema), cardiac enlargement (deviated cardiac impulse), and the presence of a third or fourth heart sound as markers of impaired left ventricular compliance. We routinely look for bruits over the carotid arteries, as the prevalence of carotid atherosclerosis is increased in patients with hypertension, as well as in the abdomen, primarily looking for renal arterial bruits heard over the epigastrium and/or flanks. These bruits are of greater significance if occurring on both systole and diastole. Finally, detailed palpation of the peripheral pulses of the arms and legs is important to look for signs of peripheral arterial disease.

To wrap up the examination, a focused neurologic examination looks for obvious cranial nerve abnormalities, motor deficits, or speech or gait abnormalities. Any further testing is based on specific symptoms or on focal findings on the screening examination.

## BLOOD PRESSURE MEASUREMENT

Because treatment decisions are based largely on BP levels, accurate BP measurement is essential. Cuff-based brachial BP is the most used method to measure BP, typically in the office setting. However, the use of 24-hour ABPM and home BP monitoring has become part of routine hypertension care in many parts of the world, as we discuss later. Finally, there has been extensive interest in the development of cuffless BP measurement devices using technologies based on pulse-wave analysis through photoplethysmography, tonometry, or bioimpedance. These are exciting technologies that will ultimately allow nonintrusive continuous or semicontinuous BP measurement and monitoring. We welcome these developments, but there are still validation issues that need to be addressed by manufacturers and regulatory agencies before these devices become widely used.[179]

### OFFICE BLOOD PRESSURE MEASUREMENT

Office BP measurement is the time-honored method for the diagnosis and management of hypertension. It is strongly associated with hypertension-related outcomes based on more than 50 years of observational and clinical trial data. Accordingly, guidance provided to clinicians for the diagnosis and treatment of hypertension by most major guidelines is based on office BP values.

Attention to BP measurement technique is essential.[180] Fig. 46.6 displays the key elements necessary for optimal BP measurement including attention to equipment and environment, patient preparation, and positioning. Averaging of several measurements is important to minimize variability, avoid over/undertreatment, and best predict clinical outcomes.

Many oscillometric automated BP devices allow automatic performance of three to five unwitnessed BP measurements. Such features (unwitnessed measurements) had been considered a better proxy for overall BP burden than conventional office BP, but current evidence does not support a difference between witnessed and unwitnessed measurements as long as both are performed in a similarly rigorous manner.[181]

Most patients should have their BP measured in the arm in the seated position. In selected situations, such as malformations, injuries, or extensive vascular disease of the upper extremities, or when comparing BP levels in the upper and lower extremities, it may be necessary to use thigh measurements with an appropriately sized thigh cuff, which should be obtained in the prone position to allow the cuff to be at the level of the heart. Mercury sphygmomanometers are now seldom available in clinical practice because of environmental concerns. Aneroid and electronic oscillometric manometers are accurate but should have periodic maintenance (every 12 months) to ensure that they are properly calibrated, as well as any time poor function is suspected. It is important that accurate BP devices be used for BP measurement. STRIDE BP (https://www.stridebp.org) is an international nonprofit organization founded by hypertension experts with the mission of improving the accuracy of BP measurement. STRIDE BP provides guidance and tools for accurate BP measurement, including an up-to-date list of validated BP monitors for use in the office and ambulatory settings.

### ORTHOSTATIC BLOOD PRESSURE MEASUREMENT

Orthostatic hypotension commonly accompanies uncontrolled hypertension, especially among older patients, where it occurs in 8% to 34% of patients.[182] Some guidelines now provide specific recommendations for measurement of standing BP to screen for orthostatic hypotension in older patients with hypertension, as well as in patients at increased risk for autonomic dysfunction, such as those with diabetes and kidney disease.[5]

The frequency of orthostatic hypotension increases with advancing age, the presence of hypertension, and the number of antihypertensive drugs. Orthostatic hypotension is a risk factor for syncope and falls. Therefore evaluation for the presence of orthostatic hypotension should be part of the assessment of risks and benefits of drug treatment.

Orthostatic vital signs (heart rate and BP) are best obtained after at least 5 minutes in the supine position, followed by immediate assumption of the standing position, when sequential measurements are taken for up to 3 minutes. The difficulties of following this protocol in a busy clinical practice are recognized, so it is acceptable to compare values in the seated position with those after standing for 1 minute; this approach results in decreased sensitivity for the detection of orthostatic hypotension but is better than no measurement at all.[183] To account for this fact, it has been proposed that a fall of 15/7 mm Hg be used for the definition of orthostatic hypotension when the test is performed using the seated BP as baseline as compared with the generally accepted definition of orthostatic hypotension as a drop in BP of more than 20/10 mm Hg that occurs after 3 minutes of standing.[184] Among patients with supine hypertension, it is recommended that the definition of systolic fall in BP for the diagnosis is a drop of more than 30 mm Hg as the level of baseline BP is directly proportional to the orthostatic BP fall.

Integration of the heart rate response to changes in BP during orthostasis can guide the differential diagnosis and further evaluation of orthostatic hypotension. A normal chronotropic response in the presence of orthostatic hypotension is defined as an orthostatic rise in heart rate more than 50% of the fall in systolic BP.[185] In the absence of medications with a negative chronotropic effect (primarily β-adrenergic blockers, central antiadrenergic agents, nondihydropyridine CCBs, and ivabradine), the lack of a tachycardic response to orthostatic hypotension is indicative of baroreflex or sympathetic autonomic dysfunction. Patients with an appropriate tachycardic response, on the other hand, likely have volume depletion or excessive vasodilation.

### OFFICE VERSUS OUT-OF-OFFICE BLOOD PRESSURE

Office BP measurement is a time-honored method to evaluate hypertension. It is easy to perform and is widely available at low cost. Home BP is also widely available, though accessibility to low-income patients is still a problem despite the availability of low-cost devices. ABPM, on the other hand, is less widely available due to costs and limited reimbursement by third-party payers in the United States.

| | |
|---|---|
| **Step 1**<br>**Facility and equipment** | • Quiet room with a comfortable temperature.<br>• Clinically validated BP measurement device; an automated device measuring BP at the brachial artery is recommended.<br>• A range of cuff sizes to fit a range of upper-arm circumferences. |
| **Step 2**<br>**Personnel performing BP measurement** | • Trained health care professional should perform the BP measurement. Annual retraining is recommended. |
| **Step 3**<br>**Prepare the patient** | • The patient should be provided with instructions to abstain from caffeine, alcohol, nicotine, and exercise for at least 30 minutes prior to the BP measurement.<br>• Eliminate discomfort such as a full bladder.<br>• Prior to the BP measurement, there should be a short rest period (3–5 minutes) without provocation (including talking, or being talked to in-person or on the phone). |
| **Step 4**<br>**The measurement procedure**<br>**[see figure below]** | • The health care professional should explain the procedure, including the number of BP measurements to be obtained.<br>• Use the arm with the higher SBP readings during an initial visit, unless a new medical condition (e.g., arm ischemia) has developed in the interim in that arm.<br>• ≥2 measurements should be obtained at least 30 seconds apart; the values should be averaged and recorded. |

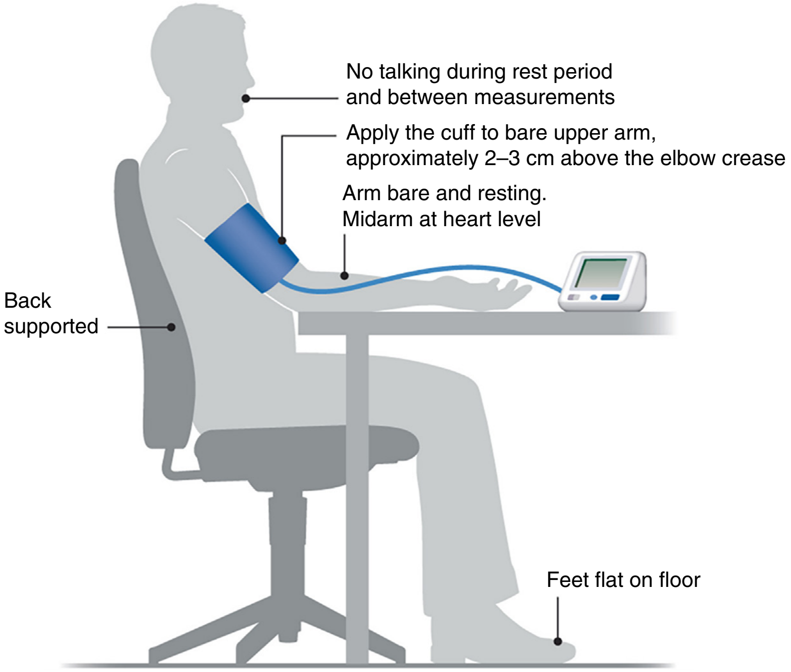


**Figure** Steps for implementing standardized clinic BP measurement. *BP*, Blood pressure; *SBP*, systolic blood pressure.

**Fig. 46.6 Steps for optimal blood pressure measurement.** (From Cheung AK, Whelton PK, Muntner P, et al. International Consensus on Standardized Clinic Blood Pressure Measurement—A Call to Action. *Am J Med.* 2023;136:438–445 e1.)

Home BP monitoring protocols and ABPM include larger numbers of readings, thus decreasing variability and improving reproducibility.[186]

In the past 30 years, ABPM and home BP have become accepted as better markers of hypertensive target-organ damage and adverse clinical outcomes. ABPM has stronger associations with several measures of LVH, albuminuria, kidney dysfunction, retinal damage, carotid atherosclerosis, and aortic stiffness than office BP, although this is not consistent among studies.[187] Likewise, home BP is a better marker than office BP for LVH and proteinuria, though it is not consistently superior for other measures of target-organ damage.

In the assessment of risks for CV endpoints, out-of-office BP has consistently outperformed office BP in studies that account for values observed in the office; in other words, no matter the office BP, out-of-office BP associates most strongly with outcomes. A systematic review by the NICE clinical guidelines group in the United Kingdom identified nine cohort studies comparing ABPM with office BP; ABPM was superior in eight and equal to office BP in one.[188] For home BP, they identified three studies comparing it with office BP; home BP was superior in two and equal in one. Lastly, two studies compared ABPM, home BP, and office BP; of these, one showed superiority of both ABPM and home BP while the other study did not show differences among any of the three methods.

In meta-analyses of studies that evaluated both office and ABPM on outcomes, only ABPM values retained significance.[189,190] Likewise, in the largest home BP cohort study that included simultaneous use of office and home BP to predict CV events and mortality, only home BP remained significantly associated with these adverse outcomes.[191] Similar observations of the superior prognostic performance of out-of-office methods exist for patients with resistant hypertension, chronic kidney disease (CKD), hemodialysis, and the general population.

In summary, evidence from prospective cohort studies convincingly demonstrates the superiority of out-of-office BP measurements as predictors of hypertension outcomes.

There are several possible explanations for the superiority of out-of-office measurements in outcomes assessment:

1. There is lower variability and better reproducibility afforded by the larger number of readings across a longer period of observation, thus making ABPM/home BP better reflections of "BP burden."
2. Home BP and ABPM can detect "white coat" and "masked" hypertension. White coat hypertension, or isolated office hypertension, is the occurrence of high BP in the office and normal BP values in the out-of-office environment. It occurs in 20% to 30% of patients with a diagnosis of office hypertension.[192] It has generally been noted that patients with white coat hypertension have similar CV outcomes as normotensive individuals.[193,194] However, data from the large International Database of Home Blood Pressure in Relation to Cardiovascular Outcome have shown a significant increase in fatal and nonfatal CV events among untreated patients with white coat hypertension diagnosed based on home BP compared with untreated normotensive persons (hazard ratio [HR], 1.42; $P$ = 0.02).[195,196] Interestingly, treated patients with hypertension who retained a white coat effect had the same overall risk as treated patients whose BP was controlled both in the office and at home (HR, 1.16; $P$ = 0.45). Moreover, an updated meta-analysis of 14 observational ABPM studies revealed an increase in risk of CV events (HR 1.7) and CV mortality (HR 2.8) among patients with white coat hypertension compared with normotensive controls, but no statistically significant increase in stroke or all-cause death. As in previous analyses, sustained hypertension was associated with a much higher risk.[197]

   In addition, white coat hypertension has been associated with an "intermediate phenotype" between normotension and hypertension as it pertains to left ventricular mass, carotid intima-media thickness, aortic pulse wave velocity, and albuminuria.[195] However, there are no data available to demonstrate that patients with white coat hypertension benefit from drug therapy. Therefore it appears that white coat hypertension may not be as benign as previously considered, and patients should be advised on general lifestyle changes to improve BP levels and overall vascular risk, especially as their risk for progressing to sustained hypertension is approximately 40% to 50% after 10 to 11 years of follow-up.[198,199] Masked hypertension, conversely, consists of normal BP in the office but high BP in the ambulatory setting, with an estimated prevalence of 10% to 15% in population studies. It has been consistently and strongly associated with increased risk for adverse CV end points and mortality, to a level that is indistinguishable from that associated with sustained hypertension.[193] The very existence of masked hypertension is troublesome, as its identification is only possible with BP measurement in the out-of-office environment. This finding has important public policy implications related to screening that remain unresolved at this time. However, in a first step to address this issue, the ACC/AHA guidelines indicate that it may be reasonable to use out-of-office BP to screen for masked hypertension in adults with office BP readings consistently between 120 and 129/80 and 84 mm Hg.

   Because white coat hypertension and masked hypertension afflict such a substantial number of patients and have diametrically different impact on outcomes, their identification improves outcome prediction in patients with hypertension.[200]
3. The ability to evaluate BP during sleep was a characteristic until now restricted to ABPM, though newer home BP monitors can be programmed for activation during sleep. In some, but not all, studies nighttime BP is a better marker of CV disease than daytime or 24-hour-average BP.[201-203] The implications of nighttime BP (compared with daytime levels) appear greater among treated patients, perhaps because antihypertensive treatment, often taken in the morning, might result in better BP control during the day than during the night.[204] The pattern of BP fluctuation between day and night also associates with prognosis. The normal circadian BP pattern includes a fall in BP of approximately 15% to 20% during sleep. Patients who lack this normal BP dip during sleep are called "nondippers" (arbitrarily defined as a sleep BP that falls by <10% compared with awake levels) and have increased target-organ damage and overall CV risk. In large observational studies, patients whose systolic BP falls by 20% or more during the night have lower fatal and nonfatal CV event rates than those whose BP decreases by <20%, while those whose BP does not fall at all during the night have significantly worse CV outcomes than all other patients.[205]
4. ABPM also provides information on BP variability throughout the day, which may add further prognostic information. Increased BP variability (expressed using several different metrics) has been associated with increased event rates, though these findings are of small magnitude when taken independently from BP values.[206] However, use of BP variability (short or long term) is not yet supported by data to guide treatment.

Despite these observations, objective evidence demonstrating that outcomes are better when patients are managed using an out-of-office method is lacking. Three randomized clinical trials have compared management of hypertension with office or out-of-office BP, one using 24-hour ABPM[207] and two using home BP.[29,208,209] All of these studies showed that more patients managed with out-of-office methods could have treatment stopped or deescalated, thus resulting in marginal cost savings. However, none of them could demonstrate the superiority of ABPM or home BP in achieving better BP control (the primary outcome of all three trials) or less LVH (evaluated in all studies as a secondary outcome).

## CLINICAL USE OF AMBULATORY AND HOME BLOOD PRESSURE MONITORING

ABPM has been in clinical use for half a century. In the United States, problems related to reimbursement have significantly limited its expansion compared with other parts of the world. Despite this limitation, there is general agreement on its value in several clinical circumstances, as outlined in Table 46.4.

ABPM is performed, typically, over a period of 24 hours, although it can be extended for longer periods (e.g., 48 hours) to provide information covering more than one wake/sleep cycle, or to cover a specific period in detail, such as a 2-day interdialytic period for a patient undergoing hemodialysis. Clinicians should use an independently validated monitor (list curated by STRIDE BP, https://www.stridebp.org).A typical measurement interval is every 20 minutes during the daytime (7 AM to 11 PM) and every 20 to 30 minutes at night (11 PM to 7 AM), though the frequency and time windows can be adjusted based on clinical needs, such as the need to identify frequent BP swings, atypical sleep patterns, etc. Patients should keep a log of activities during the day, the time of retiring to bed and waking up, and time of taking vasoactive medications (if applicable). It is preferred that the periods designated as "night" and "day" reflect the actual periods of sleep and wakefulness obtained from the patient's diary. Most patients tolerate the procedure well, although sometimes sleep is compromised (<10% of cases), and, rarely, patients have excessive bruising or discomfort from the frequent cuff inflations. Up-to-date instructions on how to perform and interpret ABPM studies are available in guideline format from the European Society of Hypertension.[210]

Home BP is performed by the patient in the home (or sometimes work) environment. It is used commonly in clinical practice and is associated with improved adherence to therapy. It has also been used successfully for treatment self-titration of BP medications and is amenable to telemedicine approaches, in which the patient can upload BP values via telephone or direct entry to a Web server so that clinicians can inspect the BP logs and make treatment decisions remotely.

Just as with office BP, it is important that the equipment fits the patient well and that measurements are obtained using the same technique as outlined earlier for office BP. Independently validated devices are listed at http://www.stridebp.org; unfortunately, many of the marketed devices have not been independently validated. The preferred devices use arm cuffs.[210] Finger cuffs are inaccurate and should not be used. Wrist cuffs often provide incorrect readings because of inappropriate technique, but if used correctly can be quite convenient and accurate, and particularly useful in obese patients. As discussed earlier, cuffless devices applied to different parts of the body are of great interest but not yet ready for general use as of this writing.

To allow management decisions, home BP monitoring is best performed using specific periods of monitoring. For most patients, a BP log obtained over 7 days before each office visit suffices, as it retains excellent reproducibility.[210] We recommended that the patient obtain readings in duplicate (approximately 1 minute apart), twice daily (in the morning before taking medications and in the evening before dinner). In selected situations, more frequent or more prolonged monitoring may be needed. For example, patients with hypotensive symptoms may benefit from BP measurements during peak action of medications, such as in the mid to late morning or late evening, depending on the time when medications are taken. Likewise, patients with labile BP can be monitored more often to capture the overall BP variability better, though we prefer to use ABPM in such patients. Detailed home BP guidelines are available from

**Table 46.4 Indications for 24-Hour Ambulatory and Home Blood Pressure Monitoring**

| Indication | Home BP Monitoring | ABPM | Comment |
|---|---|---|---|
| Identify white coat hypertension | ++ | +++ | ABPM still the gold standard when patients have home BP values that are borderline (125-135/80-85 mm Hg) |
| Identify masked hypertension | ++ | +++ | |
| Identify true-resistant hypertension | ++ | +++ | |
| Evaluate borderline office BP values without target-organ damage | ++ | +++ | |
| Evaluate nocturnal hypertension | — | +++ | |
| Evaluate labile hypertension | ++ | ++ | Home BP better for infrequent symptoms or paroxysms; ABPM better if frequent within a 24-h period |
| Evaluate hypotensive symptoms | +++ | ++ | |
| Evaluate autonomic dysfunction | + | ++ | Home BP useful to monitor orthostatic hypotension; ABPM useful to quantify supine hypertension and determine overall (average) BP levels |
| Clinical research (treatment, prognosis) | ++ | +++ | |

*ABPM,* Ambulatory blood pressure monitoring; *BP,* blood pressure.

**Table 46.5 Consensus-Based Home and Ambulatory Blood Pressure Values[a]**

| Clinic | Home BP | Daytime BP | Nighttime | 24-hour BP |
|---|---|---|---|---|
| 120/80 | 120/80 | 120/80 | 100/65 | 115/75 |
| 130/80 | 130/80 | 130/80 | 110/65 | 125/75 |
| 140/90 | 135/85 | 135/85 | 120/70 | 130/80 |
| 160/100 | 145/90 | 145/90 | 140/85 | 145/90 |

[a]And their equivalent office values.
*BP*, Blood pressure.
Data from Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *Hypertension*. 2018;71:1269–1324.

**Table 46.6 Initial Laboratory Evaluation of the Hypertensive Patient[a]**

| Test | Clinical Usefulness |
|---|---|
| Serum creatinine (and estimated glomerular filtration rate) | Assessment of renal function. Identifies parenchymal kidney disease as a possible secondary cause, as well as established TOD. |
| Serum potassium | Low potassium (of renal origin) suggests mineralocorticoid excess (primary or secondary), glucocorticoid excess, Liddle syndrome; high potassium with normal renal function suggests Gordon syndrome; low levels raise caution about the use of thiazides and loop diuretics; high levels preclude the use of ACE inhibitors, ARBs, renin inhibitors, and potassium-sparing diuretics. |
| Serum sodium | If high, suggests primary aldosteronism; if low, alerts to the need to avoid thiazide diuretics. |
| Serum bicarbonate | If high, suggests aldosterone excess (primary or secondary); if low with normal renal function, suggests Gordon syndrome (with high potassium) or primary hyperparathyroidism (with high calcium). |
| Serum calcium | If high, suggests primary hyperparathyroidism. |
| Serum glucose | Identifies prediabetes or diabetes; in the appropriate setting, suggests glucocorticoid excess, pheochromocytoma, or acromegaly. |
| Lipid profile | Identifies hyperlipidemia. |
| Hemoglobin, hematocrit | If high, in the absence of other hematologic abnormalities or underlying lung disease, suggests sleep apnea. |
| Urinalysis[b] | Proteinuria and hematuria identify a possible secondary cause (glomerulonephritis); proteinuria can also be a marker of TOD. |
| Electrocardiography | Identifies left ventricular hypertrophy, old myocardial infarction, or other ischemic changes; identifies conduction abnormalities that may preclude the use of β-blockers or nondihydropyridine CCBs. |

[a]To investigate the presence of comorbid conditions, secondary causes, or established target-organ damage.
[b]Some organizations recommend screening microalbuminuria as a more sensitive tool to identify early renal injury. The most recent guidelines do not recommend BUN measurement alone, whereas some recommend the measurement of uric acid (as a marker of cardiovascular risk) and thyroid-stimulating hormone (to more specifically screen for hypo- and hyperthyroidism).
*ACE,* Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker; *BUN,* blood urea nitrogen; *CCB,* calcium channel blocker; *TOD,* target-organ damage.

the European Society of Hypertension[210] and American College of Cardiology/American Heart Association.[4]

Normative values for the interpretation of ABPM and home BP results are available based on observed outcomes in longitudinal studies[211] (Table 46.5). For ease of use, these thresholds were matched to specific office BP levels at which the observed rate of CV events was the same, thus allowing clinicians to relate to office values that have historically driven clinical decisions. For ABPM, other measures such as the nocturnal dip, early morning surge (magnitude of BP rise during the first hours' post-awakening), BP load (percentage of time BP remains above a certain threshold, such as 140/90 mm Hg during the day and 120/80 mm Hg during the night), and overall BP variability (standard deviation of the 24-hour BP or awake BP), were not studied in relationship to hard outcomes for precise normative results. Table 46.6 also presents general home and ambulatory BP values that mirror office BP values; these represent consensus-based values and are provided to allow the clinician to relate values seen out-of-the-office with the heretofore more commonly used office readings.

## INTEGRATING OUT-OF-OFFICE BLOOD PRESSURE INTO CLINICAL PRACTICE

All current hypertension guidelines recognize the value of out-of-office BP in the diagnosis of hypertension, particularly in patients with office BP <160/100 mm Hg. The ACC/AHA guidelines also recommend the use of out-of-office BP to evaluate patients who are receiving

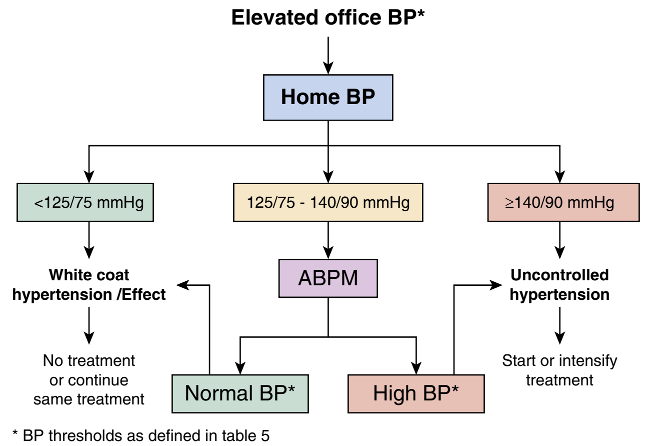


**Fig. 46.7 Practical approach to the integration of home and 24-hour ambulatory blood pressure monitoring for the evaluation and management of hypertension.**

treatment for hypertension but remain above goal in the office, with the explicit caveat that the recommendation is based on expert opinion. A finding to support this recommendation is the high prevalence (~40%–51%) of a white coat effect in patients with resistant hypertension, and the use of ABPM is formally recommended by both the U.S. and European guidelines to diagnose true-resistant hypertension.[25,212] Another caveat is that patients with office BP above 160/100 mm Hg do not need confirmation and should all be treated.

A practical approach to the use of home and ABPM in clinical practice is provided in Fig. 46.7.

## LABORATORY AND OTHER COMPLEMENTARY TESTS

Similar to the history and physical examination, laboratory tests, imaging, and other complementary tests also focus on the evaluation of comorbid conditions, established target-organ damage, and possible secondary causes. In the absence of worrisome signs or symptoms during the initial evaluation, the clinician should obtain a basic set of tests, including renal function; levels of electrolytes, calcium, glucose, and hemoglobin; lipid profile; urinalysis; and electrocardiogram (Table 46.6).

Further testing may be required in case any of these initial test results are abnormal or in case there are specific symptoms or physical findings suggesting a diagnosis (see "Secondary Hypertension" section). Likewise, patients who are resistant to treatment during follow-up have higher rates of secondary causes of hypertension, in particular sleep apnea, primary aldosteronism, and renovascular disease, thus deserving a more dedicated search for secondary causes in their evaluation.

### ECHOCARDIOGRAPHY

LVH is the most common target-organ damage in hypertension and is independently associated with worse prognosis, marked by increased risk for CV events (coronary, cerebrovascular), heart failure, and death.[213] The electrocardiogram is specific but insensitive to the detection of LVH. Not surprisingly, the prevalence of LVH among patients with hypertension is only approximately 18% based on electrocardiographic criteria, whereas this number increases to approximately 40% when more sensitive echocardiographic criteria are used. The echocardiogram also provides information on left ventricular diastolic function, which is often impaired early in the course of hypertensive heart disease and does not require the presence of LVH. Finally, it allows assessment of left ventricular systolic dysfunction, which is uncommonly present in hypertension (approximately 4%) but is associated with worse prognosis. Even though echocardiography is not recommended as a routine test in patients with hypertension, it may help to provide information to help guide treatment, such as defining the need to initiate or escalate treatment in patients with borderline office or ambulatory BP levels or to guide the choice of therapeutic agents based on associated abnormalities (e.g., heart failure with or without impaired left ventricular function).

### EVALUATION OF SODIUM AND POTASSIUM INTAKE

Because of the importance of sodium and potassium as dietary interventions in hypertension, it is often useful to quantify intake objectively. Diary recall is the most often used method in clinical practice; however, it is often problematic because many patients have difficulty defining portions. Our practice is to obtain a 24-hour urine collection to evaluate sodium and potassium on a stable diet. These ions are measured in milliequivalents per day and then converted to dietary targets in milligrams per day (1 mEq of sodium = 23 mg of sodium or 58 mg of "salt" as NaCl, and 1 mEq of potassium = 39 mg of potassium). A diuretic can be maintained as long as the dose has been stable over time. One must recognize that sodium excretion may follow a circaseptan rhythm[60] and may therefore be imprecise on a single 24-hour collection, but it is still valuable as a general guide to allow more precise dietary advice to patients.

### RENIN PROFILING

The evaluation of plasma renin activity has been proposed by Laragh as an empiric method for the evaluation and treatment of hypertension.[214] The premise for this approach is mechanistic: patients with high renin levels (>0.65 ng/mL/h, and particularly >6.5 ng/mL/h) have vasoconstriction mediated by the RAAS as the primary operative mechanism of hypertension, whereas those with suppressed renin levels (<0.65 ng/ mL/h) are volume overloaded. Accordingly, patients with high levels of renin are treated with blockers of the RAAS (ACEIs, angiotensin receptor antagonists, renin inhibitors, β-blockers), and those with low levels of renin are treated with diuretics (including aldosterone antagonists), CCBs, or α-blockers. The approach not only includes using drugs that directly address the underlying pathophysiology but also proposes removal of drugs from the opposite group as there are reports of paradoxical BP elevations in such cases.[215] A case series reported streamlined drug regimens and improved BP control in patients with resistant hypertension, and a small randomized trial of renin-guided therapy versus conventional therapy yielded greater systolic BP lowering with the renin-guided system (−29 vs. −19 mm Hg, $P$ = 0.03).[216] It is reasonable to entertain renin profiling, especially in patients who do not respond to initial therapy. In such cases, renin measurement, along with plasma aldosterone measurement, will also be useful to rule out primary aldosteronism.

### SYSTEMIC HEMODYNAMICS AND EXTRACELLULAR FLUID VOLUME

An alternative to the renin-profiling approach is to measure systemic hemodynamics and extracellular fluid volume, noninvasively. Such measurements can be achieved with several methodologies, but impedance cardiography has the advantage of simultaneously obtaining both volume (thoracic fluid content) and hemodynamic (CO, SVR) data. This approach has been used in patients with resistant hypertension with some success in two randomized trials[217,218] and in a pragmatic quality improvement study.[219] In all studies, achieved BP was lower in the hemodynamically monitored group. In one of the randomized trials, patients managed using the hemodynamic approach achieved better BP control while receiving more diuretics, while in the other, control was also better but not associated with the specific extra use of any particular drug class. Direct measurement of volume excess and hemodynamics and availability of the information at the point of care make this methodology more attractive than renin profiling. However, relatively high costs associated with the technology and lack of reimbursement in the current health care environment make widespread use of this method out of reach for most physicians and their patients.

## SECONDARY HYPERTENSION

### RISK FACTORS AND EPIDEMIOLOGY

Secondary hypertension is elevated BP due to a specific cause. In general clinical practice, there is a higher probability of secondary hypertension in patients 1. with higher levels of untreated BP, especially if control was recently lost; 2. with characteristic physical signs (for details, see later; 3. with treatment-resistant hypertension; 4. seen in tertiary referral centers (largely due to "referral bias"); and 5. "younger" patients (e.g., younger than 30 years old), though this has been a subject of controversy since the American Academy of Pediatrics limited its recommendations for work of children with hypertension only to those younger than the age of 6 years unless there is clinical evidence of a secondary cause.[16] A systematic review of the prevalence of secondary causes of hypertension in children and adolescents showed point estimates of 9% in studies in primary care and 44% in studies from specialty clinics.[220] Among factors associated with higher prevalence of secondary causes, the most prominent were a family history of secondary hypertension, low birth weight, history of premature birth, younger age (younger than 6 years), albuminuria, normal serum urate, high ambulatory BP load, and nocturnal hypertension.

The prevalence of secondary hypertension in hypertensive adults varies widely. In a large prospective study reported from a primary care setting that evaluated 1020 consecutive patients with hypertension in Japan, 9.1% of patients had a secondary cause, which was primary aldosteronism in 6%, Cushing syndrome (full-blown in 1% and preclinical in 1%), pheochromocytoma (0.6%), and renovascular hypertension (0.5%).[221]

The largest series of consecutive patients evaluated by a whole-day protocol from 1976 to 1994 in a tertiary referral center came from Syracuse, New York. In this study, 10.1% of 4429 patients with hypertension had secondary hypertension: 3.1% with renovascular hypertension, 1.4% with primary aldosteronism, 0.5% with Cushing syndrome, 0.3% with pheochromocytoma, 3% with primary hypothyroidism, and 1.8% with hypertension attributed to CKD.[222] Later series from all over the globe have suggested that primary aldosteronism (especially due to sleep-disordered breathing and obstructive sleep apnea) is far more common than previously thought, with an average prevalence of approximately 10% in population-based studies; the prevalence is roughly twice that in patients with resistant hypertension.[149]

### EVALUATION OF SECONDARY HYPERTENSION

Hypertension guidelines and clinical experience suggest a more detailed evaluation for secondary causes in patients with hypertension in persons younger than 30 years who have no family history of primary hypertension despite the aforementioned discordant recommendation of the American Academy of Pediatrics.[16] Additionally, persons older than 55 years with new-onset hypertension, sudden worsening of BP control (despite years of previously controlled BP), recurrent flash pulmonary edema, an abdominal bruit (especially with a louder diastolic component), and sudden decreases in eGFR by 30% or more after administration of an ACE inhibitor or angiotensin receptor blocker should prompt evaluation for secondary causes of hypertension. Such patients have a higher pretest probability of renovascular hypertension.

The initial set of laboratory tests recommended for newly diagnosed patients with hypertension includes serum concentrations of blood urea nitrogen and creatinine and a urinalysis, which is generally sufficient for identifying patients with underlying intrinsic kidney disease, even if the 3-month criterion for CKD is not yet met.

Either hyperthyroidism or hypothyroidism can be a cause of hypertension, which can be easily identified with serum thyroid-stimulating hormone and free thyroxine levels.

Sometimes the demographic and clinical features of the patient help direct the search for a secondary cause: Fibromuscular dysplasia is much more common in young white women, whereas atherosclerotic renovascular disease is more common in older smokers (both current and former). Some symptoms, when elicited by a careful history, are also quite suggestive (although incompletely sensitive and not very specific). For example, classically, paroxysmal "spells" occur in approximately 25% to 30% of patients with pheochromocytoma; the associated symptoms are variable across patients but commonly experienced repetitively in a given patient. Sadly, the specificity of these paroxysms, even when they include headache, sweating, and elevated BP levels, is below 5% in most large series. Similarly, the classical symptoms of proximal muscle weakness (particularly when rising from a chair or climbing stairs) reported with Cushing syndrome, or lower extremity weakness and leg cramps reported in primary aldosteronism and attributed to hypokalemia, are uncommon in today's literature and patients. Given the relatively low prevalence of secondary hypertension in adults, the decision to undertake a formal evaluation for specific causes can (and should) be individualized.

## INTRINSIC KIDNEY DISEASE

Given the high prevalence of CKD (here defined as estimated GFR [*e*GFR] < 60 mL/min/1.73 m$^2$), CKD is the most common cause of secondary hypertension in adults. CKD of any etiology can lead to hypertension, and the overall prevalence of hypertension is ~85%.[223] Patients with polycystic kidney disease and glomerulonephritides tend to be hypertensive earlier in the course of disease, whereas hypertension may occur later in the course of chronic interstitial disease.[106] Mechanisms linking CKD to hypertension (sodium avidity, endothelial dysfunction, arterial stiffness, overactivity of many vasoconstrictive systems) were discussed in detail earlier in this chapter. Management strategies for hypertension associated with CKD are identical to those used in primary hypertension except that doses and frequency of antihypertensive (and other) medications normally cleared by the kidney can depend on the degree to which kidney function is impaired (i.e., CKD stage). A detailed discussion of BP targets and treatment choices for hypertensive patients with CKD is presented later in this chapter.

## PRIMARY ALDOSTERONISM

Primary aldosteronism (i.e., persistent renin-independent aldosterone production) is a common cause of secondary hypertension, only surpassed by CKD. It is the most common cause of "classic" secondary hypertension (i.e., a condition that if identified and treated may result in cure of hypertension). Current estimates indicate that primary aldosteronism is present in 16% to 24% of U.S. hypertensive patients.[224] Interestingly, 6% to 14% of normotensive persons have evidence of mild aldosterone excess (elevated plasma aldosterone and suppressed plasma renin activity).[225] Such normotensive individuals with aldosterone excess progress to hypertension after 5 years of follow-up at rates much higher than normotensive patients without aldosterone excess (85% vs. 11%).[226]

Certain patient subgroups have higher prevalence of primary aldosteronism and should be considered for screening. Accordingly, the Endocrine Society recommends screening in patients with hypertension and hypokalemia (spontaneous or diuretic-induced), severe hypertension (BP >160/100 mmHg), resistant hypertension, family history of primary aldosteronism, hypertension complicated by cerebrovascular disease at a young age (<40 years), hypertension with an adrenal adenoma (including incidentalomas), and hypertension with obstructive sleep apnea.[227] Another group receiving attention is patients with hypertension and atrial fibrillation in the absence of structural heart disease that could explain the atrial fibrillation.[228] Although hypokalemia is one of the classic features of primary aldosteronism, it is well recognized that normokalemia is the norm. Only a small minority presents with hypokalemia, which tends to be more common in patients with aldosterone-producing adenomas (up to 40%).[228,229]

Despite these high prevalence rates, clinicians do not think of it often enough. For example, two separate analyses of large databases showed that the rate of screening for primary aldosteronism in patients with resistant hypertension (the group with highest prevalence rates) is only about 2%,[230,231] highlighting the importance of increased awareness by general practitioners.

Primary aldosteronism can be caused by one of six subtypes: most commonly 1. an aldosterone-producing adenoma, nearly always in one adrenal gland (approximately 35% of cases) or 2. bilateral adrenal hyperplasia without a detectable adenoma (approximately 60% of cases); less commonly, 3. primary (or unilateral) adrenal hyperplasia; or rarely, 4. aldosterone-producing adrenal carcinoma; 5. familial aldosteronism, which takes one of two forms: glucocorticoid-suppressible aldosteronism, due to a chimeric chromosome 8, in which the 5′-regulatory sequence for corticotropin responsiveness of 11β-hydroxylase is fused to the enzyme coding sequence for aldosterone synthase, or familial occurrences of either an aldosterone-producing adenoma or bilateral adrenal hyperplasia; or 6. ectopic production of aldosterone by an adenoma or carcinoma outside the adrenal gland. In addition, obstructive sleep apnea and sleep-disordered breathing often cause aldosteronism. This is classically described as secondary aldosteronism, but its evaluation and medical treatment are often quite similar to that of bilateral adrenal hyperplasia.

Patients with primary aldosteronism experience higher rates of target-organ damage than matched patients with primary hypertension.[232] A meta-analysis of 31 studies including 3838 patients with primary aldosteronism and 9284 patients with primary hypertension showed significantly increased risk of stroke (odds ratio 2.58), coronary disease events (odds ratio 1.77), congestive heart failure (odds ratio 2.05), de novo atrial fibrillation (odds ratio 3.52), and LVH (odds ratio 2.29).[232] Additionally, the presence of primary aldosteronism is associated with higher rates of de novo diabetes mellitus and metabolic syndrome, though these associations are likely bidirectional.

Screening for primary aldosteronism is most efficiently performed in potassium-repleted patients using the ratio of plasma aldosterone concentration to plasma renin activity (the "aldosterone to renin ratio" or ARR). For convenience, the ARR can be performed when patients are receiving antihypertensive therapy with the exception of mineralocorticoid receptor antagonists (spironolactone, eplerenone), sodium channel blockers (amiloride, triamterene), and renin inhibitors (aliskiren). The ARR can be affected by many factors including antihypertensive drug therapy, dietary sodium restriction, posture, time of day, and sample handling (Table 46.7). Most authorities recommend sustained-release verapamil, hydralazine, and peripheral $\alpha_1$-adrenoceptor antagonists as medications that have little, if any, effect on the ARR and can be used when drugs must be stopped to allow interpretation of confusing results. A positive screen must include suppressed plasma renin activity (<1 ng/mL/hour), which is the sine qua non of primary aldosteronism. The most common cutoff value for an ARR that usually leads to further investigation is 25 to 30 (when aldosterone level is measured in nanograms per deciliter and plasma renin activity in nanograms of angiotensin II per milliliter per hour). Higher thresholds increase diagnostic specificity but lead to more falsely negative tests. The recognition that plasma aldosterone levels are not accurate measures of ambient aldosterone

**Table 46.7 Factors That May Cause False-Positive or False-Negative Results of the Aldosterone/Renin Ratio**

| Factor | False-Positives | False-Negatives |
|---|---|---|
| Aldosterone relatively high | Potassium loading | |
| Renin relatively low | β-blockers; central α-adrenoceptor agonists; direct renin inhibitors; nonsteroidal antiinflammatory drugs; chronic kidney disease; sodium loading | |
| Aldosterone relatively low | | Hypokalemia |
| Renin relatively high | | Diuretics; ACE inhibitors, angiotensin receptor blockers; calcium channel blockers (dihydropyridines); acute sodium depletion |

*ACE,* Angiotensin-converting enzyme.

secretion[224] has led some experts to recommend that the diagnosis of primary aldosteronism be pursued in any patient with suppressed plasma renin activity (<0.6–1 ng/mL/min) and plasma aldosterone levels above 5 ng/dL.[225] We agree with this approach.

Clinical practice guidelines from the Endocrine Society recommend one of four confirmatory tests before proceeding to an imaging study.[227] These include the oral sodium loading followed by 24-hour urine aldosterone measurement, the rapid saline infusion test, the fludrocortisone suppression test, and the captopril challenge test. There are only a few comparative studies of these four tests; they seem to have similar performance characteristics (75%–90% sensitivity, 80%–100% specificity). Cost, patient preference, local experience, local laboratory methods, and insurance reimbursement should factor into which confirmatory test is chosen.

Our practice is to use one of the sodium-loading protocols, either orally or intravenously. The oral sodium-loading protocol involves liberalizing sodium intake >200 mEq/day for 3 days and then assaying 24-hour urine collections for sodium (to ensure loading) and aldosterone content while continuing high sodium intake during the fourth day. Traditionally, the test has been considered positive if the urinary aldosterone excretion is >12 to 14 µg/day. However, with recognition that aldosteronism is on a continuum, less emphasis is currently being placed on absolute threshold values. In other words, unless there is "complete" suppression of aldosterone production (to levels <5 mcg/day), many experts consider the test positive in that it indicates a degree of persistently inappropriate aldosterone secretion.[233] The implications of these borderline results remain uncertain, though one should consider the use of a mineralocorticoid receptor antagonist as part of the therapeutic approach to these patients.

The "saline-loading test" (2 L infused over 4 hours) is confirmatory if the postinfusion plasma aldosterone concentration is >5 ng/mL. Patients with aldosterone concentrations between 5 and 10 ng/mL are considered indeterminate and should be retested, typically with an alternative confirmatory test. Sodium-loading protocols should be avoided or done cautiously in patients with heart failure, uncontrolled hypertension, advanced kidney disease, or untreated hypokalemia.

After the diagnosis of primary aldosteronism is confirmed or strongly suspected based on biochemical results, a computed tomographic scan of the adrenals can be undertaken, which is quite useful in detecting large masses that might be adrenal carcinomas. Adrenal carcinomas are typically larger in size (>4 cm diameter), have an inhomogeneous character (often with internal hemorrhage), have internal calcifications (in approximately 40%), have irregular borders (often due to micrometastases), and show enhancement after intravenous contrast medium is administered. Aldosterone-producing adenomas are most commonly small (<2 cm diameter), hypodense (<10 Hounsfield units), unilateral nodules. Idiopathic aldosteronism usually has normal-appearing adrenal glands, but sometimes nodular changes and/or general enlargement are visible in one or both adrenals. Magnetic resonance imaging is no better at detecting these abnormalities than computed tomography.

At some centers, patients with hypertension with proven primary aldosteronism younger than 35 to 40 years of age with a single typical hypodense nodule in one adrenal gland are directly offered adrenalectomy. However, because of the frequent occurrence of adrenal incidentalomas, particularly among older women, computed tomographic scans identify unilateral adrenal disease with a sensitivity of only 78% and specificity of only 75%, the Endocrine Society recommends adrenal venous sampling for most surgical candidates.[227] Despite being invasive, expensive, technically challenging, and requiring an experienced and well-coordinated team to minimize complications, it has sensitivity and specificity of 95% and 100% for detecting unilateral aldosterone production. It is most often performed at 8 AM, with continuous cosyntropin administration and simultaneous adrenal vein cortisol level measurement. Most centers use a 4:1 cutoff value of the cortisol-corrected aldosterone ratio (i.e., the ratio between the aldosterone/cortisol ratios on each side) to define lateralization.

New imaging techniques are under development to try to supplant the need for AVS. The first test to be compared with AVS in a clinical trial is the [$^{11}$C]metomidate positron emission tomography computed tomography (MTO) scan.[234] In a study of 143 patients with biochemical evidence of primary aldosteronism undergoing both MTO and AVS, MTO was noninferior to AVS in predicting clinical response to adrenalectomy (as an indicator of lateralization). However, there were substantial concerns that will require further studies before clinical use of the radiotracer, such as the infrequent concordance between AVS and MTO (only 30%) and occasional (2.5%) complete discordance between the two methods (AVS lateralizing to one side, MTO to the other on the same patient).[234]

Testing for familial forms of primary aldosteronism is recommended for those who are younger than 20 years of age at diagnosis and for those with a family history of primary aldosteronism or stroke at an early age (typically <40 years of age). Genetic testing for glucocorticoid-remediable aldosteronism (familial hyperaldosteronism, type I, the most common monogenic cause of hypertension) is available at reasonable cost and coverage by many insurance payers, so the diagnosis can be genetically confirmed. When not possible, confirmation of suppression of aldosterone production by dexamethasone should be used to confirm the diagnosis. Familial aldosteronism type II and type III are genetically heterogeneous, despite being autosomal dominant in most affected cases. Genetic testing is available from some specialized laboratories to identify some of the described genes resulting in these rare phenotypes.

Treatment of primary aldosteronism differs according to subtype. Laparoscopic adrenalectomy is the treatment of choice for patients who lateralize on AVS (aldosterone-producing adenoma or unilateral adrenal hyperplasia). Although nearly all return to normokalemia, hypertension is "cured" (i.e., follow-up BP levels of <140/90 mm Hg without antihypertensive drug therapy) in only approximately 50%.[227] Cure is more likely in patients who are younger, women, nonobese, with short duration of hypertension, prior BP control with only one or two agents, and who had adequate preoperative response to a mineralocorticoid receptor antagonist. Typically, plasma aldosterone concentration and plasma renin activity are measured shortly after successful surgery, and potassium supplementation and aldosterone antagonists are discontinued. Intravenous saline is often required, as the remaining adrenal gland recovers its normal function (which may take a few weeks, often requiring increased sodium intake). The nonsurgical option for patients who are high-risk surgical candidates or decline adrenalectomy and for those with bilateral adrenal hyperplasia is spironolactone, which demonstrated significantly better efficacy than eplerenone in an international randomized clinical trial in hypertensive subjects with primary aldosteronism.[235] It is important to recognize that incomplete interruption of aldosterone excess likely results in worse clinical outcomes. A meta-analysis of studies comparing adrenalectomy with medical therapy showed a 75% higher risk of cardiovascular events among patients treated medically.[236] Unfortunately, the studies were not controlled and have significant limitations. The use of highly selective aldosterone synthase inhibitors, alone or in combination with aldosterone receptor antagonists, is under active investigation in patients with primary aldosteronism.[237] Most clinicians use dexamethasone or prednisone at bedtime (over twice-daily hydrocortisone) for glucocorticoid-remediable aldosteronism, but the doses are kept low to avoid iatrogenic Cushing syndrome.

## APPARENT MINERALOCORTICOID EXCESS STATES AND HYPERCORTISOLISM

Several conditions, most of them rare, can present with similar features as primary aldosteronism (typically hypertension and hypokalemia) and suppressed plasma renin activity but have suppressed plasma aldosterone levels.

The most common such condition is hypercortisolism (Cushing syndrome). Most cases of Cushing syndrome today are iatrogenic (due to prescribed oral corticosteroids). Once the use of exogenous corticosteroids is ruled out, identification of cause of glucocorticoid excess is indicated. The pathophysiology of hypertension in Cushing syndrome overlaps somewhat with mineralocorticoid excess states since excess cortisol often overwhelms the capacity of 11β-hydroxysteroid dehydrogenase type 2 to convert cortisol to inactive cortisone. This allows cortisol to bind and activate the mineralocorticoid receptor, producing unopposed mineralocorticoid activity.[238] In addition, there is increased hepatic production of angiotensinogen leading to activation of the renin-angiotensin system and resultant increases in peripheral vascular resistance and vascular remodeling.[238]

The full-blown syndrome of truncal obesity with striae, hirsutism, acne, hyperglycemia, hypertension, hypokalemia, and muscular weakness is less commonly seen today. After an appropriate screening test (urinary free cortisol, late-night salivary cortisol, or overnight dexamethasone suppression test)[239,240] yields positive results, it is necessary to define the cause of the syndrome. Pituitary adenomas are most common, followed by adrenal ACTH-independent glucocorticoid hypersecretion or, rarely, ectopic ACTH production or micronodular or macronodular adrenal hyperplasia.[240] Measurement of plasma ACTH allows distinction between ACTH-dependent (pituitary adenomas, ectopic ACTH production) and ACTH-independent processes. It is recommended that further studies (imaging or biochemical) be guided by an endocrinologist, who will also guide specific therapy, surgical or medical.

Congenital adrenal hyperplasia is due to one of several autosomal recessive genetic mutations in genes coding for enzymes involved in adrenal steroidogenesis, most often resulting in mineralocorticoid and androgen excess. Although most common in infants and children, some patients (especially those with milder loss-of-function mutations) go undiagnosed until adulthood. The two forms of adrenal hyperplasia associated with hypertension are 11-β hydroxylase and 17-hydroxylase deficiencies. In 11-β hydroxylase deficiency, which presents with virilization, hypertension is caused by accumulation of the mineralocorticoid precursor 11-deoxycorticosterone. In 17-hydroxylase deficiency, which presents with feminization of the genitalia or primary amenorrhea, hypertension is precipitated by increased production of the mineralocorticoid agonists 11-deoxycorticosterone and corticosterone.[238]

Rare causes of apparent mineralocorticoid excess include deoxycorticosterone-producing tumors (which are usually quite large and often malignant), primary cortisol resistance, or 11β-hydroxysteroid dehydrogenase type 2 deficiency (of which only rare cases are caused by mutations in the gene coding for the enzyme; most cases are acquired and associated with the use of glycyrrhetinic acid acid-containing licorice, licorice-containing chewing tobacco or nutritional supplements, or naringinic acid). Mutations in the mineralocorticoid receptor may cause Geller syndrome. In this extremely rare disorder, the mineralocorticoid receptor is activated by progesterone leading to a syndrome of hypertension exacerbated by pregnancy. We may also include Liddle syndrome in this list. Liddle syndrome presents with hypertension, hypokalemia, and inappropriate kaliuresis associated with low plasma aldosterone and renin activity. It is caused by mutations in the β- or γ-subunits of the renal amiloride-sensitive epithelial sodium channel.

## RENOVASCULAR HYPERTENSION

Hypertension due to renal artery stenosis (fibromuscular dysplasia or atherosclerotic disease) is an important cause of secondary hypertension. It is discussed in detail in Chapter 47.

## PHEOCHROMOCYTOMA

Pheochromocytomas and paragangliomas (PPGLs) are rare neuroendocrine chromaffin tumors (estimated incidence: 2–8 cases per million per year) that are derived from the adrenal medulla and extraadrenal sympathetic or parasympathetic ganglia, respectively.[241] Diagnosis and management of these tumors is critical because: 1. they can cause fatal hypertensive crises and associated complications, 2. 15% to 25% of PCC/PGL can become metastatic, 3. specific therapy can be curative if diagnosed early, and 4. approximately 40% of these tumors are due to inherited germline mutations.[242]

Catecholamine-secreting tumors are found in 0.2% to 0.6% of patients with hypertension, may be more common in hypertensive children (1.7%), and are often found incidentally, or only at autopsy. The clinical presentation of patients with these tumors is quite variable, as symptoms may occur constantly, in paroxysms or, less often, patients can be totally asymptomatic.[243]

Patients classically present with a triad of symptoms including headache, palpitations, and sweating with hypertension, which may be persistently elevated, labile, or normal. The presentation is highly variable, so a high index of suspicion is required, even when faced with common presenting conditions (e.g., heart failure or aortic dissection, which can be precipitated, if not exacerbated, by a functioning tumor). Some of the inherited germline mutations associated with PPGLs have characteristic physical signs (e.g., café au lait spots and neurofibromas in von Hippel Landau [vHL]); hyperparathyroidism, multiple mucosal neuromas, intestinal ganglioneuromas, and often a marfanoid habitus with multiple endocrine neoplasia (MEN)-associated tumors.[242]

Tumors in the adrenal gland are termed “pheochromocytomas,” all other extraadrenal tumors are termed paragangliomas (PGL), and tumors occurring in the head and neck region are termed head and neck paragangliomas (HNPGLs). Adrenal and extraadrenal tumors are usually derived from sympathetic ganglia, which hypersecrete catecholamines, whereas HNPGLs are usually derived from parasympathetic ganglia and are often nonsecretory. Ninety percent of tumors are found in or in close proximity to an adrenal gland. Paragangliomas can occur anywhere along the sympathetic ganglia but most commonly in or near the organ of Zuckerkandl (at the aortic bifurcation) or near the bladder (which gives rise to the triad of symptoms that occur related to micturition).

PPGLs are the most commonly associated tumors with a germline mutation in a known susceptibility gene more than any other solid tumor type[244] and due to the high prevalence of associated germline mutations, all patients with proven PPGL should undergo genetic testing. There are more than 12 pathogenic germline mutations now associated with PPGL that can be clinically tested. MEN syndromes are autosomal dominant and are caused by activating mutations in the RET protooncogene. Approximately 50% of MEN2 patients will develop pheochromocytoma, which can be bilateral in about half of cases and can present synchronously or metachronously. MEN2A accounts for 95% of cases and has a high penetrance for medullary thyroid carcinoma (MTC) or C-cell hyperplasia (nearly 100%), pheochromocytoma (50%), and multiglandular parathyroid hyperplasia or adenoma (20%–30%), leading to primary hyperparathyroidism, whereas MEN2B and the rare familial medullary thyroid cancer subtype account for the rest. MEN2B typically does not develop primary hyperparathyroidism.[241] The mean age at diagnosis of pheochromocytoma is between 30 and 40 years old, and the malignancy rate is <5%.[244] Most MEN-associated pheochromocytomas are metanephrine predominant.[245]

vHL is an autosomal dominant hereditary cancer syndrome, with a 95% penetrance by age 60, characterized by the development of a variety of tumors such as hemangioblastomas of the CNS including the retina, renal cell carcinoma (RCC), pancreatic neuroendocrine tumors, pheochromocytomas (60% will have bilateral tumors), endolymphatic sac tumors, and visceral cysts (including renal and pancreatic).[241] Patients with VHL type 1 (typically those with truncating mutations) have a lower risk of developing pheochromocytomas and lower risk of developing RCC. Patients with VHL type 2 disease (typically those with missense mutations) have a higher risk of developing pheochromocytomas. The overall frequency of PPGL is 20% overall but increases to 60% in patients with VHL type 2.[241] The risk of metastatic disease is about 5%, and most tumors are pheochromocytomas with PGLs occurring rarely. Most VHL-associated PPGL are normetanephrine predominant, due to a lack of expression of phenylethanolamine-N-methyltransferase, the enzyme that converts norepinephrine to epinephrine.[245]

NF1 is an autosomal dominant tumor-predisposition syndrome characterized by café au lait spots, neurofibromas, axillary or inguinal freckling, and optic gliomas; Lisch nodules (benign iris hamartomas); and bone lesions. About 10% to 15% of patients with NF1 will develop pheochromocytomas, which can be bilateral, and the risk of metastatic disease is around 12%.[241] NF1-associated PCCs tend to have metanephrine predominance.[245]

Hereditary paraganglioma-pheochromocytoma syndromes are caused by autosomal dominant mutations in succinate dehydrogenase (SDH), complex II of the mitochondrial respiratory chain. Mutations can occur in any of the subunit genes *(SDHA, SDHB, SDHC, SDHD)* or cofactor (SDHAF2 ). SDHB is the most commonly mutated subunit leading to PPGL. Mutation carriers develop extraadrenal PGL but can also develop pheochromocytomas and HNPGL. SDHB mutation carriers have the highest risk of developing metastatic disease and can also develop renal cell carcinomata. SDHD is also commonly mutated and is associated with multiple primary tumors often in the head and neck area but can also develop pheochromocytomas and rarely metastatic disease. SDH mutations in the other subunits occur less often, and the risk of malignancy is lower. Given the higher risk for metastatic disease in patients with SDHB mutations and reported cases of early-onset of disease during childhood, it is proposed to start screening patients with known mutations at age 6–10 for asymptomatic SDHB mutation carriers and at age 10–15 for carriers of mutations in *SDHA, SDHC,* and *SDHD* genes.[246] Biochemical testing should be performed at least every 2 years during childhood (i.e., age <18) and every year during adulthood, and full-body magnetic resonance imaging (MRI) from the skull base to pelvis should be performed every 2 to 3 years from the time the variant is known. Some suggest that a 68Ga-DOTATE positron emission tomography/computed tomography (CT) scan can be used instead of MRI, if available.[241]

If PPGL is suspected, biochemical testing for plasma-free metanephrines and/or urinary-fractionated metanephrines and catecholamines should be performed. Plasma-free metanephrines have at least equal sensitivity and specificity (98% and 92%, respectively) as 24-hour urine collections and are more convenient for patients.[247] Abnormal results are at least two to four times above the upper limit of normal. Indeterminate results may be due to interfering factors and include medications (e.g., tricyclic antidepressants, selective serotonin reuptake inhibitors, serotonin and norepinephrine reuptake inhibitors, and α-adrenergic blockers), drugs (e.g., cocaine, marijuana, stimulants, and caffeine), or position (i.e., sitting up rather than lying down for the plasma tests).

To improve cost-effectiveness and reduce radiation exposure, imaging studies for pheochromocytoma and related tumors are generally not ordered until after biochemical evidence of catecholamine overproduction is obtained. Cross-sectional imaging to localize the tumor using CT scan or MRI starting with the abdomen/pelvis is recommended as 80% to 85% of PPGL will be in this region.[247] If metastatic disease is suspected, functional imaging with DOTATATE positron emission tomography/CT or MIBG scans can be useful.[241]

When planning for surgical resection, patients should absolutely undergo α-1 blockade before resection. Surgical resection is the treatment of choice for PCC/PGL, but the hypersecretion of catecholamines can be life threatening perioperatively; the use of α-blockade and modern anesthesia have reduced surgical morbidity and mortality.[248] α-1 adrenergic blockers are the mainstay of perioperative management in patients with PCC/PGL. Competitive and noncompetitive α-blockers are used in perioperative management. Phenoxybenzamine is a noncompetitive inhibitor that covalently binds to α-1 and α-2 receptors irreversibly. Therefore the extra release of catecholamines intraoperatively with manipulation of the tumor mass cannot overcome the inhibition of phenoxybenzamine as de novo synthesis of α receptors is required. This helps to prevent intraoperative hypertensive crisis; however, this irreversible binding can lead to hypotension after the tumor is resected, thereby removing the source of catecholamine production. Vasopressor support and intravenous fluids may be required for 24 to 48 hours postoperatively to maintain BP. Side effects of phenoxybenzamine include orthostasis and nasal congestion.[249] Other α-blockers used include the selective α-1 receptor blockers doxazosin, terazosin, and prazosin. These are competitive inhibitors with relatively shorter durations of action and therefore can be overcome by the extra catecholamine release intraoperatively which can lead to hypertensive crisis. However, the shorter half-life makes these inhibitors less likely to result in hypotension after tumor removal.

Other antihypertensive medications are used occasionally in the perioperative blockade. CCBs, such as nicardipine or amlodipine, work by inhibiting norepinephrine-mediated transmembrane calcium influx into smooth muscle. Β-blockers should never be used alone in patients with PPGL as the unopposed α-adrenergic effect can cause severe vasoconstriction leading to hypertensive crisis. Β-blockers, such as selective β-1 antagonists like metoprolol, do have a role in perioperative blockade after full α-blocking has been achieved as a known side effect of a full α-blockade is reflex tachycardia. At this point, β-blockers can be added to reduce the tachycardia.[249]

Most surgeons favor laparoscopic procedures for small adrenal pheochromocytomas or paragangliomas in accessible locations. Biochemistries should be checked 6 weeks after surgery to ensure levels have returned to normal and should be checked annually in all patients due to the risk of recurrence. All patients should be referred for genetic testing. If biochemistries remain elevated and metastatic disease is suspected, patients should be referred to centers of excellence with experience for further evaluation.

## THYROID DYSFUNCTION

The literature about thyroid dysfunction being associated with hypertension is inconsistent. Many patients with hyperthyroidism have wide pulse pressures (and therefore elevated systolic BP levels) and high pulse rates, but this is seldom missed, especially in younger patients. After diagnosis, a nonselective β-blocker such as propranolol may be specifically useful, as it treats tachycardia and hypertension and possibly inhibits peripheral conversion of thyroxine to triiodothyronine.

The role of hypothyroidism as a potential cause of hypertension (especially isolated diastolic) is less clear, despite the experience in upstate New York, in which 3% of patients with hypertension reverted to normotension after treatment of hypothyroidism. The hypertension in hypothyroidism is predominantly diastolic and usually in the range of stage 1 (i.e., diastolic BP <99 mm Hg). In children and adolescents, especially in areas of iodide deficiency, a positive association between serum TSH and BP has been seen. A pooling of seven population-based European data sets showed similar findings in adults, but there was no consistent relationship with either a 5-year change in BP or incident hypertension.

## HYPERPARATHYROIDISM, CALCIUM INTAKE, VITAMIN D, AND HYPERTENSION

Although hypercalcemia and hypertension associated with hyperparathyroidism often improve after appropriate treatment, causal relations among calcium and vitamin D intake, serum parathyroid hormone concentrations, and BP have not been consistently seen in large populations.[250] As a result, most current guidelines recommend a serum calcium (and not parathyroid hormone) determination during blood testing for patients with an initial diagnosis of hypertension.

### COARCTATION OF THE AORTA

Although most discrete isthmic constrictions of the aorta occur in or near the ductus arteriosus, there is growing awareness that this fifth most common form of congenital CV disorders constitutes a spectrum of aortic and vasculopathic disorders and is not always "cured" by surgical procedures that relieve the obstruction. Most patients with the condition are hypertensive[251,252] and are diagnosed in infancy or childhood, but some escape detection until adulthood. Many cases are identified by suggestive physical findings (e.g., murmur, BP lower in the legs than the arms, and radial-femoral pulse delay), some after imaging studies done for other reasons (e.g., rib notching or a "3" sign on chest radiograph, the latter of which results from indentation of the aorta, with prestenotic and poststenotic dilation), and others during investigation of associated abnormalities (e.g., bicuspid aortic valve).

Echocardiography is highly recommended for diagnosis and localization of the coarctation, although some patients (especially adults and those with associated anomalies) may require cardiac catheterization. Most pediatric patients undergo percutaneous catheter balloon dilation with stent placement; this can be followed by definitive surgical correction later, if needed. In a systematic review, 25% to 68% of patients with a coarctation had persistent hypertension despite satisfactory procedure results, with age at the time of surgery, age at follow-up, and the type of intervention being strong predictors of persistent hypertension.[253]

### ACROMEGALY

Hypertension occurs in more than 40% of patients with excessive growth hormone release causing acromegaly, and it can be exacerbated by concomitant sleep apnea.[254] Most such patients are easily identified by symptoms or signs of acral bony overgrowth, particularly in children or adolescents before epiphyseal closure, although some patients ignore or tolerate these changes for a decade or more. The vast majority (98%) of cases are caused by a pituitary adenoma; serum insulin-like growth factor-1 is the most useful initial laboratory screening test, although other tests (including the response of plasma growth hormone levels to an oral 75-g glucose load and prolactin levels) are often performed. Successful treatment of acromegaly usually lowers BP, but hypertension often persists, especially in older and overweight patients.[255] No specific antihypertensive drug therapy seems to be more effective than others, but because acromegaly is an unusual cause of resistant hypertension, antihypertensive drug therapy is usually effective.

### SLEEP DEPRIVATION AND SLEEP-DISORDERED BREATHING

Poor sleep quality, if chronic, yields the same symptomology of paroxysmal hypertension and elevated BP, especially during the afternoon and evening hours. Poor sleep quality is not just the result of obstructive sleep apnea (OSA) but a host of sleep disorders including restless leg syndrome and insomnia of various causes.[256] Many times, these different sleep disorders coexist and prevent the patient from achieving proper sleep.

The mechanism of poor sleep quality contributing to elevated BP and paroxysmal bouts of high BP relates to activation of both the sympathetic and renin-angiotensin-aldosterone system.[257,258] Sympathetic activity is also increased in sleep deprivation, restless leg syndrome, and OSA.[257,258]

Patients without OSA who suffer from sleep deprivation, defined as less than a minimum of six hours of uninterrupted sleep, also have increased sympathetic activity. In this case, it is a consequence of reduced time in non–rapid eye movement (NREM) or slow wave sleep that also affects the nocturnal dip in BP.[258] This supports the hypothesis that disturbed NREM sleep quantity or quality is a mechanism by which sleep deprivation or restless leg syndrome leads to an increase in sympathetic tone.

OSA should be suspected in patients (men more so than women) who report severe snoring, daytime somnolence, witnessed nocturnal choking or gasping, and have a "crowded oropharynx" (limited or no visualization of the soft palate) on physical examination. Formal diagnosis is based on an ambulatory sleep study or in-center polysomnography.

In contrast to reduced sleep time and quality, the increase in sympathetic activity associated with OSA is a function of intermittent hypoxia as the acute rise in BP parallels the severity of oxygen desaturation at night.[259] Indeed, increased sympathetic activity is seen in both animal models subjected exclusively to intermittent hypoxia.[260] It is also important to note that OSA-associated hypertension is only slightly reduced by continuous positive airway pressure (CPAP) treatment.[261] However, patients who have more severe hypertension, more severe OSA, higher daytime sleepiness scores, and greater adherence to CPAP (average use >4 hours per night) tend to have greater responses.[262] Additionally, certain drug classes appear to have benefits in controlling BP in patients with OSA, including mineralocorticoid receptor antagonists and β-blockers.[263]

## DRUG-INDUCED HYPERTENSION

Patients presenting with hypertension should be questioned about exposure to substances that can raise BP (Table 46.8). These include drugs of abuse and over-the-counter and prescription medications. Oral contraceptive pills (OCP) can cause hypertension, though modern low-estrogen pills do it at rates much lower than with older preparations. Stopping the OCP cures hypertension after several weeks to months in most, but not all, women. Nonsteroidal antiinflammatory drugs (NSAIDs) result in a modest average hypertensive effect (up to ~5 mm Hg), but some patients can have larger, clinically significant BP elevation. NSAID-induced hypertension may also present as loss of BP control in patients taking a diuretic or a blocker of the renin-angiotensin system, whereas CCBs tend to be less affected in NSAID users.

Sympathomimetic amines (legal or illegal) usually cause hypertension acutely following their ingestion. Alcohol has an acute hypotensive effect, but chronic use in large amounts (>4-5 drink-equivalents per day) is associated with increased BP. Glucocorticoids and mineralocorticoids can produce a dose-dependent rise in BP. Glucocorticoids with low mineralocorticoid activity (dexamethasone, budesonide) induce lesser pressor responses.

Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) can increase BP modestly, but some patients receiving SNRIs

**Table 46.8 Drugs Commonly Associated With Hypertension**

| |
|---|
| **Oral Contraceptives** |
| Nonsteroidal antiinflammatory drugs (NSAIDs, selective, and nonselective) |
| Sympathomimetics: pseudoephedrine, phenylpropanolamine, phentermine, cocaine, amphetamines (prescription or illegal). Yohimbine ($\alpha$-2 antagonist) |
| Selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs) |
| Monoamine oxidase inhibitors (MAOIs) |
| Cyclosporine and tacrolimus |
| Erythropoietin and darbepoetin |
| Corticosteroids, mineralocorticoids (fludrocortisone) |
| Anti-VEGF antibodies (bevacizumab, ramucirumab) and certain tyrosine kinase inhibitors with anti-VEGF activity (e.g., sorafenib, sunitinib, and axitinib) |
| Proteasome inhibitors (carfilzomib) |
| 11-$\beta$ hydroxysteroid dehydrogenase type 2 inhibitors: licorice, posaconazole |
| Ethanol |

From Peixoto AJ. Secondary hypertension. In: Gilbert SJ, Weiner DE, Bomback AS, Perazella MA, Rifkin DE, eds. *National Kidney Foundation's Primer on Kidney Diseases*. Philadelphia: Elsevier; 2023.

can have a severe hypertensive response. Interestingly, when used for hypertensive patients with depression, BP often improves as depressive symptoms improve. Angiogenesis inhibitors, such as anti-VEGF antibodies (bevacizumab, ramucirumab) and tyrosine kinase inhibitors with impact on the VEGF pathway (sorafenib, sunitinib, axitinib, and others), can produce hypertension that often persists despite discontinuation. Because hypertension during the use of these drugs often correlates with better oncologic outcomes (likely a reflection of successful antiangiogenic effect), treatment is usually continued unless reasonable BP control is not achievable or if severe kidney injury develops.

## HYPERTENSIVE URGENCY AND EMERGENCY

A hypertensive emergency is the combination of elevated BP levels (with no specific diagnostic BP level, although current guidelines use 180/110–120 mm Hg as the threshold for definition of severe BP elevation) and signs or symptoms of acute, ongoing target-organ damage involving the brain (stroke, intracerebral hemorrhage, hypertensive encephalopathy), heart (acute decompensated heart failure, acute coronary syndromes), large vessels (aortic dissection), or microvasculature ("malignant hypertension" presenting as acute kidney injury and/or microangiopathic hemolytic anemia and thrombocytopenia).[264] Pregnant women with preeclampsia/eclampsia and patients with pheochromocytoma may present with evidence of acute target-organ damage despite lower BP levels.

Patients with hypertensive emergencies should be admitted to an intensive care unit and given parenteral infusions of short-acting antihypertensive agents to control the BP and restore autoregulation in vascular beds. Conversely, patients who present with elevated BP levels, but no acute, ongoing target-organ damage, are diagnosed with a "hypertensive urgency." They should be managed as any other patient with uncontrolled BP and do not require escalation of care (such as referral to an emergency department).[264] This practice is justified by the low short-term morbidity related to hypertensive urgencies.[264] However, these patients do have worse long-term outcomes,[265] thus justifying close follow-up and diligent attempts to control the BP.

The initial evaluation of a severely hypertensive patient includes a thorough inspection of the optic fundi (looking for acute hemorrhages, exudates, or papilledema); a mental status assessment; a careful cardiac, pulmonary, and neurologic examination; a quick search for clues that might indicate secondary hypertension; and laboratory studies to assess kidney function (metabolic panel, urinalysis) and rule out myocardial injury (electrocardiogram, troponin) and microangiopathy (complete blood count).[264]

Several options for intravenous drug treatment exist. Most conditions can be managed with a vasodilator (typically nicardipine, clevidipine, or nitroprusside) with or without a $\beta$-blocker (typically labetalol or esmolol). A discussion of the pharmacology of these agents is beyond the scope of this chapter but can be found in recent reviews.[264] The tempo of BP reduction varies according to the type of hypertensive emergency. Table 46.9 provides a summary of the current recommendations for management of different hypertensive emergencies. As noted, different syndromes call for different paces of BP reduction and often different drug choices. Hypertensive encephalopathy, ischemic stroke, and malignant hypertension should be managed with slower BP reduction to avoid excessively rapid reductions that overwhelm cerebral autoregulatory limits (i.e., an inability of cerebral arterioles to vasodilate enough to allow adequate cerebral perfusion at "normal" BP levels; this occurs because of loss of autoregulation under states of chronic absence of BP control).[264] However, certain conditions call for faster BP reduction to avert further organ injury that is potentially lethal despite the risk of decreased cerebral perfusion. These include intracerebral hemorrhage, acute heart failure, acute coronary syndromes, and most prominently, aortic dissection.

A hypertensive emergency involving the kidney commonly is followed by a further deterioration in kidney function even when BP is lowered properly. The most important predictor of the need for acute dialysis is not the BP level, but instead the degree of kidney dysfunction (both *e*GFR and degree of albuminuria). Some physicians prefer fenoldopam to nicardipine or nitroprusside in this setting because of its lack of toxic metabolites and specific renal vasodilating effects. The need for acute dialysis often is precipitated by BP reduction in patients with preexisting stage 3 to 5 CKD, but many patients are able to avoid dialysis (and some even discontinue it) in the long term if BP is carefully and well controlled during follow-up.

Hypertensive crises resulting from catecholamine excess states (e.g., pheochromocytoma, monoamine oxidase inhibitor crisis, and cocaine intoxication) are most appropriately managed with an intravenous $\alpha$-blocker (e.g., phentolamine), with a $\beta$-blocker with $\alpha$-$\beta$ activity

**Table 46.9 Recommended Treatment Approaches to Hypertensive Emergencies**

| Acute Target-Organ Damage | Timing for Acute BP Reduction[a] | Preferred Intravenous Drugs[b] |
|---|---|---|
| Diffuse microvascular injury ("malignant hypertension")[c] | Decrease BP by 20%-25% during first hour and to 160/100 mm Hg by 2-6 h | Labetalol, nicardipine, nitroprusside |
| Hypertensive encephalopathy | Decrease BP by 20%-25% during first hour and to 160/100 mm Hg by 2-6 h | Labetalol, nicardipine, nitroprusside; avoid hydralazine |
| Acute intracerebral hemorrhage | If systolic BP is 150-220 mm Hg, decrease systolic BP to 140-150 mm Hg within 1 h, particularly in patients without known hypertension and those with underlying vascular abnormalities, such as aneurysms or arteriovenous malformations. In patients with large hematoma volume and evidence of increased intracranial pressure, BP management should be more liberal (keep systolic BP <180 mm Hg). Lowering systolic BP below 140 mm Hg may be harmful. | Labetalol, nicardipine, devidipine, nitroprusside; avoid hydralazine |
| Acute ischemic stroke | If thrombolytic therapy is indicated, decrease BP to <185/110 mm Hg before giving thrombolytic agents and maintain BP <180/105 mm Hg for the first 24 h. If thrombolytic therapy is not indicated and there is no acute target-organ damage other than stroke, the strategy depends on BP. If BP is <220/120 mm Hg, no intervention is indicated for the first 48-72 h. If BP is 2220/120 mm Hg or there is other acute target-organ damage, such as heart failure or myocardial infarction, decrease BP by 15% within 1 h. | Labetalol, nicardipine, clevidipine, nitroprusside; avoid hydralazine |
| Acute coronary syndromes | Decrease systolic BP to <140 mm Hg within 1 h; keep diastolic BP >60 mm Hg | Nitroglycerin, labetalol, esmolol, metoprolol; avoid hydralazine |
| Acute heart failure | Decrease systolic BP to <140 mm Hg within 1 h | Nitroglycerin, nitroprusside, loop diuretics needed in most cases; enalaprilat or hydralazine may be useful; avoid β-blockers |
| Aortic dissection | Decrease both systolic BP to <120 mm Hg and heart rate to <60 beats/min within 20 min | Esmolol (or labetalol) plus one of nicardipine, clevidipine, nitroprusside, or nitroglycerin; both a beta-blocker (unless bradycardia is already present) and a vasodilator should be used |

[a]Initiation, reinstatement, or adjustments of long-acting oral antihypertensive medications should take place during the first 6 to 12 hours. Oral drugs are chosen on the basis of conventional guidelines. Intravenous medications can be tapered on the basis of observed blood pressure levels after the addition of oral agents.
[b]Nitroprusside continues to be a recommended agent by most guidelines. Given its potential toxicity, it should be avoided as a first choice when other options are available.
[c]Diffuse microvascular injury is identified as high-grade retinopathy (hemorrhages, exudates, or papilledema), acute kidney injury, or microangiopathic hemolytic anemia or thrombocytopenia present alone or in combination.
*BP,* Blood pressure.

(e.g., labetalol) added later, if needed. Many patients with severe hypertension caused by sudden withdrawal of antihypertensive agents (e.g., clonidine) are easily managed by giving one acute dose of the missed drug.

Hypertensive crises during pregnancy must be managed in a more careful and conservative manner because of the presence of the fetus. Magnesium sulfate, labetalol, and hydralazine are the drugs of choice for acute management until delivery is performed to assist in the management of hypertension in pregnancy.

Hypertensive urgencies (elevated BP, but without acute ongoing target-organ damage) ought to be treated.[266] Although comparison of less tight versus tight control showed no effect on rates of preeclampsia, less tight control demonstrated a higher risk of thrombocytopenia and elevated liver enzyme levels, markers of disease severity. In addition, tight control (<135/85 mm Hg) may have decreased the risk of preterm birth.[266,267]

A common cause of hypertensive urgency is anxiety, especially in older people, and can result in high readings.[268] The BP in many such patients, however, spontaneously falls during a 30-minute period of quiet rest and slow breathing exercises.[269] Conversely, immediate-release nifedipine capsules should NOT be used as they can cause precipitous hypotension, stroke, myocardial infarction, and death. According to the U.S. Food and Drug Administration, immediate-release nifedipine "should be used with great caution, if at all." In such instances, true "hypotension" (e.g., systolic BP < 90 mm Hg) may not be observed, yet the BP may fall below the autoregulatory threshold (which is likely different for every patient, and unknown to the treating physician until it is surpassed), precipitating ischemia.

Clonidine, captopril, labetalol, several other short-acting antihypertensive drugs, and even amlodipine, have been used in this setting, but none has a clear advantage over the others, and each is usually effective in most patients, though there is no evidence of benefit and there is possible harm related to excessive BP reduction following treatment of hypertensive urgencies with oral medications.[270] The most important aspect of managing a hypertensive urgency is to refer the patient to a good source of ongoing care for hypertension, where adherence to antihypertensive therapy during long-term follow-up will be more likely.

## CLINICAL OUTCOME TRIALS

Hundreds of clinical studies have evaluated the efficacy and safety of the eight different classes of antihypertensive medications. All BP-lowering agents need to have at least two appropriately powered, placebo-controlled studies to meet specific U.S. Food and Drug Administration criteria for approval as antihypertensive agents. This chapter does not focus on the details of these studies but rather on data supporting BP reductions with certain BP-lowering classes and the impact of CKD progression, as well as trials evaluating CV outcomes in patients with kidney disease. This section is not an exhaustive evaluation of all CV endpoint trials but rather a focus on trials associated with renal outcomes or CV outcome trials with substudies in patients with kidney disease.

Meta-analyses of all commonly used antihypertensive drug classes demonstrate that, regardless of the agent used, reduction in BP corresponds to reduction in CV events if BP reduction is achieved[271-273] This reduction in CV risk, however, is predominantly seen in people with stage 2 hypertension with less data to support risk reduction in stage 1 hypertension.

Risk reduction is derived predominantly from reduced incidence of stroke, myocardial infarction, and heart failure. In all trials published to date, the intervention group with the best BP control experienced the best outcomes. An exception to this observation was the Avoiding Cardiovascular Events Through Combination Therapy in Patients Living with Systolic Hypertension (ACCOMPLISH), a CV outcome trial with more than 11,000 people.[274] In this trial, both groups had similar BP control and both randomized to the same ACE inhibitor (benazepril), yet the group initially randomized to a single-pill combination of benazepril with a calcium antagonist had a 20% CV risk reduction compared with the ACE inhibitor plus diuretic group. Patients randomized to the benazepril-amlodipine combination also experienced attenuation of CKD progression.[275]

Almost all persons with an eGFR of <60 mL/min/1.73 $m^2$ and hypertension will require two or more medications to achieve a BP goal of <140/90 mm Hg. Single-pill combinations, including the matching of an RAAS blocker with either a calcium antagonist or diuretic, are preferred agents.[276,277] These combinations, when given, generally in an additive fashion, reduce CV events and CKD progression. Other combinations that are efficacious for reducing BP but not tested in clinical trials include β-blockers with dihydropyridine calcium blockers and dihydropyridine CCBs with diuretics.[276]

There have been a number of trials assessing both CV outcomes and changes in CKD progression. All these trials assume adherence with antihypertensive medications. However, according to one report, only 71% of persons with hypertension in the United States are on treatment and only 48% have their BP under what had been previously considered to be adequate (<140/90 mm Hg) control. Moreover, two separate studies, one in the United Kingdom and the other in Germany, both evaluating medication adherence, showed that only approximately 45% of patients who claimed to be taking BP-lowering medication actually were, as assessed by urine analysis of drug metabolites.[278,279] Although there has been significant reduction in the age-adjusted death rate for stroke and coronary artery disease since the early 1980s as a result of better BP control (and better treatment of other risk factors including hypercholesterolemia), heart disease and stroke remain the first and third leading causes of death in Western countries. This finding emphasizes the importance of identifying and treating patients with hypertension. The following section discusses outcome trials focused on BP reduction that evaluated CKD progression, as well as CV outcomes in persons with CKD.

### BLOOD PRESSURE CONTROL AND CHRONIC KIDNEY DISEASE PROGRESSION

There is clear evidence that partial blockade of the RAAS slows nephropathy progression among those with stage 3 or higher proteinuric kidney disease. A comparative study of outcome trials demonstrated, while showing no significant difference among the classes, indicated that ARBs are better tolerated and with fewer side effects.[280]

It is also clear that a lower level of BP (i.e., <130/80 mm Hg) does not slow nephropathy progression. Three prospective randomized long-term CKD outcome trials in nondiabetic kidney disease failed to show a benefit on slowing nephropathy progression among the groups with lower BP.[281-284] Hence the updated KDIGO BP guidelines that recommend a BP goal below 120/80 mm Hg in patients with CKD[285] are justified on the basis of cardiovascular rather than kidney benefits, wherein there was no evidence of heterogeneity of benefit in the Systolic Blood Pressure Intervention Trial (SPRINT) by baseline eGFR.[283] Moreover, longer-term follow-up of SPRINT trial participants failed to show a reduction in ESKD events among patients randomized to lower systolic BP (<120 mm Hg) targets.[286] The previous goal of <130/80 mm Hg has much more evidence of support and is endorsed in the presence of high albuminuria (>300 mg/g).

### TRIALS IN NONDIABETIC CHRONIC KIDNEY DISEASE THAT CONTRIBUTED TO BLOOD PRESSURE GOAL

#### MODIFICATION OF DIET IN RENAL DISEASE STUDY

The Modification of Diet in Renal Disease (MDRD) Study was the first appropriately powered study to test whether a lower BP goal was associated with slower progression of CKD. The lower BP goal (target mean arterial pressure ≤92 mm Hg for patients ≤60 years old or 98 mm Hg for patients ≥61 years old) was associated with a significant reduction in proteinuria and a slower subsequent decline in GFR, albeit not significant compared with the higher pressure group with a mean arterial pressure of 102 to 107 mm Hg.

### THE AFRICAN AMERICAN STUDY OF KIDNEY DISEASE AND HYPERTENSION

The effect of antihypertensive therapy on the progression of CKD secondary to hypertension is more controversial. In the Multiple Risk Factor Intervention Trial (MRFIT), where thiazide diuretics and β-blockers were primarily used to control BP, slowing of decline in kidney function was not seen in African American men but was seen in all other racial groups studied.[287]

In the African American Study of Kidney Disease and Hypertension (AASK), the use of an ACEI (ramipril) was found to be more effective at slowing CKD progression compared with either the dihydropyridine CCB, amlodipine, or metoprolol.[24] This trial in >1000 African Americans failed to show superior protection with BP reduced to levels below 130/80 mm Hg compared with conventional BP targets of 140/90 mm Hg in subjects with hypertensive nephrosclerosis, even with a total of a 10-year follow-up.[282] Masked hypertension may be a confounder of these outcomes. A subanalysis of more than 50% of the AASK participants who had 24-hour ABPM showed inadequate 24-hour BP control in approximately 36% of the cohort.[288] Masked hypertension and failure of nocturnal dipping were the two most common reasons for poor out-of-office BP control.

Further studies were performed to assess whether change in antihypertensive dose timing corrected the failure of nocturnal dipping, and the results were negative.[289] In contrast, studies performed in Spain in Caucasian patients with CKD stages 2 to 3b demonstrate an improvement in dipping status when dosing BP medications at night.[289]

### RAMIPRIL EFFICACY IN NEPHROPATHY TRIAL–2 (REIN-2)

This multicenter, randomized controlled trial was conducted in Italy among patients with proteinuria associated with nondiabetic kidney disease who were receiving background treatment with the ACEI ramipril (2.5–5 mg/day).[281] The aim was to assess the effect of intensified versus conventional BP control on progression to end-stage kidney disease. Subjects were randomly assigned to either conventional (diastolic <90 mm Hg; $n$ = 169) or intensified (systolic/diastolic <130/80 mm Hg; $n$ = 169) BP control. To achieve the intensified BP level, patients received add-on therapy with dihydropyridine CCB, felodipine (5–10 mg/day). The primary outcome measure was time to end-stage kidney disease over 36 month follow-up. The authors found that over a median follow-up of 19 months, 38 out of 167 (23%) patients assigned to intensified BP control and 34 out of 168 (20%) allocated conventional control progressed to end-stage kidney disease (HR, 1.00; 95% confidence interval [CI], 0.61–1.64; $P$ = 0.99). Hence there was no benefit of aggressive BP lowering in slowing this nondiabetic nephropathy.

### BENAZEPRIL RENAL OUTCOME STUDIES

In a 3-year randomized placebo-controlled trial involving 583 patients with CKD resulting from a variety of etiologies, 300 patients were randomized to benazepril and 283 were randomized to placebo. Kidney function was classified according to the baseline measured creatinine clearance (CrCl): 227 patients had CrCl 46–60 mL/min), and 356 had CrCl 30–45 mL per minute). The primary endpoint was doubling of serum creatinine or need for dialysis. At 3 years, there was a 53% risk reduction of reaching the primary endpoint in the benazepril group, compared with placebo.[290] Similar findings were observed in a clinical trial of benazepril versus placebo among patients with more advanced CKD (CrCl below 30 mL/min). Treatment with benazepril resulted in a 43% relative reduction in the risk of doubling of serum creatinine, progression to ESKD, or death (the study's primary outcome).[291]

## DIABETES

There are no randomized trials of BP control and CKD progression among patients with diabetes. The lower-goal BP that has been recommended in patients with diabetes resulted from post hoc analyses of trials that evaluated CV outcomes in subjects with diabetic kidney disease. All these studies showed a benefit on CV risk reduction and some in slowing CKD progression in the group randomized to the lower pressures.

The Hypertension Optimal Treatment (HOT) trial was the first CV outcome trial that evaluated different levels of diastolic BP (80 mm Hg, 85 mm Hg, and 90 mm Hg) in 18,790 patients with hypertension, from 26 countries, mean age of 61.5 years. In the subgroup of patients with diabetes mellitus, a post hoc analysis showed a 51% reduction in major CV events in the group randomized to a diastolic BP ≤80 mm Hg compared with the target group of ≤90 mm Hg ($P$ for trend = 0.005).[292] Since the trial failed to achieve statistical significance on the primary endpoint, this subgroup analysis should be interpreted with caution and the result nominally significant.

The Ongoing Telmisartan Alone and in combination with Ramipril Global Endpoint Trial (ONTARGET) and the International Verapamil-Trandolapril Study (INVEST), like the Action to Control Cardiovascular Risk in Diabetes (ACCORD) trial, failed to show a benefit on CV outcomes.[293-295] Taken together, these three studies suggest no additional benefit of BP lowering below 130/80 mm Hg on CV risk reduction compared with 130 to 139/80 to 85 mm Hg.

However, two more recent trials suggest that there may be value to more intensive treatment goals in diabetes. In the Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients (STEP) trial, 8511 hypertensive patients aged 60 to 80 years were randomized to a systolic BP target of <130 mm Hg or a target between 130 and 150 mm Hg.[296]

The 2017 American Diabetes Association Position Paper of Blood Pressure Goals in Diabetes supports a BP goal of <140/90 but acknowledges that most patients will achieve a greater cardiovascular risk reduction at BP levels of <130/80 mm Hg and thus suggests that most patients with diabetes strive for this goal.[297] Given the interim developments, the 2024 American Diabetes Association Standards of Care in Diabetes propose a target BP goal of <130/80 mm Hg if it can be safely attained.[298] Several studies demonstrate that some classes of antihypertensive drugs should be used preferentially in patients with diabetes who have nephropathy (i.e., blockers of the RAAS system). However, this class of agents does not possess any specific advantages over other antihypertensive classes in people with diabetes who do not have nephropathy. Moreover, there is no evidence that blockers of the RAAS system benefit people with normotension with or without microalbuminuria from developing declines in kidney function.[297,299,300]

Many post hoc analyses demonstrate that diuretics and β-blockers both worsen glycemic control among patients with hypertension and diabetes and increase the development of new-onset diabetes in patients with hypertension and impaired fasting glucose. Thiazides worsen glycemic status through hypokalemia and other mechanisms related to increased visceral adiposity. β-Blockers that result in vasoconstriction, such as metoprolol and atenolol, worsen insulin sensitivity. The vasodilating β-blockers, such as carvedilol and nebivolol, have neutral effects on glycemic control and increase insulin sensitivity.[301]

Post hoc analyses of two different CV outcome trials note that CV event rates were not higher in patients randomized to thiazides despite a higher incidence of new-onset diabetes. An analysis of the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT) subgroup with diabetes failed to show a higher CV event rate in the thiazide group, even though they had the most pronounced worsening of glycemic control. In a 12-year follow-up of the ALLHAT, participants on chlorthalidone with incident diabetes had consistently lower, nonsignificant risk for CV disease mortality versus no diabetes or participants on amlodipine or lisinopril with incident diabetes.[302] Moreover, participants with incident diabetes had elevated coronary heart disease risk compared with those with no diabetes, but those on chlorthalidone had significantly lower risk for events compared with those on lisinopril. Thus thiazide-related incident diabetes has less adverse long-term CV disease impact than incident diabetes that develops while on other antihypertensive medications.

## CHRONIC KIDNEY DISEASE ENDPOINT TRIALS IN PATIENTS WITH DIABETES

While many studies have evaluated changes in albuminuria and eGFR, four CKD outcome trials have been statistically powered for meaningful outcomes on CKD progression; those trials are discussed in this section.

### The Captopril Trial

The first clinical outcome trial to put RAAS blocking agents on the proverbial map for slowing CKD progression was this trial in 1993. It was a randomized, controlled trial comparing captopril with placebo in 472 patients with hypertension and type 1 diabetes in whom urinary protein excretion was >500 mg/day and the serum creatinine concentration <2.5 mg/dL.[303] The primary endpoint was a doubling of the baseline serum creatinine concentration. Captopril reduced doubling of serum creatinine concentration by 48% over placebo (25 patients, captopril group, vs. 43 patients, placebo group; $P = 0.007$). The benefit was even higher in patients with a serum creatinine concentration above 2 mg/dL. The mean rate of decline in creatinine clearance was 11% ± 21% per year in the captopril group and 17% ± 20% per year in the placebo group ($P = 0.03$). Captopril treatment resulted in a 50% reduction in the risk for the combined endpoints of death, dialysis, and transplantation. There was a 4 mm Hg significantly lower systolic BP in the captopril group; however, the benefit was independent of this lower BP. To date, this is the only kidney endpoint trial with an ACEI in advanced diabetic kidney disease.

### Reduction of Endpoints in Non–insulin dependent diabetes mellitus with the Angiotensin II Antagonist Losartan

The Reduction of Endpoints in Non–Insulin-Dependent Diabetes Mellitus with the Angiotensin II Antagonist Losartan (RENAAL) study was a randomized, double-blind study in 1513 patients comparing losartan (50 to 100 mg once daily) with placebo, in addition to conventional antihypertensive treatment wherein participants were followed for a mean of 3.4 years.[304] The primary outcome was the composite of a doubling of the baseline serum creatinine concentration, end-stage kidney disease, or death. Secondary endpoints included a composite of morbidity and mortality from CV causes, proteinuria, and the rate of progression of renal disease. Losartan reduced the incidence of a doubling of the serum creatinine concentration (risk reduction, 25%; $P = 0.006$) and end-stage kidney disease (risk reduction, 28%; $P = 0.002$) but had no effect on the rate of death. The benefit exceeded that attributable to changes in BP. The composite of morbidity and mortality from CV causes was similar in the two groups, although the rate of first hospitalization for heart failure was significantly lower with losartan (risk reduction, 32%; $P = 0.005$). Proteinuria declined by 35% with losartan compared to placebo ($P < 0.001$).

### Irbesartan Diabetic Nephropathy Trial

In this study, 1715 patients with hypertension with nephropathy due to type 2 diabetes were randomized to treatment with irbesartan (300 mg daily), amlodipine (10 mg daily), or placebo.[301] The target BP was 135/85 mm Hg or less in all groups. The primary composite endpoint was a doubling of the baseline serum creatinine level, development of end-stage kidney disease, or death from any cause. Treatment with irbesartan resulted in a 20% decrease in risk of the primary composite endpoint compared with placebo ($P = 0.02$) and a 23% lower risk than the amlodipine group ($P = 0.006$). Randomization to irbesartan also resulted in lower risks of doubling of the serum creatinine level was 33% compared with placebo ($P = 0.003$) and 37% compared with amlodipine ($P < 0.001$). Like the RENAAL trial, there were no significant differences in the rates of death from any cause or in the CV composite endpoint. Thus irbesartan, like losartan, is effective in protecting against the progression of nephropathy due to type 2 diabetes.

The next trials examined whether the *combination* of an ACEI and ARB would further slow nephropathy progression associated with type 2 diabetes.

### Veterans Affairs Nephropathy in Diabetes Trial

The Veterans Affairs Nephropathy in Diabetes (VA NEPHRON D) trial studied 1448 male veterans with type 2 diabetes, hypertension, and nephropathy. Participants were initially given losartan 100 mg per day and then randomized to receive either lisinopril (at a dose of 10–40 mg per day) or placebo.[306] The primary endpoint was the first occurrence of a change in the eGFR (a decline of ≥30 mL/min/1.73 $m^2$ if the initial eGFR was ≥60 mL/min/1.73 $m^2$ or a decline of ≥50% if the initial eGFR was <60 mL/min/1.73 $m^2$), end-stage kidney disease, or death. Safety outcomes included mortality, hyperkalemia, and acute kidney injury. The study was stopped early secondary to safety concerns with a median follow-up of 2.2 years. There was no difference in the primary endpoints between groups ($P = 0.30$), although results weighed numerically in favor of the combination group (hazard ratio 0.88, 95% CI [0.70, 1,12]). There was no benefit with respect to mortality or CV events. Combination therapy increased the risk for hyperkalemia (6.3 events per 100 person-years vs. 2.6 events per 100 person-years with monotherapy; $P < 0.001$) and acute kidney injury (12.2 vs. 6.7 events per 100 person-years; $P < 0.001$).

A second study in persons with diabetic nephropathy to examine combination therapy was the Aliskiren Trial in Type 2 Diabetes Using Cardio-Renal Endpoints (ALTITUDE). This trial had as a primary endpoint a CV outcome rather than CKD progression,[307] although CKD was part of the primary composite outcome. ALTITUDE was a double-blind randomized trial of 8561 patients allocated to aliskiren (300 mg daily) or placebo as an adjunct to background ACEI or ARB. The primary endpoint was a composite of the time to CV death or a first occurrence of cardiac arrest with resuscitation; nonfatal myocardial infarction; nonfatal stroke; unplanned hospitalization for heart failure; end-stage kidney disease; death attributable to kidney failure; the need for renal replacement therapy with no dialysis or transplantation available or initiated; or doubling of the baseline serum creatinine.

Like the VA NEPHRON D trial, ALTITUDE was terminated prematurely after the second interim efficacy analysis, at a median of 32.9 months. The primary reasons for stopping were hyperkalemia and hypotension in the combined aliskiren + ACEI or ARB group. Moreover, at the time of stopping there was no difference in the primary endpoint between groups ($P = 0.12$), nor was there a difference in the secondary kidney endpoints. Systolic and diastolic BP levels were lower with aliskiren (between-group differences, 1.3 and 0.6 mm Hg, respectively), and the mean reduction in the urine albumin-to-creatinine ratio was greater in the aliskiren group.

### HEAD-TO-HEAD COMPARISONS OF ARBS AND OTHER DRUG CLASSES IN PATIENTS WITH HYPERTENSION WITH TYPE 2 DIABETES WITH OVERT NEPHROPATHY

Two different ARBs were compared in a double-blind, prospective trial of 860 patients with type 2 diabetes whose BP levels were >130/80 mm Hg or who were receiving antihypertensive medication and had a morning spot urine albumin-to-creatinine ratio of 700 or more. Patients were randomized to telmisartan or losartan.[308] The primary endpoint was the difference in the urine albumin-to-creatinine ratio between the groups at 52 weeks. The geometric mean and the mean of the urine albumin-to-creatinine ratio fell in both groups at 52 weeks, but both were significantly greater for telmisartan compared with losartan. Mean systolic BP reductions were not significantly different between groups at the end of the trial. This trial suggests that longer-acting telmisartan is superior to losartan in reducing albuminuria in patients with hypertension with diabetic nephropathy, despite a similar reduction in BP. Keep in mind that there are no adequately powered renal outcome studies with telmisartan. However, there is a renal substudy of the ONTARGET study that shows a similar effect in albuminuria reduction between ramipril and telmisartan with the combined group having the greatest reduction in albuminuria.[293]

A second trial that compared the effects of BP lowering on albuminuria reduction in diabetic nephropathy was the Gauging Albuminuria Reduction with Lotrel in Diabetic Patients with Hypertension (GUARD) trial.[309] This double-blind, randomized controlled trial of 322 patients with hypertension, albuminuria, and type 2 diabetes tested the hypothesis that combining an ACEI with either a thiazide or a CCB will result in similar reductions in BP and albuminuria. Groups were randomized to either benazepril plus amlodipine or benazepril plus hydrochlorothiazide for 1 year. The trial employed a noninferiority design. Both combinations significantly reduced the urine albumin-to-creatinine ratio and sitting BP of the entire cohort. The percentage of patients progressing to macroalbuminuria was similar for both groups. When only patients with microalbuminuria and hypertension were examined, a larger percentage of patients receiving the diuretic and ACEI normalized their albuminuria. In contrast, BP reduction, particularly the diastolic component, favored the combination with amlodipine. This finding was also seen in the large CV outcome trial ACCOMPLISH.

### ADVANCE TRIAL

In the Action in Diabetes and Vascular Disease: Preterax and Diamicron Modified Release Controlled Evaluation (ADVANCE) factorial trial, the combination of perindopril and indapamide reduced mortality among patients with type 2 diabetes.[310] While this trial did not randomize to different levels of BP control, it did maintain systolic BP control to levels below 140 mm Hg. After a 6-year posttrial follow-up, surviving participants were invited for reassessment. The primary endpoints were death from any cause and major macrovascular events. The authors found the baseline characteristics similar among the 11,140 patients who originally underwent randomization and the 8494 patients who participated in the posttrial follow-up for a median of 5.9 years (blood pressure–lowering comparison) or 5.4 years (glycemic control comparison). Between-group differences in BP and glycated hemoglobin levels during the trial were no longer evident by the first posttrial visit. The reductions in the risk for death from any cause and of death from CV causes that had been observed in the group receiving active BP-lowering treatment during the trial were attenuated but remained significant at the end of the posttrial follow-up period ($P = 0.04$). Thus in this long-term diabetes trial, the mortality benefits originally observed among patients assigned to BP-lowering therapy were attenuated but still evident years later.

## CLINICAL TRIALS IN OLDER ADULTS

All major clinical trials involving persons older than 65 years are summarized in Table 46.12. Most patients recruited in these trials before Hypertension in the Very Elderly Trial (HYVET) were younger than 80 years of age, limiting information about octogenarians.[13] Results in these trials showed a reduction in the incidence of both stroke and CV morbidity. Previous clinical trials showed that lowering BP in older adults has no effect on overall mortality or on the incidence of fatal or nonfatal myocardial infarction.[252,311] The results of the older subgroup of SPRINT with a mean age of 79 years, however, showed a clear benefit on all cardiovascular outcomes, especially heart failure risk, among those treated to a lower BP.[14] More recently, the Strategy of Blood Pressure Intervention in the Elderly Hypertensive Patients (STEP) trial enrolled 8511 patients of 60 to 80 years.[296] Patients were randomized to a target BP goal of 110 to 129 mm Hg (intensive group) or 130 to 149 mm Hg. After a median follow-up of 3.3 years, assignment to the intensive target resulted in a significant 26% reduction in the primary cardiovascular composite endpoint compared with the control group (HR 0.74, CI 0.60–0.92), with significant reductions in some of the individual components of the primary endpoint (stroke, acute coronary syndrome, and acute heart failure).[296] Before SPRINT and STEP, there were five trials performed in older persons but only two were randomized to different BP levels that were below a systolic of 140 mm Hg. The trials were SHEP, Systolic Hypertension in Europe (Syst-Eur), HYVET, Japanese Trial to Assess Optimal Systolic Blood Pressure in Elderly Hypertensive Patients (JATOS), and Valsartan in Elderly Isolated Systolic Hypertension (VALISH)[212] (see Table 46.12). Both randomized BP trials with lower BP levels were carried out in Japan. The results from these two studies showed no additional benefit of achieving a BP of less than 140/90 mm Hg compared to less than 150/90 mm Hg. Despite these older studies, the strong results of SPRINT and STEP justify the use of intensive BP goals (<130/80 mm Hg) also in older individuals as long as well tolerated.

A concern in patients with isolated systolic hypertension is the reduction of diastolic BP after initiation of antihypertensive therapy. A low diastolic pressure carries a theoretical risk of reducing coronary perfusion and possibly increasing CV risk. The relation between diastolic BP and CV death in observational studies, particularly myocardial infarction, is like a J-shaped curve; thus excessive reduction in diastolic pressures should be avoided in patients with coronary artery disease who are being treated for hypertension.[312,313] A BP of 119/84 mm Hg was identified as a nadir when treating patients with coronary artery disease[314]; however, among patients with CKD, a nadir of 131/71 mm Hg was identified as a level below which higher CV mortality was present.[314] However, an analysis of outcomes in SPRINT stratified by baseline diastolic BP did not show any significant impact of baseline diastolic BP on the benefits (or risks) of intensive systolic BP reduction.[315]

When treating patients with isolated systolic hypertension, the JNC 7 suggests a minimum posttreatment diastolic BP of 60 mm Hg overall or perhaps 65 mm Hg in patients with known coronary artery disease, unless symptoms that could be attributed to hypoperfusion occur at higher pressures. This was seen in the findings of the SHEP trial,[316] where older patients with lower diastolic BP have higher CV event rates.

Additionally, in some trials involving older individuals, there is concern about thiazide use, hyponatremia, and worsening of kidney function in subjects with hypertension. In the European Working Party on High Blood Pressure in the Elderly trial, a significantly higher incidence of impaired kidney function was found in those receiving thiazides compared with placebo.[317] In the SHEP trial, serum creatinine levels increased significantly in patients treated with thiazides compared with placebo. In ALLHAT, the chlorthalidone-treated group showed worse kidney function than either the amlodipine- or lisinopril-treated groups at both the 2- and 4-year endpoints.[318]

The HYVET trial in 2008 changed how hypertension in older adults is managed. It randomly assigned 3845 patients who were 80 years old and above and had systolic BP above 160 mm Hg to either indapamide or placebo. Results of HYVET provided evidence that BP lowering with treatment using antihypertensive medications is associated with definite CV benefits. The use of indapamide supplemented by perindopril showed reductions in the incidence of stroke, congestive heart failure, and CV fatal events.

There are no specific trials powered to assess CKD progression in response to BP lowering in older patients. There are prespecified and post hoc analyses of trials where the mean age is above 65 examining changes in CKD progression. A notable example is the prespecified analysis of the ACCOMPLISH trial, where more than 40% of participants with CKD were age 70 or greater. This trial showed the benefit of the RAAS blocker–CCB combination over the RAAS blocker–thiazide for slowing progression and time to dialysis.[275] A post hoc analysis of the ALLHAT study in participants older than 65 years also showed that BP control slowed progression.[318] In this trial we have no information about albuminuria, however.

## MANAGEMENT OF HYPERTENSION IN OLDER ADULTS WITH CHRONIC KIDNEY DISEASE

### Nonpharmacologic intervention

In the Trial of Nonpharmacologic Interventions in the Elderly (TONE),[319] the combination of weight loss and sodium restriction showed a drop of 5.3 ± 1.2 mm Hg in the systolic BP and 3.4 ± 0.8 mm Hg diastolic BP in obese, older patients with hypertension. The goal of sodium restriction was 1.8 g/24 hours, and the goal for weight reduction was 10 lb. Other lifestyle changes recommended for the older patient include watching potassium intake to keep the level of serum potassium in a safe range. A strategy of community-wide provision of a lower-sodium/higher-potassium diet was tested in a cluster randomized trial in China involving patients with a history of stroke or age 60 years or older.[320] An increase in dietary potassium of 20 mEq/day resulted in a modest fall in BP (3.4/0.7 mm Hg) and a 13% reduction in major cardiovascular events.[320] Physical activity, especially aerobic or dynamic resistance exercise, should be engaged for 90 to 150 minutes per week.[17] Intake of NSAIDs should also be decreased to a minimum, as these patients are more likely to take NSAIDs for arthritis and pain. These drugs are known to cause elevations in BP by inhibiting the production of vasodilatory prostaglandins and may increase BP by as much as 6 mm Hg.[321,322] Their BP-raising effects can be blunted by both CCBs and, to a lesser extent, diuretics.[323]

### Pharmacologic intervention

The primary agents employed in the treatment of hypertension include thiazide and thiazide-type agents, ACE inhibitors, ARBs, and CCBs. While many other drugs/drug classes are available, confirmation that these agents decrease adverse clinical outcomes to a similar extent as the primary agents are either lacking or safety and tolerability may relegate their role to use as secondary agents.[275] In particular, there is inadequate evidence to support the initial use of β-blockers for hypertension in the absence of specific cardiovascular comorbidities (e.g., angina or heart failure). In considering the initial drug treatment of high BP, several different strategies may be contemplated. Many patients can be started on a single agent, but consideration should be given to starting with two drugs for those >20/10 mm Hg above the goal of 130/80 mm Hg. Many patients started on a single agent will subsequently require ≥2 drugs from different pharmacologic classes to reach their BP goals. Drug regimens with complementary activity, where a second antihypertensive is used to block compensatory responses to the initial agent or affect a different pressor mechanism, can result in additive lowering of BP. Use of combination therapy may also allow for treatment with lower doses of individual agents, which serves to minimize adverse effects and improve adherence. Several two- and three-fixed-dose drug combinations of antihypertensive drug therapy are available in generic form with complementary mechanisms of action among the components.

The full dose of the drug is the highest pharmacologic dose of drug available, while the maximum dose is the highest dose that the person can tolerate without side effects. If the person is having no therapeutic response or has significant side effects, a drug from another class should be substituted. In fact, most older patients require two or more drugs to achieve the recommended goals in clinical trials.

Either a thiazide or CCB may be an initial drug, or a thiazide should be one of the first two agents when starting combination drugs, When the BP is more than 20/10 mm Hg above goal, the recommendation is to initiate with two antihypertensive medications, with one of the choices being a thiazide. Single-pill combinations that incorporate logical doses of two agents may enhance convenience and compliance.[211] However, there is still a need for individualization of treatment; thus treatment options should be carefully considered in older adults. The benefits of lowering BP must be weighed against the risks of side effects and the concomitant morbidity of the patient.

Thiazide and thiazide-like agents including hydrochlorothiazide, chlorthalidone, indapamide, and bendrofluazide, as well as CCBs, are recommended for initiating therapy.[8,12] Thiazides can cause an initial reduction of intravascular volume, peripheral vascular resistance, and BP in more than 50% of patients and are well tolerated and inexpensive.[324,325] However, they can cause hypokalemia, hypomagnesemia, and hyponatremia and are therefore not recommended in patients with baseline electrolyte abnormalities or those with a history of hyponatremia. Serum potassium concentrations should be monitored, and supplementation given if needed.

CCBs are well suited for older patients whose hypertensive profile is based on increasing arterial dysfunction secondary to decreased atrial and ventricular compliance. This class of drugs dilates coronary and peripheral arteries in doses that do not severely affect myocardial contractility.[326,327] Most adverse effects, like ankle edema, headache, and postural hypotension, relate to vasodilation. Ankle edema is not secondary to sodium retention, as CCBs are natriuretic when given initially, but profound vasodilation with poor venous return in older people is the major contributor.

First-generation (nondihydropyridine) drugs including verapamil and diltiazem should be avoided in patients with left ventricular dysfunction. Nondihydropyridines can precipitate heart blocks in older adults with underlying conduction defects.

RAAS blockers, such as ACEIs, ARBs, and direct renin inhibitors, may be used in older adults.[328] Theoretically, as aging occurs, there is a reduction in angiotensin levels; thus ACEIs may not be an effective medication for hypertension in older adults. However, several clinical trials have shown otherwise. The use of ACEIs is beneficial in the reduction of morbidity and mortality in patients with myocardial infarction, reduced systolic function, heart failure, and reduction in the progression of diabetic kidney disease and hypertensive nephrosclerosis.[329] Lastly, it appears that use of a blocker of the RAAS may provide greater benefit for CV and renal risk reduction than use of a diuretic, based on data from ACCOMPLISH, a large outcome trial of 11,506 persons with a mean age of 68 years. While data on kidney disease in older adults are scarce, ACCOMPLISH did provide some evidence worthy of being tested in a prospective trial (i.e., that a CCB–ACEI combination led to fewer patients going on dialysis than a diuretic–ACEI combination).[275,329] Note that this outcome could not be explained by differences in BP of 1.2 mm Hg systolic. It must be noted that those older than 70 years of age tend to drink small amounts of fluid, and hence this makes them more vulnerable to decline in kidney function by RAAS blockade, as already discussed. Thus it is recommended that they increase their fluid intake to prevent volume depletion.

The clinical benefits of β-blockers as monotherapy in uncomplicated older patients are poorly documented. They may have a role in combination therapy, especially in conjunction with thiazides. β-Blockers have established roles in patients with hypertension complicated by certain arrhythmias, migraine headaches, senile tremors, coronary artery disease, or heart failure.[330,331] Nebivolol, a selective $\beta_1$-blocker with NO properties, does not show any associated symptoms of depression, sexual dysfunction, dyslipidemia, and hyperglycemia in older adults, unlike earlier generations of β-blockers.[332]

Potassium-sparing diuretics are useful when combined with other agents. Aldosterone-blocking agents like spironolactone and eplerenone reduce vascular stiffness and systolic BP.[333,334] They are also useful for patients with hypertension with heart failure or primary aldosteronism. Gynecomastia and sexual dysfunction are the limiting adverse events, reactions that occur in men using spironolactone but are less frequent with eplerenone. The epithelial sodium transport antagonists (amiloride, triamterene) are most useful when combined with other agents; several commercially available and widely used combination agents combine HCTZ with triamterene, amiloride, and spironolactone.

Other agents, such as α-blockers, centrally acting drugs (e.g., clonidine), and nonspecific vasodilators (e.g., minoxidil) should not be used as first- or second-line agents in an older adult with hypertension. Instead, they are reserved as part of a combination regimen to maximize BP control after other agents have been deployed.

### BLOOD PRESSURE MANAGEMENT IN PATIENTS UNDERGOING DIALYSIS

Elevations in BP in dialysis patients are almost exclusively due to excessive volume. Hence hypertension control related directly to volume management is a more common problem in hemodialysis than in peritoneal dialysis. Nonpharmacologic interventions targeting sodium and volume excess are fundamental for hypertension control in this cohort. If BP remains elevated after appropriate treatment of sodium and volume excess, drug treatment is needed. The patient's comorbidities and specific characteristics of each agent, such as dialysability, should be taken into account.[335] The European Renal and Cardiovascular Medicine (EURECA-m) working group provided the most recent approach for BP management in dialysis patients.[335]

There are no large multicenter randomized trials in patients undergoing dialysis to evaluate different levels of BP on CV outcomes. However, there is a prospective, randomized trial that evaluated the effects of an ACEI versus a β-blocker on LVH and as a secondary endpoint, CV mortality in patients undergoing dialysis.[336] The purpose of the study was to determine in 200 hypertensive patients undergoing hemodialysis with echocardiographic LVH whether a β-blocker or an ACEI used for BP-lowering results in a greater regression of LVH. Subjects were randomly assigned to either open-label lisinopril ($n$ = 100) or atenolol ($n$ = 100), each administered three times per week after dialysis. Monthly monitored home BP was controlled to <140/90 mm Hg with medications, dry weight adjustment, and sodium restriction. The primary outcome was the change in left ventricular mass index from baseline to 12 months. The results demonstrated that there was no between-group difference in 44-hour ambulatory BP. However, monthly measured home BP was consistently higher in the lisinopril group despite the need for both a greater number of antihypertensive agents and a greater reduction in dry weight. An independent data safety monitoring board recommended termination because of CV safety. Serious CV events were more prominent in the lisinopril group (20 events, atenolol vs. 43 events, lisinopril; $P$ = 0.001). Combined serious adverse events of myocardial infarction, stroke, and hospitalization for heart failure or CV death were much lower in the atenolol group. Thus it appears that β-blocker therapy may be superior to RAAS blockade in patients undergoing hemodialysis to reduce CV morbidity and all-cause hospitalizations.

Retrospective analysis of patients on hemodialysis supports other β-blockers such as carvedilol for also lowering CV events in patients who were hypertensive and undergoing hemodialysis. Japanese investigators further corroborate the results supporting β-blocker use in patients undergoing dialysis. The effect of β-blocker use on mortality among a cohort of patients undergoing hemodialysis was evaluated in a database analysis from the Dialysis Outcomes and Practice Patterns Study phase II of 2286 randomly selected patients on hemodialysis in Japan.[337] The main outcome measure was all-cause mortality. The authors found β-blocker use was low (i.e., only 247 patients [10.8%] were administered β-blockers, and 1828 patients [80%] were not). In multivariable, fully adjusted models, treatment with β-blockers was also independently associated with reduced all-cause mortality (HR, 0.48; $P$ = 0.02).[337]

Data from the U.S. Renal Data System demonstrates a U-shaped relationship between achieved systolic BP in patients undergoing dialysis and CV outcomes, suggesting that BP should be consistently above 120 mm Hg and below 150 mm Hg.[338] A meta-analysis by Agarwal and Sinha[339] of antihypertensive medications in patients undergoing dialysis demonstrates that regardless of class, reducing BP reduces CV events. Data from epidemiologic studies consistently show that in patients undergoing maintenance dialysis, low BP values are associated with higher death rates when compared with normal to moderately high values. In the only randomized clinical trial to test BP targets in hemodialysis patients, assignment to predialysis systolic BP between 115 and 140 mm Hg (vs. 155–165 mm Hg) did not result in any improvement in LVH or cardiovascular events and caused more adverse events (hospitalizations, vascular access thromboses and intradialytic hypotension).[340]

While there is no generalizable approach to manage BP in patients receiving dialysis, the following points are vital to have an accurate assessment of BP: 1. The most representative BP is the one taken the morning after dialysis (or, if available, interdialytic 48-hour BP monitoring; and 2. there should be a minimum of two and ideally three readings obtained 1 to 2 minutes apart during those morning readings and then averaged.[341] Given that heart failure and sudden death are the most common causes of death in patients receiving dialysis, β-blockers may have a beneficial role in the BP-lowering armamentarium unlike the general population. However, the most relevant factor is ensuring euvolemia. The clinical examination guides the general assessment of volume status, but it appears that bioimpedance, blood volume monitoring, and inferior vena cava ultrasound are effective in improving the definition of dry weight and could be considered, particularly in patients with uncontrolled BP despite conventional dry weight probing using the clinical examination.[218,342]

## MANAGEMENT OF PRIMARY HYPERTENSION IN CHRONIC KIDNEY DISEASE

There are many guidelines written on the management of hypertension in the general population, and those have been summarized at the beginning of this chapter. Both the Kidney Disease Outcomes Quality Initiative (KDOQI) and the KDIGO guidelines focus on the evidence for certain BP levels and classes of medication in patients with CKD. These guidelines have also been discussed in this chapter. To summarize the key findings, the strongest level of evidence supports a BP goal <140/90 mm Hg and the use of RAAS-blocking agents in people with stage 3 or higher CKD who have severely increased albuminuria. Nevertheless, there is weaker evidence to support RAAS blocker use in people with CKD in general, as well as weak evidence supporting a BP target below 130/80 mm Hg for those who have a urine albumin level of 1 g or more and an eGFR of <60 mL/min/1.73 m$^2$.[343,344] These BP levels and recommendations

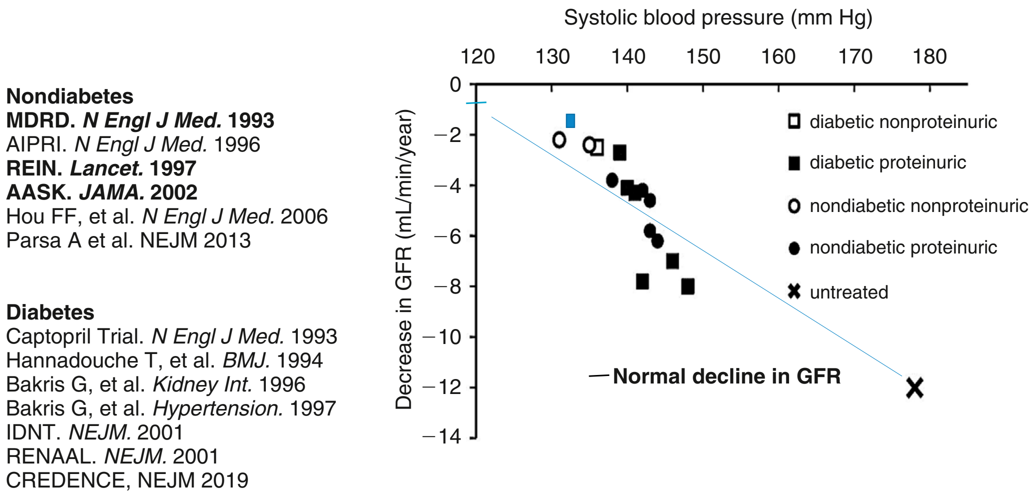


**Fig. 46.8 Relationship between achieved blood pressure and decline in kidney function from primary renal endpoint trials.** *GFR,* Glomerular filtration rate. (From Kalaitzidis R, Bakris GL. Update on reducing the development of diabetic kidney disease and cardiovascular death in diabetes. In Daugirdas JT, ed. *Handbook of Chronic Kidney Disease Management*. Philadelphia: Elsevier; 2018.)

for RAAS blockers are focused on the primary outcome of slowing CKD progression. A summary of all trials and changes in eGFR decline are in Fig. 46.8.

The most recent AHA/ACC BP guidelines have modified the approach slightly with a focus on CV risk reduction rather than a focus on renal preservation since the BP range of 125 to 130 mm Hg has not been shown to harm the kidneys yet reduce further CV events.[283] An algorithm proposed for BP management in CKD is shown in Fig. 46.9.

Nuances in the management of BP in patients with CKD are necessary as these patients have problems not seen in the general population. First, risk for hyperkalemia is a problem, especially in certain subgroups of patients. A review of clinical trials where hyperkalemia developed when managing hypertension in CKD found three risk predictors. If a patient was already receiving an appropriate diuretic for level of kidney function, then these were reliable risk predictors for hyperkalemia (i.e., $[K^+]_s$ >5.5 mEq/L): 1. eGFR of <45 mL/min/1.73 m$^2$, 2. serum potassium level above 4.5 mEq/L, and 3. body mass index of <25.[345] Hyperkalemia often limits our ability to use RAAS blockers in patients with advanced CKD. Newer agents to manage hyperkalemia have opened the door to safer management and expanded research.[346,347] Moreover, with two different agents available (patiromer and sodium zirconium cyclosilicate), both well tolerated, we now have the ability to continue ACEIs or ARBs or even mineralocorticoid receptor antagonists. The agents act as "therapeutic enablers." For instance, in the setting of resistant hypertension in patients with CKD, concomitant treatment with patiromer allows continued use of spironolactone in >80% of patients with eGFR as low as 25 mL/min/1.73m$^2$.[348-350]

Another nuance in BP management in the patient with CKD is the critical importance of sodium restriction to <2400 mg/day and reduced alcohol consumption, as well as regular physical activity.[351,352]

To demonstrate the importance of exercise, a study of 296 dialysis patients was randomized to normal physical activity (control; $n$ = 145) or walking exercise ($n$ = 151) over a 6-month period. Those who were in the active group using a simple, personalized, home-based, low-intensity exercise program managed by dialysis staff improved physical performance and quality of life.[352]

Studies evaluating the effect of sodium intake on BP control in people with stage 4 CKD show that approximately every 400 mg above a sodium intake base of 3000 mg/day requires an additional BP medication to maintain BP control.[351] Moreover, failure to reduce sodium intake suppresses the RAAS system and hence reduces efficacy of RAAS blockers. Thus failure to reduce sodium intake is a cause of resistant hypertension as noted earlier in the chapter. Additionally, BP targets need to be clearly defined and finally, the appropriate antihypertensive agents indicated as initial therapy by evidence-based guidelines should be used.

The role of initial combination therapy in patients with CKD and championed for almost 2 decades since the guidelines recommend that all people with a BP that is 20/10 mm Hg above the goal be initiated on single-pill combination therapy to enhance adherence and efficacy.[276,353] Studies evaluating initial monotherapy versus single-pill combination among those with an eGFR >60 mL/min/1.73 m$^2$ uniformly show an advantage of achieving BP goal more quickly and with better tolerability.[354,355]

These new guidelines and concepts have been integrated into an algorithm that was in the original National Kidney Foundation consensus report.[356] It is meant to provide an approach to pharmacologically meaningful agents that, when combined, actually further reduce BP. Table 13 provides an approved list of combination agents noting which are meaningful for optimal BP reduction and which are not.

## ANTIHYPERTENSIVE MEDICATIONS

All information about antihypertensive medications used for general management of hypertension may be found in a compendium of these medications published by the American Society of Hypertension.[357] This section focuses on antihypertensive medications in patients with CKD, dosage adjustments, and other related issues.

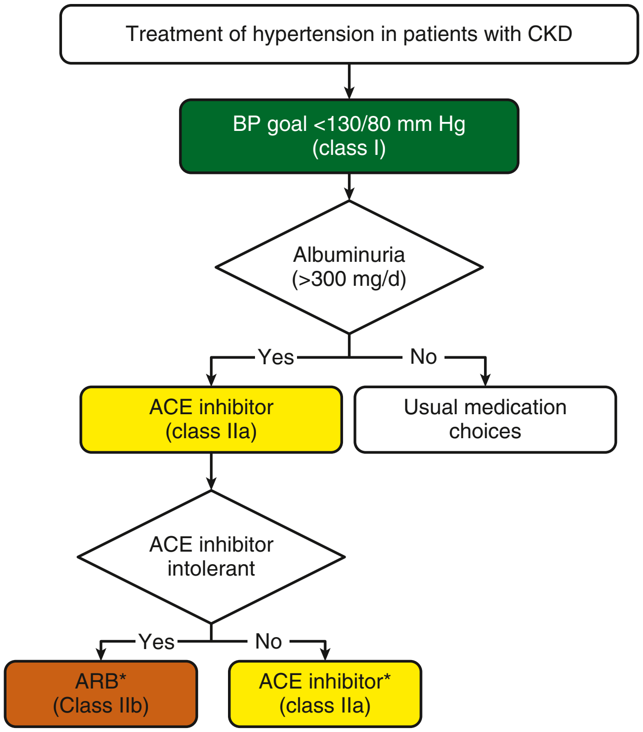


**Fig. 46.9 Management of hypertension in patients with chronic kidney disease *(CKD)* according to American Heart Association (ACC)/American Heart Association (AHA) guidelines.** Colors correspond to class of recommendation: *green,* class I (strong); *yellow,* class IIa (moderate); *orange,* class IIIb (weak). *CKD stage 3 or higher or stage 1 or 2 with albuminuria ≥300 mg/day or ≥300 mg/g creatinine. It should be noted that treating blood pressure *(BP)* in CKD patients to <140/90 mm Hg with an angiotensin receptor blocker or ACE inhibitor as first-line therapy is recommended for slowing progression of CKD by KDIGO guidelines. To further reduce CV risk, levels of <130/80 mm Hg need to be achieved. The use of combination therapy with diltiazem in the subgroup with high albuminuria has an additive reduction, in addition, to lower BP further compared with other calcium channel blockers. *ACE,* Angiotensin-converting enzyme; *ARB,* angiotensin receptor blocker. (From Whelton PK, Carey RM, Aronow WS, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA Guideline for the Prevention, Detection, Evaluation and Management of High Blood Pressure in Adults: A Report of the American College of Cardiology/American Heart Association and Task Force on Clinical Practice Guidelines. *Hypertension.* 2018;71(6):e13–e115.)

## ANGIOTENSIN-CONVERTING ENZYME INHIBITORS AND ANGIOTENSIN RECEPTOR BLOCKERS

The most compelling indication to use ACE inhibitors comes from the AASK trial, where ramipril was superior to either amlodipine or metoprolol for slowing CKD progression.[24] However, no other ACEIs have been tested in hypertensive nephropathy. Moreover, while an earlier systematic review and meta-analysis by the Cochrane group found that there are insufficient data to claim that ACEIs prevent or delay nephropathy in early stage (G1 or G2) CKD, a meta-analysis does support a benefit when used in stage G3 disease.[358,359] Keep in mind that all ACE inhibitors are dialyzable except for trandolapril.

## ANGIOTENSIN RECEPTOR BLOCKERS

Most of the data on ARBs in CKD are in diabetic nephropathy. Given the mechanism of action and the benefit seen in diabetic nephropathy, discussed elsewhere in this book, there is no reason to think these agents would not be as effective in hypertension management in nondiabetic kidney disease including hypertensive nephropathy. There are no adequately powered trials that directly compare progression of CKD between ARBs and ACEIs. In general, one major difference between ACEIs and ARBs is that ARBs are better tolerated than ACEIs due to lower incidence of cough, angioedema, taste disturbances, and hyperkalemia.[360,361]

It is important to understand that in patients with advanced nephropathy, increases in serum creatinine level of up to 30% are commonly seen within 1 to 2 weeks of starting ACEIs or ARBs.[362-364] This increase in serum creatinine level is especially true if BP was previously not controlled and now is within goal. However, in patients younger than age 66 years with baseline serum creatinine values of more than 3.5 mg/dL, a rise in serum creatinine level of up to 30% within the first 4 months of starting ACEIs or ARBs correlated with slower long-term decline in kidney function over a mean follow-up period of 3 or more years.[362,363] For acute, sustained increases in serum creatinine levels of more than 35%, patients should be evaluated for 1. volume depletion *(the most common cause),* 2. decompensated congestive heart failure, or 3. bilateral renal artery stenosis. Hyperkalemia should be managed with avoidance of high-potassium foods, appropriately dosing diuretics, and stopping agents known to increase potassium levels, such as NSAIDs.[345,356]

## DIRECT RENIN INHIBITORS

Aliskiren is currently the first and only approved oral direct renin inhibitor for reduction of BP. There are no direct outcome data available describing the use of aliskiren in patients with hypertensive nephropathy. However, there are data from 24-hour ABPM studies evaluating the effect of this agent alone and in combination on BP control. Aliskiren was tested alone and in combination with valsartan in people with stage 2 CKD from hypertension to evaluate BP control. The 24-hour ABPM showed additive BP reduction with the combination and improvement in nocturnal dipping in people with stage 2 nephropathy. Moreover, there were no problems with elevated potassium levels or reduced kidney function that emerged in the combination group.[365] Lastly, aliskerin was assessed in a cardiovascular/renal outcome trial, The Aliskiren Trial in Type 2 Diabetes Using Cardiorenal Endpoints.[307] This was a double-blind trial that randomly assigned 8561 patients to aliskiren (300 mg daily) or placebo as an adjunct to an angiotensin-converting enzyme inhibitor or an angiotensin receptor blocker. The primary endpoint was a composite of the time to cardiovascular death or a first occurrence of cardiac arrest with resuscitation; nonfatal myocardial infarction; nonfatal stroke; unplanned hospitalization for heart failure; end-stage renal disease, death attributable to kidney failure, or the need for renal replacement therapy with no dialysis or transplantation available or initiated; or doubling of the baseline serum creatinine level. The trial was stopped prematurely after the second interim efficacy analysis with a median follow-up of 32.9 months for potential harm. The proportion of patients with hyperkalemia (serum potassium level, ≥6 mmol/L) was significantly higher in the aliskiren group than in the placebo group (11.2% vs. 7.2%), as was the proportion with reported hypotension (12.1% vs. 8.3%)

($P < 0.001$ for both comparisons). The authors concluded that aliskiren added to standard therapy with renin-angiotensin system blockade in patients with type 2 diabetes who are at high risk for cardiovascular and renal events is not supported by these data and may even be harmful.

## ALDOSTERONE ANTAGONISTS

Aldosterone antagonists such as spironolactone are recommended for treating hypertension in patients with advanced heart failure and post-myocardial infarction, but their role in kidney disease is unclear.[366,367] While there are good data supporting their use in people with stages G1 and G2 CKD with resistant hypertension, there are no outcome data in advanced CKD; hence this class is not discussed in this chapter.[368]

## DIURETICS

Thiazide-like agents (chlorthalidone and indapamide) have a strong evidence base for reducing CV risk even in patients with stage 3 CKD.[369] However, their role in kidney outcome studies is unclear. Post hoc analyses of ALLHAT demonstrated that chlorthalidone was as good as ACEIs for slowing CKD progression in stage 3 nephropathy.[318] This post hoc analysis of ALLHAT included both those with hypertensive nephropathy and diabetic nephropathy, so the data should be considered less than definitive.

Data from the updated international guidelines recommend thiazide agents in a supportive role following RAAS blockade and calcium antagonists in patients with hypertension and CKD.[11,38] There are major differences among chlorthalidone, indapamide, and hydrochlorothiazide. Chlorthalidone has a much longer half-life compared with hydrochlorothiazide at the same milligram dose.[370] This difference in duration of action by chlorthalidone translated into an additional 7–mm Hg reduction in systolic BP when substituted for hydrochlorothiazide.[369,371,372] However, in a large randomized pragmatic trial comparing the use of chlorthalidone versus hydrochlorothiazide in 13,523 patients previously treated with hydrochlorothiazide, there was no difference between the two groups for the primary composite endpoint of nonfatal myocardial infarction, stroke, heart failure resulting in hospitalization, urgent coronary revascularization for unstable angina, and non–cancer-related death.[373]

An important development in the field was the publication of the Chlorthalidone in Chronic Kidney Disease (CLICK) study.[374] Until recently, most guidelines have discouraged the use of thiazide diuretics in CKD because of presumed lesser efficacy for BP management. This was not based on strong evidence. In CLICK, however, treatment of patients with advanced CKD (mean eGFR 32 mL/min/$m^2$) with chlorthalidone for 12 weeks resulted in an 11 mm Hg decrease in 24-hour systolic BP from baseline to 12 weeks compared with a 0.5 mm Hg decrease in the placebo group when added to a mean of 3.4 other antihypertensive medications. This study positions thiazides (chlorthalidone in particular) as an important tool in the management of hypertension in advanced CKD.

## CALCIUM CHANNEL BLOCKERS

When used in patients with hypertensive nephropathy, both dihydropyridine and nondihydropyridine CCBs are effective in lowering BP and CV events in high-risk populations.[375] These agents are particularly efficacious for CV risk reduction when combined with an ACEI as shown by the results of ACCOMPLISH.[274] In this trial, patients at high risk for a CV event and CKD progression treated with a single-pill combination of an ACEI with amlodipine had a 20% relative risk reduction in CV events and slower CKD progression when compared with an ACEI plus hydrochlorothiazide.

Also, a prespecified post hoc analysis of ACCOMPLISH evaluated CKD outcomes showing that the DCCB plus ACEI inhibitor decreased CKD progression to a greater extent than the ACEI and diuretic (HR, 0.52; $P < 0.001$).[275] Note that approximately 60% of the patients in this substudy were thought to have hypertensive nephropathy.

As an antihypertensive class, all CCBs lower severely increased albuminuria but the dihydropyridine CCBs do not reduce albuminuria to the same degree as the non-dihydropyridine CCBs.[375,376] The mechanism of this difference is due to changes in glomerular permeability that occur in patients with advanced nephropathy[376,377] and has translated into worse CKD outcomes when compared with those treated with RAAS blockers.[378] Note that when used with a RAAS blocker, dihydropyridine CCBs are much less likely to cause edema and worsen severely increased albuminuria.[355,379] Lastly, the combination of a nondihydropyridine CCB with a dihydropyridine CCB is known to have additive BP-lowering effects in part because of differences in inhibition on the L-channel.[380]

In summary, both dihydropyridine and nondihydropyridine CCBs are effective BP-reducing agents in patients without albuminuric kidney disease. However, in those with advanced albuminuric nephropathy, nondihydropyridine CCBs are preferred but DCCBs should be used in combination with a RAAS blocker to maximally slow progression of nephropathy.

## β-ADRENERGIC BLOCKERS

β-Blockers are no longer considered a first-line option to lower BP according to current clinical practice guidelines. Some patients with hypertensive nephropathy have an increase in sympathetic activity and high CV event rates including patients receiving dialysis and those with heart failure and postmyocardial infarction and should benefit from β-blockers, however.[381-383] Despite being reasonably effective at lowering BP, physicians are often reluctant to use β-blockers because of significant adverse metabolic effects and bradycardia.[384]

The vasodilating and metabolically neutral β-blockers have expanded the role of these agents, especially in patients with diabetes and hypertensive nephropathy. The combined α- and β-blocker, carvedilol, and the $\beta_1$-vasodilating agent nebivolol have neutral glycemic and lipid parameters.[301,385] There are major pharmacologic differences between labetalol (also a combined α- and β-blocker) and carvedilol; specifically, there are no cardiovascular outcomes with labetalol and pharmacologically the ratio of β- to α-blockade is 4:1 for labetalol (400 mg) and 5:1 for carvedilol (25 mg); however, there is much more α at 50 mg of carvedilol.[386] Moreover, labetolol has a short half-life and ideally should be given three to four times daily. Carvedilol has much more outcome data in heart disease and is better tolerated with less liver dysfunction than labetalol.[387] Thus in patients with hypertensive nephropathy, vasodilating β-blockers can be used and are excellent add-on agents to reduce CV risk and achieve BP goals.

Other classes of BP-lowering medications such as α-adrenergic antagonists, vasodilators such as hydralazine and minoxidil, as well as central α-agonists are not discussed, as there are no CV or CKD outcome data on their usefulness in HN.

### DIRECT-ACTING VASODILATORS

Hydralazine and minoxidil are direct-acting vasodilators that have different mechanisms of action but the same clinical consequences. They both increase sodium reabsorption dramatically by the proximal tubule, increase sympathetic tone, and as a result, should never be used long term without concomitant use of loop diuretics and β-blockers.[388-390] Moreover, hydralazine is well known to increase genesis of peroxynitrite, which damages cells and reduces NO.[391] Thus nitrates should be given with hydralazine to reduce this risk.[392]

## RESISTANT HYPERTENSION

Resistant hypertension is currently defined as the failure to achieve a goal BP of <130/80 mm Hg in patients who are adherent with maximally tolerated doses of a minimum of three antihypertensive drugs, one of which must be a diuretic (typically a thiazide) appropriate for the level of kidney function.[152] The increasing prevalence of obesity and hypertension in the general population has resulted in this disorder's gaining attention in the past decade. Large-scale population-based studies such as the NHANES have specifically examined the prevalence and incidence of resistant hypertension and associated risk factors. These findings suggest the prevalence of resistant hypertension is approximately 8% to 12% of adult patients with hypertension (6–9 million people).[393] The increasing prevalence of resistant hypertension contrasts with the improvement in BP control rates during the same period. Studies also show that patients with resistant hypertension who are older than 55 years, of Black race, with high body mass index, diabetes, or CKD have an increased risk for CV events compared with patients with nonresistant hypertension. The effects of white coat hypertension and pseudoresistant hypertension have not been factored into many of the prevalence studies, and hence the true prevalence is not known. The white coat effect contributes greatly to the high perceived incidence of resistant hypertension, as was evidenced by the >60% screen failure due to white coat hypertension in the Renal Denervation in Patients with Uncontrolled Hypertension (SYMPLICITY HTN-3) trial of renal denervation.[111,394]

Thus resistant hypertension is a diagnosis frequently seen by nephrologists and is most commonly due to nonadherence with medication, as well as volume overload secondary to impaired kidney function and nonadherence with a low-sodium diet. Once a diagnosis of resistant hypertension is documented using a 24-hour ABPM along with direct observation of therapy (i.e., observing the patient take all medications at the time of placement of the 24-hour BP monitor), then a fourth drug is indicated preferably a mineralocorticoid inhibitor, as spironolactone has consistently demonstrated significant benefit in controlling BP in patients with resistant hypertension.[395-398] A detailed discussion of this topic is beyond the scope of this chapter, but the reader is referred to the most recent consensus report by the American Heart Association/American College of Cardiology.[399] The reader should know that there are several different agents being developed for resistant hypertension, along with the approval of device-based renal denervation. Current evidence for these approaches is summarized in the referenced articles.[400,401]

*Visit Elsevier eBooks+ (eBooks.Health.Elsevier.com) for a complete set of references.*

## KEY REFERENCES


1. Collaboration NCDRF. Worldwide trends in hypertension prevalence and progress in treatment and control from 1990 to 2019: a pooled analysis of 1201 population-representative studies with 104 million participants. *Lancet*. 2021;398(10304):957–980. https://doi.org/10.1016/S0140-6736(21)01330-1.
4. Whelton PK, C R, A W, Casey Jr DE, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association and Task Force on Clinical Practice Guidelines. *Hypertension*. 2017;71(6).
5. Mancia G, Kreutz R, Brunstrom M, et al. 2023 ESH guidelines for the management of arterial hypertension the task force for the management of arterial hypertension of the European society of hypertension: endorsed by the international society of hypertension (ISH) and the European renal association (ERA). *J Hypertens*. 2023;41(12):1874–2071.
45. Guyton AC. Kidneys and fluids in pressure regulation. Small volume but large pressure changes. *Hypertension*. 1992;19(1 suppl):I2–I8. https://doi.org/10.1161/01.hyp.19.1_suppl.i2.
91. Crowley SD, Gurley SB, Oliverio MI, et al. Distinct roles for the kidney and systemic tissues in blood pressure regulation by the renin-angiotensin system. *J Clin Invest*. 2005;115(4):1092–1099.
113. Cluett JL, Blazek O, Brown AL, et al. Renal denervation for the treatment of hypertension: a scientific statement from the American heart association. *Hypertension*. 2024;81(10):e135–e148.
136. Schlaich MP, Bellet M, Weber MA, et al. Dual endothelin antagonist aprocitentan for resistant hypertension (PRECISION): a multicentre, blinded, randomised, parallel-group, phase 3 trial. *Lancet*. 2022;400(10367):1927–1937.
151. Kaess BM, Rong J, Larson MG, et al. Aortic stiffness, blood pressure progression, and incident hypertension. *JAMA*. 2012;308(9):875–881.
172. Nakai M, Ribeiro RV, Stevens BR, et al. Essential hypertension is associated with changes in gut microbial metabolic pathways: a multisite analysis of ambulatory blood pressure. *Hypertension*. 2021;78(3):804–815.
179. Stergiou GS, Mukkamala R, Avolio A, et al. Cuffless blood pressure measuring devices: review and statement by the European society of hypertension working group on blood pressure monitoring and cardiovascular variability. *J Hypertens*. 2022;40(8):1449–1460.
180. Cheung AK, Whelton PK, Muntner P, et al. International consensus on standardized clinic blood pressure measurement - a call to action. *Am J Med*. 2023;136(5):438–445 e1.
185. Norcliffe-Kaufmann L, Kaufmann H, Palma JA, et al. Orthostatic heart rate changes in patients with autonomic failure caused by neurodegenerative synucleinopathies. *Ann Neurol*. 2018;83(3):522–531.
194. Cohen JB, Lotito MJ, Trivedi UK, Denker MG, Cohen DL, Townsend RR. Cardiovascular events and mortality in white coat hypertension: a systematic review and meta-analysis. *Ann Intern Med*. 2019;170(12):853–862.
196. Asayama K, Stolarz-Skrzypek K, Yang WY, et al. What did we learn from the international databases on ambulatory and home blood pressure in relation to cardiovascular outcome? *Hypertens Res*. 2023;46(4):934–949.

200. Zhang DY, Guo QH, An DW, Li Y, Wang JG. A comparative meta-analysis of prospective observational studies on masked hypertension and masked uncontrolled hypertension defined by ambulatory and home blood pressure. *J Hypertens*. 2019;37(9):1775–1785.
210. Stergiou GS, Palatini P, Parati G, et al. 2021 European Society of Hypertension practice guidelines for office and out-of-office blood pressure measurement. *J Hypertens*. 2021;39(7):1293–1302.
220. Nugent JT, Jiang K, Funaro MC, et al. Does this child with high blood pressure have secondary hypertension?: the rational clinical examination systematic review. *JAMA*. 2023;329(12):1012–1021.
224. Brown JM, Siddiqui M, Calhoun DA, et al. The unrecognized prevalence of primary aldosteronism: a cross-sectional study. *Ann Intern Med*. 2020;173(1):10–20.
230. Jaffe G, Gray Z, Krishnan G, et al. Screening rates for primary aldosteronism in resistant hypertension: a cohort study. *Hypertension*. 2020;75(3):650–659.
231. Cohen JB, Cohen DL, Herman DS, Leppert JT, Byrd JB, Bhalla V. Testing for primary aldosteronism and mineralocorticoid receptor antagonist use among U.S. Veterans: a retrospective cohort study. *Ann Intern Med*. 2021;174(3):289–297.
236. Jing Y, Liao K, Li R, et al. Cardiovascular events and all-cause mortality in surgically or medically treated primary aldosteronism: a Meta-analysis. *J Renin-Angiotensin-Aldosterone Syst*. 2021;22(1):14703203211003781.
237. Laffin LJ, Rodman D, Luther JM, et al. Aldosterone synthase inhibition with lorundrostat for uncontrolled hypertension: the target-HTN randomized clinical trial. *JAMA*. 2023;330(12):1140–1150.
241. Turin CG, Crenshaw MM, Fishbein L. Pheochromocytoma and paraganglioma: germline genetics and hereditary syndromes. *Endocr Oncol*. 2022;2(1):R65–R77. https://doi.org/10.1530/EO-22-0044.
264. Peixoto AJ. Acute severe hypertension. *N Engl J Med*. 2019;381(19):1843–1852. https://doi.org/10.1056/NEJMcp1901117.
265. Vlcek M, Bur A, Woisetschlager C, Herkner H, Laggner AN, Hirschl MM. Association between hypertensive urgencies and subsequent cardiovascular events in patients with hypertension. *J Hypertens*. 2008;26(4):657–662.
266. Garovic VD, Dechend R, Easterling T, et al. Hypertension in pregnancy: diagnosis, blood pressure goals, and pharmacotherapy: a scientific statement from the American heart association. *Hypertension*. 2022;79(2):e21–e41.
274. Jamerson K, Weber MA, Bakris GL, et al. Benazepril plus amlodipine or hydrochlorothiazide for hypertension in high-risk patients. *N Engl J Med*. 2008;359(23):2417–2428.
275. Bakris GL, Sarafidis PA, Weir MR, et al. Renal outcomes with different fixed-dose combination therapies in patients with hypertension at high risk for cardiovascular events (ACCOMPLISH): a prespecified secondary analysis of a randomised controlled trial. *Lancet*. 2010;375(9721):1173–1181.
278. Jung O, Gechter JL, Wunder C, et al. Resistant hypertension? Assessment of adherence by toxicological urine analysis. *J Hypertens*. 2013;31(4):766–774.
283. Cheung AK, Rahman M, Reboussin DM, et al. Effects of intensive BP control in CKD. *J Am Soc Nephrol*. 2017;28(9):2812–2823.
285. Kidney disease: improving global outcomes blood pressure work G. KDIGO 2021 clinical practice guideline for the management of blood pressure in chronic kidney disease. *Kidney Int*. 2021;99(3S):S1–S87.
286. Pajewski NM, Beddhu S, Bress AP, et al. The legacy effect of intensive versus standard BP control on the incidence of needing dialysis or kidney transplantation. *J Am Soc Nephrol*. 2024;35(12):1737–1745.
293. Mann JF, Anderson C, Gao P, et al. Dual inhibition of the renin-angiotensin system in high-risk diabetes and risk for stroke and other outcomes: results of the ONTARGET trial. *J Hypertens*. 2013;31(2):414–421.
294. Group AS, Cushman WC, Evans GW, et al. Effects of intensive blood-pressure control in type 2 diabetes mellitus. *N Engl J Med*. 2010;362(17):1575–1585.
296. Zhang W, Zhang S, Deng Y, et al. Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med*. 2021;385(14):1268–1279.
298. American Diabetes Association Professional Practice C. 10. Cardiovascular disease and risk management: standards of care in diabetes-2024. *Diabetes Care*. 2024;47(suppl 1):S179–S218.
303. Lewis EJ, Hunsicker LG, Bain RP, Rohde RD. The effect of angiotensin-converting-enzyme inhibition on diabetic nephropathy. The Collaborative Study Group. *N Engl J Med*. 1993;329(20):1456–1462.
304. Brenner BM, Cooper ME, de Zeeuw D, et al. Effects of losartan on renal and cardiovascular outcomes in patients with type 2 diabetes and nephropathy. *N Engl J Med*. 2001;345(12):861–869. https://doi.org/10.1056/NEJMoa011161.
305. Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. *N Engl J Med*. 2001;345(12):851–860.
306. Fried LF, Emanuele N, Zhang JH, et al. Combined angiotensin inhibition for the treatment of diabetic nephropathy. *N Engl J Med*. 2013;369(20):1892–1903.
315. Beddhu S, Chertow GM, Cheung AK, et al. Influence of baseline diastolic blood pressure on effects of intensive compared with standard blood pressure control. *Circulation*. 2018;137(2):134–143.
318. Rahman M, Pressel S, Davis BR, et al. Renal outcomes in high-risk hypertensive patients treated with an angiotensin-converting enzyme inhibitor or a calcium channel blocker vs a diuretic: a report from the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *Arch Intern Med*. 2005;165(8):936–946.
320. Neal B, Wu Y, Feng X, et al. Effect of salt substitution on cardiovascular events and death. *N Engl J Med*. 2021;385(12):1067–1077.
335. Sarafidis PA, Persu A, Agarwal R, et al. Hypertension in dialysis patients: a consensus document by the European renal and cardiovascular medicine (EURECA-m) working group of the European renal association-European dialysis and transplant association (ERA-EDTA) and the hypertension and the kidney working group of the European society of hypertension (ESH). *Nephrol Dial Transplant*. 2017;32(4):620–640. official publication of the European Dialysis and Transplant Association - European Renal Association.
339. Agarwal R, Sinha AD. Cardiovascular protection with antihypertensive drugs in dialysis patients: systematic review and meta-analysis. *Hypertension*. 2009;53(5):860–866.
340. Miskulin DC, Gassman J, Schrader R, et al. BP in dialysis: results of a pilot study. *J Am Soc Nephrol*. 2018;29(1):307–316.
370. Agarwal R, Sinha AD, Tu W. Chlorthalidone for hypertension in advanced CKD. Reply. *N Engl J Med*. 2022;386(14):1384.
371. Ishani A, Cushman WC, Leatherman SM, et al. Chlorthalidone vs. Hydrochlorothiazide for hypertension-cardiovascular events. *N Engl J Med*. 2022;387(26):2401–2410.
378. Agodoa LY, Appel L, Bakris GL, et al. Effect of ramipril vs amlodipine on renal outcomes in hypertensive nephrosclerosis: a randomized controlled trial. *JAMA*. 2001;285(21):2719–2728.
396. Williams B, MacDonald TM, Morant S, et al. Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2): a randomised, double-blind, crossover trial. *Lancet*. 2015;386(10008):2059–2068. https://doi.org/10.1016/S0140-6736(15)00257-3.
399. Carey RM, Calhoun DA, Bakris GL, et al. Resistant hypertension: detection, evaluation, and management: a scientific statement from the American Heart Association. *Hypertension*. 2018;72(5):e53–e90.

## REFERENCES


1. Collaboration NCDRF. Worldwide trends in hypertension prevalence and progress in treatment and control from 1990 to 2019: a pooled analysis of 1201 population-representative studies with 104 million participants. *Lancet*. 2021;398(10304):957–980.
2. Diseases GBD, Injuries C. Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet*. 2020;396(10258):1204–1222.
3. Lewington S, Clarke R, Qizilbash N, Peto R, Collins R, Prospective Studies C. Age-specific relevance of usual blood pressure to vascular mortality: a meta-analysis of individual data for one million adults in 61 prospective studies. *Lancet*. 2002;360(9349):1903–1913.
4. Whelton PK, C R, A W, Casey Jr DE, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association and Task Force on Clinical Practice Guidelines. *Hypertension*. 2017;71(6).
5. Mancia G, Kreutz R, Brunstrom M, et al. 2023 ESH guidelines for the management of arterial hypertension the task force for the management of arterial hypertension of the European society of hypertension: endorsed by the international society of hypertension (ISH) and the European renal association (ERA). *J Hypertens*. 2023;41(12):1874–2071.
6. Muntner P, Miles MA, Jaeger BC, et al. Blood pressure control among US adults, 2009 to 2012 through 2017 to 2020. *Hypertension*. 2022;79(9):1971–1980.
7. Franklin SS, Jacobs MJ, Wong ND, L'Italien GJ, Lapuerta P. Predominance of isolated systolic hypertension among middle-aged and elderly US hypertensives: analysis based on National Health and Nutrition Examination Survey (NHANES) III. *Hypertension*. 2001;37(3):869–874.
8. Ehret GB, Munroe PB, International Consortium for Blood Pressure Genome-Wide Association S, et al. Genetic variants in novel pathways influence blood pressure and cardiovascular disease risk. *Nature*. 2011;478(7367):103–109.
9. McGrath BP, Kundu P, Daya N, et al. Isolated diastolic hypertension in the UK biobank: comparison of ACC/AHA and ESC/NICE guideline definitions. *Hypertension*. 2020;76(3):699–706.
10. McEvoy JW, Yang WY, Thijs L, et al. Isolated diastolic hypertension in the IDACO study: an age-stratified analysis using 24-hour ambulatory blood pressure measurements. *Hypertension*. 2021;78(5):1222–1231.
11. Global Burden of Disease Cancer C, Fitzmaurice C, Allen C, et al. Global, regional, and national cancer incidence, mortality, years of life lost, years lived with disability, and disability-adjusted life-years for 32 cancer groups, 1990 to 2015: a systematic analysis for the global burden of disease study. *JAMA Oncol*. 2017;3(4):524–548.
12. Elliott WJ. Management of hypertension in the very elderly patient. *Hypertension*. 2004;44(6):800–804.
13. Beckett NS, Peters R, Fletcher AE, et al. Treatment of hypertension in patients 80 years of age or older. *N Engl J Med*. 2008;358(18):1887–1898.
14. Williamson JD, Supiano MA, Applegate WB, et al. Intensive vs standard blood pressure control and cardiovascular disease outcomes in adults aged >/=75 Years: a randomized clinical trial. *JAMA*. 2016;315(24):2673–2682.
15. Ogden CL, Carroll MD, Lawman HG, et al. Trends in obesity prevalence among children and adolescents in the United States, 1988-1994 through 2013-2014. *JAMA*. 2016;315(21):2292–2299.
16. Flynn JT, Kaelber DC, Baker-Smith CM, et al. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. *Pediatrics*. 2017;140(3).
17. Whelton PKCR, Aronow WS, Casey Jr DE, et al. ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation and management of high blood pressure in adults: a report of the American College of cardiology/American heart association and task force on clinical practice guidelines. *Hypertension*. 2018;71(10):e136–e139.
18. Benjamin EJ, Blaha MJ, Chiuve SE, et al. Heart disease and stroke statistics-2017 update: a report from the American heart association. *Circulation (New York, N Y)*. 2017;135(10):e146–e603.
19. Yoon SS, Carroll MD, Fryar CD. Hypertension prevalence and control among adults: United States, 2011-2014. *NCHS Data Brief*. 2015;(220):1–8. https://www.ncbi.nlm.nih.gov/pubmed/26633197.
20. Turnbull F, Blood Pressure Lowering Treatment Trialists C. Effects of different blood-pressure-lowering regimens on major cardiovascular events: results of prospectively-designed overviews of randomised trials. *Lancet*. 2003;362(9395):1527–1535.
21. Officers A. Coordinators for the ACRGTA, lipid-lowering treatment to prevent heart attack T. Major outcomes in high-risk hypertensive patients randomized to angiotensin-converting enzyme inhibitor or calcium channel blocker vs diuretic: the antihypertensive and lipid-lowering treatment to prevent heart attack trial (ALLHAT). *JAMA*. 2002;288(23):2981–2997.
22. Kochanek K. Deaths: final data for 2014, natl vital statist. *Rep*. 2016;65(4).
23. System USRD. USRDS 2023 Annual Data Report: Incidence, Prevalence, Patient Characteristics, and Treatment Modalities. (http://www.usrds.org/2016/view/v2_01.aspx).
24. Wright Jr JT, Bakris G, Greene T, et al. Effect of blood pressure lowering and antihypertensive drug class on progression of hypertensive kidney disease: results from the AASK trial. *JAMA*. 2002;288(19):2421–2431.
25. Mancia Chairperson G, Kreutz Co-Chair R, Brunstrom M, et al. 2023 ESH guidelines for the management of arterial hypertension the task force for the management of arterial hypertension of the European society of hypertension endorsed by the European renal association (ERA) and the international society of hypertension (ISH). *J Hypertens*. 2023;41(12):1874–2071.
26. Jaffe MG, Lee GA, Young JD, Sidney S. Go AS. Improved blood pressure control associated with a large-scale hypertension program. *JAMA*. 2013;310(7):699–705.
27. Casey Jr DE, Thomas RJ, Bhalla V, et al. 2019 AHA/ACC clinical performance and quality measures for adults with high blood pressure: a report of the American College of cardiology/American heart association task force on performance measures. *Circ Cardiovasc Qual Outcomes*. 2019;12(11):e000057.
28. Weber MA, Julius S, Kjeldsen SE, et al. Blood pressure dependent and independent effects of antihypertensive treatment on clinical events in the VALUE Trial. *Lancet*. 2004;363(9426):2049–2051.
29. Staessen JA, Den Hond E, Celis H, et al. Antihypertensive treatment based on blood pressure measurement at home or in the physician's office: a randomized controlled trial. *JAMA*. 2004;291(8):955–964.
30. Kostis JB, Cabrera J, Cheng JQ, et al. Association between chlorthalidone treatment of systolic hypertension and long-term survival. *JAMA*. 2011;306(23):2588–2593.
31. Dahl LK. Possible role of chronic excess salt consumption in the pathogenesis of essential hypertension. *Am J Cardiol*. 1961;8:571–575.
32. Beaglehole R, Salmond CE, Hooper A, et al. Blood pressure and social interaction in Tokelauan migrants in New Zealand. *J Chron Dis*. 1977;30(12):803–812.
33. Prior IA, Evans JG, Harvey HP, Davidson F, Lindsey M. Sodium intake and blood pressure in two Polynesian populations. *N Engl J Med*. 1968;279(10):515–520.
34. DeClue JW, Guyton AC, Cowley Jr AW, Coleman TG, Norman Jr RA, McCaa RE. Subpressor angiotensin infusion, renal sodium handling, and salt-induced hypertension in the dog. *Circ Res*. 1978;43(4):503–512.
35. Young DB, Guyton AC. Steady state aldosterone dose-response relationships. *Circ Res*. 1977;40(2):138–142.
36. Guyton AC, Coleman TG, Young DB, Lohmeier TE, DeClue JW. Salt balance and long-term blood pressure control. *Annu Rev Med*. 1980;31:15–27.
37. Coleman TG, Guyton AC. Hypertension caused by salt loading in the dog. 3. Onset transients of cardiac output and other circulatory variables. *Circ Res*. 1969;25(2):153–160.
38. Douglas BH, Guyton AC, Langston JB, Bishop VS. Hypertension caused by salt loading. Ii. Fluid volume and tissue pressure changes. *Am J Physiol*. 1964;207:669–671.
39. Manning Jr RD, Coleman TG, Guyton AC, Norman Jr RA, McCaa RE. Essential role of mean circulatory filling pressure in salt-induced hypertension. *Am J Physiol*. 1979;236(1):R40–R47.
40. Johnson RJ, Herrera-Acosta J, Schreiner GF, Rodriguez-Iturbe B. Subtle acquired renal injury as a mechanism of salt-sensitive hypertension. *N Engl J Med*. 2002;346(12):913–923.

41. Rodriguez-Iturbe B, Romero F, Johnson RJ. Pathophysiological mechanisms of salt-dependent hypertension. *Am J Kidney Dis.* 2007;50(4):655–672.
42. Elijovich F, Kirabo A, Laffer CL. Salt sensitivity of blood pressure in Black people: the need to sort out ancestry versus epigenetic versus social determinants of its causation: salt series. *Hypertension.* 2024;81(3):456–467.
43. Pratt JH, Jones JJ, Miller JZ, Wagner MA, Fineberg NS. Racial differences in aldosterone excretion and plasma aldosterone concentrations in children. *N Engl J Med.* 1989;321(17):1152–1157.
44. Flynn JT, Kaelber DC, Baker-Smith CM, et al. Subcommittee on screening and management of high blood pressure in children. Clinical practice guideline for screening and management of high blood pressure in children and adolescents. *Pediatrics.* 2017;140(3):e20171904. Pediatrics 2018;142(3).
45. Guyton AC. Kidneys and fluids in pressure regulation. Small volume but large pressure changes. *Hypertension.* 1992;19(1 suppl):I2–I8.
46. Guyton AC. Blood pressure control–special role of the kidneys and body fluids. *Sci Technol Humanit.* 1991;252(5014):1813–1816.
47. Hall MHJ. Pathogenesis of hypertension. In: Bakris GS, M J, eds. *Hypertension : A Companion to Braunwald's Heart Disease.* 3rd ed. Philadelphia, PA: Elsevier; 2018:33–51.
48. Ivy JR, Bailey MA. Pressure natriuresis and the renal control of arterial blood pressure. *J Physiol.* 2014;592(18):3955–3967.
49. O'Connor PM, Cowley Jr AW. Modulation of pressure-natriuresis by renal medullary reactive oxygen species and nitric oxide. *Curr Hypertens Rep.* 2010;12(2):86–92.
50. Evans RG, Majid DS, Eppel GA. Mechanisms mediating pressure natriuresis: what we know and what we need to find out. *Clin Exp Pharmacol Physiol.* 2005;32(5–6):400–409.
51. Ehret GB, Caulfield MJ. Genes for blood pressure: an opportunity to understand hypertension. *Eur Heart J.* 2013;34(13):951–961.
52. Evangelou E, Warren HR, Mosen-Ansorena D, et al. Genetic analysis of over 1 million people identifies 535 new loci associated with blood pressure traits. *Nat Genet.* 2018;50(10):1412–1425.
53. Warren HR, Evangelou E, Cabrera CP, et al. Genome-wide association analysis identifies novel blood pressure loci and offers biological insights into cardiovascular risk. *Nat Genet.* 2017;49(3):403–415.
54. Lander E, Kruglyak L. Genetic dissection of complex traits: guidelines for interpreting and reporting linkage results. *Nat Genet.* 1995;11(3):241–247.
55. Orias M, Tabares AH, Peixoto AJ. Hypothesis: it is time to reconsider phenotypes in hypertension. *J Clin Hypertens.* 2010;12(5):350–356.
56. McEniery CM, Yasmin WS, et al. Increased stroke volume and aortic stiffness contribute to isolated systolic hypertension in young adults. *Hypertension.* 2005;46(1):221–226.
57. Romero CA, Alfie J, Galarza C, et al. Hemodynamic circulatory patterns in young patients with predominantly diastolic hypertension. *J Am Soc Hypertens.* 2013;7(2):157–162.
58. Kotchen TA, Cowley Jr AW, Liang M. Ushering hypertension into a new era of precision medicine. *JAMA.* 2016;315(4):343–344.
59. Mattson DL, Liang M. Hypertension: from GWAS to functional genomics-based precision medicine. *Nat Rev Nephrol.* 2017;13(4):195–196.
60. Kelly TN, Sun X, He KY, et al. Insights from a large-scale whole-genome sequencing study of systolic blood pressure, diastolic blood pressure, and hypertension. *Hypertension.* 2022;79(8):1656–1667.
61. Yu B, Pulit SL, Hwang SJ, et al. Rare exome sequence variants in CLCN6 reduce blood pressure levels and hypertension risk. *Circ Cardiovasc Genet.* 2016;9(1):64–70.
62. Spence JD, Rayner BL. Hypertension in blacks: individualized therapy based on renin/aldosterone phenotyping. *Hypertension.* 2018;72(2):263–269.
63. Jain G, Ong S, Warnock DG. Genetic disorders of potassium homeostasis. *Semin Nephrol.* 2013;33(3):300–309.
64. Boyden LM, Choi M, Choate KA, et al. Mutations in kelch-like 3 and cullin 3 cause hypertension and electrolyte abnormalities. *Nature.* 2012;482(7383):98–102.
65. Geller DS, Farhi A, Pinkerton N, et al. Activating mineralocorticoid receptor mutation in hypertension exacerbated by pregnancy. *Sci Technol Humanit.* 2000;289(5476):119–123.
66. Lifton RP, Dluhy RG, Powers M, et al. A chimaeric 11 beta-hydroxylase/aldosterone synthase gene causes glucocorticoid-remediable aldosteronism and human hypertension. *Nature.* 1992;355(6357):262–265.
67. Lifton RP, Gharavi AG, Geller DS. Molecular mechanisms of human hypertension. *Cell.* 2001;104(4):545–556.
68. Maass PG, Aydin A, Luft FC, et al. PDE3A mutations cause autosomal dominant hypertension with brachydactyly. *Nat Genet.* 2015;47(6):647–653.
69. Titze J, Bauer K, Schafflhuber M, et al. Internal sodium balance in DOCA-salt rats: a body composition study. *Am J Physiol Ren Physiol.* 2005;289(4):F793–F802.
70. Titze J, Shakibaei M, Schafflhuber M, et al. Glycosaminoglycan polymerization may enable osmotically inactive Na+ storage in the skin. *Am J Physiol Heart Circ Physiol.* 2004;287(1):H203–H208.
71. Titze J, Luft FC. Speculations on salt and the genesis of arterial hypertension. *Kidney Int.* 2017;91(6):1324–1335.
72. Machnik A, Dahlmann A, Kopp C, et al. Mononuclear phagocyte system depletion blocks interstitial tonicity-responsive enhancer binding protein/vascular endothelial growth factor C expression and induces salt-sensitive hypertension in rats. *Hypertension.* 2010;55(3):755–761.
73. Machnik A, Neuhofer W, Jantsch J, et al. Macrophages regulate salt-dependent volume and blood pressure by a vascular endothelial growth factor-C-dependent buffering mechanism. *Nat Med.* 2009;15(5):545–552.
74. Marko L, Dorr A, Linz P, et al. Effect of sunitinib treatment on skin sodium accumulation in patients with renal cancer: a pilot study. *Hypertension.* 2022;79(5):e103–e105.
75. Kopp C, Linz P, Wachsmuth L, et al. (23)Na magnetic resonance imaging of tissue sodium. *Hypertension.* 2012;59(1):167–172.
76. Schneider MP, Raff U, Kopp C, et al. Skin sodium concentration correlates with left ventricular hypertrophy in CKD. *J Am Soc Nephrol.* 2017;28(6):1867–1876.
77. Rakova N, Juttner K, Dahlmann A, et al. Long-term space flight simulation reveals infradian rhythmicity in human Na(+) balance. *Cell Metab.* 2013;17(1):125–131.
78. Sun J, Zhao M, Miao S, Xi B. Polymorphisms of three genes (ACE, AGT and CYP11B2) in the renin-angiotensin-aldosterone system are not associated with blood pressure salt sensitivity: a systematic meta-analysis. *Blood Press.* 2016;25(2):117–122.
79. Dickson ME, Sigmund CD. Genetic basis of hypertension: revisiting angiotensinogen. *Hypertension.* 2006;48(1):14–20.
80. Kahlon T, Carlisle S, Otero Mostacero D, Williams N, Trainor P, DeFilippis AP. Angiotensinogen: more than its downstream products: evidence from population studies and novel therapeutics. *JACC Heart Fail.* 2022;10(10):699–713.
81. Desai AS, Webb DJ, Taubel J, et al. Zilebesiran, an RNA interference therapeutic agent for hypertension. *N Engl J Med.* 2023;389(3):228–238.
82. Danser AH. The role of the (Pro)renin receptor in hypertensive disease. *Am J Hypertens.* 2015;28(10):1187–1196.
83. Rosendahl A, Niemann G, Lange S, et al. Increased expression of (pro)renin receptor does not cause hypertension or cardiac and renal fibrosis in mice. *Lab invest.* 2014;94(8):863–872.
84. Prieto MC, Reverte V, Mamenko M, et al. Collecting duct prorenin receptor knockout reduces renal function, increases sodium excretion, and mitigates renal responses in ANG II-induced hypertensive mice. *Am J Physiol Ren Physiol.* 2017;313(6):F1243–F1253.
85. Ramkumar N, Kohan DE. The nephron (pro)renin receptor: function and significance. *Am J Physiol Ren Physiol.* 2016;311(6):F1145–F1148.
86. Gonzalez-Villalobos RA, Janjoulia T, Fletcher NK, et al. The absence of intrarenal ACE protects against hypertension. *The Journal of clinical investigation.* 2013;123(5):2011–2023.
87. Textor SC, Brunner HR, Gavras H. Converting enzyme inhibition during chronic angiotensin II infusion in rats. Evidence against a nonangiotensin mechanism. *Hypertension.* 1981;3(2):269–276.
88. Yang T, Xu C. Physiology and pathophysiology of the intrarenal renin-angiotensin system: an update. *J Am Soc Nephrol.* 2017;28(4):1040–1049.
89. Reudelhuber TL. Where hypertension happens. *J clin invest.* 2013;123(5):1934–1936.
90. Crowley SD, Gurley SB, Herrera MJ, et al. Angiotensin II causes hypertension and cardiac hypertrophy through its receptors in the kidney. *Proc Natl Acad Sci U S A.* 2006;103(47):17985–17990.

91. Crowley SD, Gurley SB, Oliverio MI, et al. Distinct roles for the kidney and systemic tissues in blood pressure regulation by the renin-angiotensin system. *J Clin Invest*. 2005;115(4):1092–1099.
92. McCurley A, Jaffe IZ. Mineralocorticoid receptors in vascular function and disease. *Mol Cell Endocrinol*. 2012;350(2):256–65.
93. Tikellis C, Bernardi S, Burns WC. Angiotensin-converting enzyme 2 is a key modulator of the renin-angiotensin system in cardiovascular and renal disease. *Curr Opin Nephrol Hypertens*. 2011;20(1):62–68.
94. Ferrario CM. ACE2: more of Ang-(1-7) or less Ang II? *Curr Opin Nephrol Hypertens*. 2011;20(1):1–6.
95. Fraga-Silva RA, Costa-Fraga FP, Murca TM, et al. Angiotensin-converting enzyme 2 activation improves endothelial function. *Hypertension*. 2013;61(6):1233–1238.
96. Hernandez Prada JA, Ferreira AJ, Katovich MJ, et al. Structure-based identification of small-molecule angiotensin-converting enzyme 2 activators as novel antihypertensive agents. *Hypertension*. 2008;51(5):1312–1317.
97. Wysocki J, Ye M, Rodriguez E, et al. Targeting the degradation of angiotensin II with recombinant angiotensin-converting enzyme 2: prevention of angiotensin II-dependent hypertension. *Hypertension*. 2010;55(1):90–98.
98. Haschke M, Schuster M, Poglitsch M, et al. Pharmacokinetics and pharmacodynamics of recombinant human angiotensin-converting enzyme 2 in healthy human subjects. *Clin Pharmacokinet*. 2013;52(9):783–792.
99. Self WH, Shotwell MS, Gibbs KW, et al. Renin-angiotensin system modulation with synthetic angiotensin (1-7) and angiotensin II type 1 receptor-biased ligand in adults with COVID-19: two randomized clinical trials. *JAMA*. 2023;329(14):1170–1182.
100. Li XC, Zhang J, Zhuo JL. The vasoprotective axes of the renin-angiotensin system: physiological relevance and therapeutic implications in cardiovascular, hypertensive and kidney diseases. *Pharmacol Res*. 2017;125(Pt A):21–38 (In eng).
101. Mancia G, Grassi G. The autonomic nervous system and hypertension. *Circ Res*. 2014;114(11):1804–1814.
102. DiBona GF. Sympathetic nervous system and hypertension. *Hypertension*. 2013;61(3):556–560.
103. Grassi G, Cattaneo BM, Seravalle G, Lanfranchi A, Mancia G. Baroreflex control of sympathetic nerve activity in essential and secondary hypertension. *Hypertension*. 1998;31(1):68–72.
104. Smith PA, Graham LN, Mackintosh AF, Stoker JB, Mary DA. Relationship between central sympathetic activity and stages of human hypertension. *Am J Hypertens*. 2004;17(3):217–222.
105. Johnson RJ, Gordon KL, Suga S, Duijvestijn AM, Griffin K, Bidani A. Renal injury and salt-sensitive hypertension after exposure to catecholamines. *Hypertension*. 1999;34(1):151–159.
106. Desir GV, Peixoto AJ. Renalase in hypertension and kidney disease. *Nephrol Dial Transplant*. 2014;29(1):22–28. official publication of the European Dialysis and Transplant Association - European Renal Association.
107. Ton QV, Hammes SR. Recent insights on circulating catecholamines in hypertension. *Curr Hypertens Rep*. 2014;16(12):498.
108. Zoccali C, Mallamaci F, Parlongo S, et al. Plasma norepinephrine predicts survival and incident cardiovascular events in patients with end-stage renal disease. *Circulation*. 2002;105(11):1354–1359.
109. Bangalore S, Sawhney S, Messerli FH. Relation of beta-blocker-induced heart rate lowering and cardioprotection in hypertension. *J Am Coll Cardiol*. 2008;52(18):1482–1489.
110. Julius S, Palatini P, Kjeldsen SE, et al. Usefulness of heart rate to predict cardiac events in treated patients with high-risk systemic hypertension. *Am J Cardiol*. 2012;109(5):685–692.
111. Bakris GL, Townsend RR, Liu M, et al. Impact of renal denervation on 24-hour ambulatory blood pressure: results from SYMPLICITY HTN-3. *J Am Coll Cardiol*. 2014;64(11):1071–1078.
112. Azizi M, Mahfoud F, Weber MA, et al. Effects of renal denervation vs sham in resistant hypertension after medication escalation: prespecified analysis at 6 Months of the RADIANCE-HTN TRIO randomized clinical trial. *JAMA Cardiol*. 2022;7(12):1244–1252.
113. Cluett JL, Blazek O, Brown AL, et al. Renal denervation for the treatment of hypertension: a scientific statement from the American heart association. *Hypertension*. 2024;81(10):e135–e148.
114. Hall JE, Mouton AJ, da Silva AA, et al. Obesity, kidney dysfunction, and inflammation: interactions in hypertension. *Cardiovasc Res*. 2021;117(8):1859–1876.
115. Dyer AR, Elliott P. The INTERSALT study: relations of body mass index to blood pressure. INTERSALT Co-operative Research Group. *J Hum Hypertens*. 1989;3(5):299–308. https://www.ncbi.nlm.nih.gov/pubmed/2810326.
116. Hall JE, do Carmo JM, da Silva AA, Wang Z, Hall ME. Obesity-induced hypertension: interaction of neurohumoral and renal mechanisms. *Circ Res*. 2015;116(6):991–1006.
117. Dorresteijn JA, Visseren FL, Spiering W. Mechanisms linking obesity to hypertension. *Obes Rev*. 2012;13(1):17–26.
118. Flynn C. Increased aldosterone: mechanism of hypertension in obesity. *Semin Nephrol*. 2014;34(3):340–348.
119. Schinner S, Willenberg HS, Krause D, et al. Adipocyte-derived products induce the transcription of the StAR promoter and stimulate aldosterone and cortisol secretion from adrenocortical cells through the Wnt-signaling pathway. *Int J Obes*. 2007;31(5):864–870.
120. Ehrhart-Bornstein M, Lamounier-Zepter V, Schraven A, et al. Human adipocytes secrete mineralocorticoid-releasing factors. *Proc Natl Acad Sci U S A*. 2003;100(24):14211–14216.
121. Asferg CL, Nielsen SJ, Andersen UB, et al. Relative atrial natriuretic peptide deficiency and inadequate renin and angiotensin II suppression in obese hypertensive men. *Hypertension*. 2013;62(1):147–153.
122. Wolk R, Kara T, Somers VK. Sleep-disordered breathing and cardiovascular disease. *Circulation*. 2003;108(1):9–12.
123. Ichiki T, Burnett Jr JC. Atrial natriuretic peptide - old but new therapeutic in cardiovascular diseases. *Circ J*. 2017;81(7):913–919.
124. Curry FR. Atrial natriuretic peptide: an essential physiological regulator of transvascular fluid, protein transport, and plasma volume. *J Clin Invest*. 2005;115(6):1458–1461.
125. Hu BC, Li Y, Liu M, et al. Blood pressure and urinary sodium excretion in relation to 16 genetic polymorphisms in the natriuretic peptide system in Chinese. *Endocr J*. 2014;61(9):861–874.
126. Newton-Cheh C, Larson MG, Vasan RS, et al. Association of common variants in NPPA and NPPB with circulating natriuretic peptides and blood pressure. *Nat Genet*. 2009;41(3):348–353.
127. Maimaitiming S, Roussel R, Hadjadj S, et al. Association of common variants in NPPA and NPPB with blood pressure does not translate into kidney damage in a general population study. *J Hypertens*. 2010;28(6):1230–1233.
128. Armaly Z, Assady S, Abassi Z. Corin: a new player in the regulation of salt-water balance and blood pressure. *Curr Opin Nephrol Hypertens*. 2013;22(6):713–722.
129. Li H, Zhang Y, Wu Q. Role of corin in the regulation of blood pressure. *Curr Opin Nephrol Hypertens*. 2017;26(2):67–73.
130. Spieker LE, Flammer AJ, Lüscher TF. The vascular endothelium in hypertension. *Handb Exp Pharmacol*. 2006;(176 Pt 2):249–283 (In eng).
131. Taddei S, Virdis A, Mattei P, Arzilli F, Salvetti A. Endothelium-dependent forearm vasodilation is reduced in normotensive subjects with familial history of hypertension. *J Cardiovasc Pharmacol*. 1992;20(suppl 12):S193–S195.
132. Dharmashankar K, Widlansky ME. Vascular endothelial function and hypertension: insights and directions. *Curr Hypertens Rep*. 2010;12(6):448–455.
133. Popolo A, Autore G, Pinto A, Marzocco S. Oxidative stress in patients with cardiovascular disease and chronic renal failure. *Free Radic Res*. 2013;47(5):346–356.
134. Kohan DE, Barton M. Endothelin and endothelin antagonists in chronic kidney disease. *Kidney Int*. 2014;86(5):896–904.
135. Lazich I, Bakris GL. Endothelin antagonism in patients with resistant hypertension and hypertension nephropathy. *Contrib Nephrol*. 2011;172:223–234.
136. Schlaich MP, Bellet M, Weber MA, et al. Dual endothelin antagonist aprocitentan for resistant hypertension (PRECISION): a multicentre, blinded, randomised, parallel-group, phase 3 trial. *Lancet*. 2022;400:10367.
137. Mazzali M, Hughes J, Kim YG, et al. Elevated uric acid increases blood pressure in the rat by a novel crystal-independent mechanism. *Hypertension*. 2001;38(5):1101–1106.
138. Feig DI. Serum uric acid and the risk of hypertension and chronic kidney disease. *Curr Opin Rheumatol*. 2014;26(2):176–185.
139. Feig DI, Soletsky B, Johnson RJ. Effect of allopurinol on blood pressure of adolescents with newly diagnosed essential hypertension: a randomized trial. *JAMA*. 2008;300(8):924–932.

140. Borgi L, McMullan C, Wohlhueter A, Curhan GC, Fisher ND, Forman JP. Effect of uric acid-lowering agents on endothelial function: a randomized, double-blind, placebo-controlled trial. *Hypertension*. 2017;69(2):243–248.
141. Jia G, Aroor AR, Whaley-Connell AT, Sowers JR. Fructose and uric acid: is there a role in endothelial function? *Curr Hypertens Rep*. 2014;16(6):434.
142. Rosset R, Surowska A, Tappy L. Pathogenesis of cardiovascular and metabolic diseases: are fructose-containing sugars more involved than other dietary calories? *Curr Hypertens Rep*. 2016;18(6):44.
143. van Goor H, van den Born JC, Hillebrands JL, Joles JA. Hydrogen sulfide in hypertension. *Curr Opin Nephrol Hypertens*. 2016;25(2):107–113.
144. Yang G, Wu L, Jiang B, et al. H2S as a physiologic vasorelaxant: hypertension in mice with deletion of cystathionine gamma-lyase. *Sci Technol Humanit*. 2008;322(5901):587–590.
145. Quyyumi AA, Patel RS. Endothelial dysfunction and hypertension: cause or effect? *Hypertension*. 2010;55(5):1092–1094.
146. Shimbo D, Muntner P, Mann D, et al. Endothelial dysfunction and the risk of hypertension: the multi-ethnic study of atherosclerosis. *Hypertension*. 2010;55(5):1210–1216.
147. Taddei S, Bruno RM. Endothelial dysfunction in hypertension: achievements and open questions. *J Hypertens*. 2016;34(8):1492–1493.
148. Quinn U, Tomlinson LA, Cockcroft JR. Arterial stiffness. (In eng) *JRSM Cardiovasc Dis*. 2012;1(6).
149. Zieman SJ, Melenovsky V, Kass DA. Mechanisms, pathophysiology, and therapy of arterial stiffness. *Arterioscler Thromb Vasc Biol*. 2005;25(5):932–943.
150. Mitchell GF. Arterial stiffness: insights from Framingham and Iceland. *Curr Opin Nephrol Hypertens*. 2015;24(1):1–7.
151. Kaess BM, Rong J, Larson MG, et al. Aortic stiffness, blood pressure progression, and incident hypertension. *JAMA*. 2012;308(9):875–881.
152. Chen W, Li S, Fernandez C, et al. Temporal relationship between elevated blood pressure and arterial stiffening among middle-aged Black and white adults: the bogalusa heart study. *Am J Epidemiol*. 2016;183(7):599–608. https://doi.org/10.1093/aje/kwv274.
153. Ben-Shlomo Y, Spears M, Boustred C, et al. Aortic pulse wave velocity improves cardiovascular event prediction: an individual participant meta-analysis of prospective observational data from 17,635 subjects. *J Am Coll Cardiol*. 2014;63(7):636–646.
154. Rodriguez-Iturbe B, Pons H, Quiroz Y, Johnson RJ. The immunological basis of hypertension. *Am J Hypertens*. 2014;27(11):1327–1337.
155. Harrison DG. Inflammation and immunity in hypertension. In: Bakris GS, M J, eds. *Hypertension : A Companion to Braunwald's Heart Disease*. 3rd ed. Philadelphia, PA: Elsevier; 2018:60–69.
156. Barbaro NR, Foss JD, Kryshtal DO, et al. Dendritic cell amiloride-sensitive channels mediate sodium-induced inflammation and hypertension. *Cell Rep*. 2017;21(4):1009–1020.
157. Guzik TJ, Hoch NE, Brown KA, et al. Role of the T cell in the genesis of angiotensin II induced hypertension and vascular dysfunction. *J Exp Med*. 2007;204(10):2449–2460.
158. Herrera J, Ferrebuz A, MacGregor EG, Rodriguez-Iturbe B. Mycophenolate mofetil treatment improves hypertension in patients with psoriasis and rheumatoid arthritis. *J Am Soc Nephrol*. 2006;17(12 suppl 3):S218–S225.
159. Rodriguez-Iturbe B, Pons H, Quiroz Y, Lanaspa MA, Johnson RJ. Autoimmunity in the pathogenesis of hypertension. *Nat Rev Nephrol*. 2014;10(1):56–62.
160. Madhur MS, Lob HE, McCann LA, et al. Interleukin 17 promotes angiotensin II-induced hypertension and vascular dysfunction. *Hypertension*. 2010;55(2):500–507.
161. Jantsch J, Schatz V, Friedrich D, et al. Cutaneous Na+ storage strengthens the antimicrobial barrier function of the skin and boosts macrophage-driven host defense. *Cell Metab*. 2015;21(3):493–501.
162. Kleinewietfeld M, Manzel A, Titze J, et al. Sodium chloride drives autoimmune disease by the induction of pathogenic TH17 cells. *Nature*. 2013;496(7446):518–522.
163. Davis GK, Fehrenbach DJ, Madhur MS. Interleukin 17A: key player in the pathogenesis of hypertension and a potential therapeutic target. *Curr Hypertens Rep*. 2021;23(3):13.
164. Kirabo A, Fontana V, de Faria AP, et al. DC isoketal-modified proteins activate T cells and promote hypertension. *J Clin Invest*. 2014;124(10):4642–4656.
165. Yu M, Moreno C, Hoagland KM, et al. Antihypertensive effect of glucagon-like peptide 1 in Dahl salt-sensitive rats. *J Hypertens*. 2003;21(6):1125–1135.
166. Wong HJ, Toh KZX, Teo YH, et al. Effects of glucagon-like peptide-1 receptor agonists on blood pressure in overweight or obese patients: a meta-analysis of randomized controlled trials. *J Hypertens*. 2025;43(2):290–300.
167. Richards EM, Pepine CJ, Raizada MK, Kim S. The gut, its microbiome, and hypertension. *Curr Hypertens Rep*. 2017;19(4):36. https://doi.org/10.1007/s11906-017-0734-1.
168. Yang T, Santisteban MM, Rodriguez V, et al. Gut dysbiosis is linked to hypertension. *Hypertension*. 2015;65(6):1331–1340.
169. Karbach SH, Schonfelder T, Brandao I, et al. Gut microbiota promote angiotensin II-induced arterial hypertension and vascular dysfunction. *J Am Heart Assoc*. 2016;5(9).
170. Pluznick JL, Protzko RJ, Gevorgyan H, et al. Olfactory receptor responding to gut microbiota-derived signals plays a role in renin secretion and blood pressure regulation. *Proc Natl Acad Sci U S A*. 2013;110(11):4410–4415.
171. Nakai H, Murosaki S, Yamamoto Y, Furutani M, Matsuoka R, Hirose Y. Safety and efficacy of using heat-killed Lactobacillus plantarum L-137: high-dose and long-term use effects on immune-related safety and intestinal bacterial flora. *J Immunotoxicol*. 2021;18(1):127–135.
172. Nakai M, Ribeiro RV, Stevens BR, et al. Essential hypertension is associated with changes in gut microbial metabolic pathways: a multisite analysis of ambulatory blood pressure. *Hypertension*. 2021;78(3):804–815.
173. Ferguson JF, Aden LA, Barbaro NR, et al. High dietary salt-induced dendritic cell activation underlies microbial dysbiosis-associated hypertension. *JCI Insight*. 2019;5(13). https://doi.org/10.1172/jci.insight.126241.
174. Khalesi S, Sun J, Buys N, Jayasinghe R. Effect of probiotics on blood pressure: a systematic review and meta-analysis of randomized, controlled trials. *Hypertension*. 2014;64(4):897–903. https://doi.org/10.1161/HYPERTENSIONAHA.114.03469.
175. Zampaglione B, Pascale C, Marchisio M, Cavallo-Perin P. Hypertensive urgencies and emergencies. Prevalence and clinical presentation. *Hypertension*. 1996;27(1):144–147. https://doi.org/10.1161/01.hyp.27.1.144.
176. Peixoto AJ. A young patient with a family history of hypertension. *Clin J Am Soc Nephrol*. 2014;9(12):2164–2172. https://doi.org/10.2215/CJN.02240314.
177. Muiesan ML, Salvetti M, Paini A, et al. Ocular fundus photography with a smartphone device in acute hypertension. *J Hypertens*. 2017;35(8):1660–1665. https://doi.org/10.1097/HJH.0000000000001354.
178. Shanmugam MP, Mishra DK, Madhukumar R, Ramanjulu R, Reddy SY, Rodrigues G. Fundus imaging with a mobile phone: a review of techniques. *Indian J Ophthalmol*. 2014;62(9):960–962. https://doi.org/10.4103/0301-4738.143949.
179. Stergiou GS, Mukkamala R, Avolio A, et al. Cuffless blood pressure measuring devices: review and statement by the European society of hypertension working group on blood pressure monitoring and cardiovascular variability. *J Hypertens*. 2022;40(8):1449–1460.
180. Cheung AK, Whelton PK, Muntner P, et al. International consensus on standardized clinic blood pressure measurement–a call to action. *Am J Med*. 2023;136(5):438–445.
181. Kollias A, Stambolliu E, Kyriakoulis KG, Gravvani A, Stergiou GS. Unattended versus attended automated office blood pressure: systematic review and meta-analysis of studies using the same methodology for both methods. *J Clin Hypertens*. 2019;21(2):148–155. https://doi.org/10.1111/jch.13462.
182. Shibao C, Lipsitz LA, Biaggioni I, American Society of Hypertension Writing G. Evaluation and treatment of orthostatic hypotension. *J Am Soc Hypertens*. 2013;7(4):317–324.
183. Freeman R, Wieling W, Axelrod FB, et al. Consensus statement on the definition of orthostatic hypotension, neurally mediated syncope and the postural tachycardia syndrome. *Clin Auton Res*. 2011;21(2):69–72.
184. Shaw BH, Garland EM, Black BK, et al. Optimal diagnostic thresholds for diagnosis of orthostatic hypotension with a 'sit-to-stand test'. *J Hypertens*. 2017;35(5):1019–1025.
185. Norcliffe-Kaufmann L, Kaufmann H, Palma JA, et al. Orthostatic heart rate changes in patients with autonomic failure caused by neurodegenerative synucleinopathies. *Ann Neurol*. 2018;83(3):522–531.

186. Stergiou GS, Baibas NM, Gantzarou AP, et al. Reproducibility of home, ambulatory, and clinic blood pressure: implications for the design of trials for the assessment of antihypertensive drug efficacy. *Am J Hypertens.* 2002;15(2 Pt 1):101–104. https://doi.org/10.1016/s0895-7061(01)02324-x.
187. Bliziotis IA, Destounis A, Stergiou GS. Home versus ambulatory and office blood pressure in predicting target organ damage in hypertension: a systematic review and meta-analysis. *J Hypertens.* 2012;30(7):1289–1299. https://doi.org/10.1097/HJH.0b013e3283531eaf.
188. McManus RJ, Caulfield M, Williams B, National Institute for H, Clinical E. NICE hypertension guideline 2011: evidence based evolution. *Bmj.* 2012;344:e181.
189. Conen D, Bamberg F. Noninvasive 24-h ambulatory blood pressure and cardiovascular disease: a systematic review and meta-analysis. *J Hypertens.* 2008;26(7):1290–1299.
190. Piper MA, Evans CV, Burda BU, Margolis KL, O'Connor E, Whitlock EP. Diagnostic and predictive accuracy of blood pressure screening methods with consideration of rescreening intervals: a systematic review for the U.S. Preventive Services Task Force. *Ann Intern Med.* 2015;162(3):192–204.
191. Niiranen TJ, Hanninen MR, Johansson J, Reunanen A, Jula AM. Home-measured blood pressure is a stronger predictor of cardiovascular risk than office blood pressure: the Finn-Home study. *Hypertension.* 2010;55(6):1346–1351.
192. Kollias A, Ntineri A, Stergiou GS. Is white-coat hypertension a harbinger of increased risk? *Hypertens Res.* 2014;37(9):791–795.
193. Pierdomenico SD, Cuccurullo F. Prognostic value of white-coat and masked hypertension diagnosed by ambulatory monitoring in initially untreated subjects: an updated meta analysis. *Am J Hypertens.* 2011;24(1):52–58.
194. Cohen JB, Lotito MJ, Trivedi UK, Denker MG, Cohen DL, Townsend RR. Cardiovascular events and mortality in white coat hypertension: a systematic review and meta-analysis. *Ann Intern Med.* 2019;170(12):853–862.
195. Stergiou GS, Asayama K, Thijs L, et al. Prognosis of white-coat and masked hypertension: international Database of HOme blood pressure in relation to Cardiovascular Outcome. *Hypertension.* 2014;63(4):675–682.
196. Asayama K, Stolarz-Skrzypek K, Yang WY, et al. What did we learn from the international databases on ambulatory and home blood pressure in relation to cardiovascular outcome? *Hypertens Res.* 2023;46(4):934–949.
197. Briasoulis A, Androulakis E, Palla M, Papageorgiou N, Tousoulis D. White-coat hypertension and cardiovascular events: a meta-analysis. *J Hypertens.* 2016;34(4):593–599.
198. Mancia G, Bombelli M, Facchetti R, et al. Long-term risk of sustained hypertension in white-coat or masked hypertension. *Hypertension.* 2009;54(2):226–232.
199. Sivén S, Niiranen T, Kantola I, Jula A. 1B. 03: white-coat and masked hypertension as risk factors for progression to sustained hypertension: the Finn-home study. *J Hypertens.* 2015;33:e5–e6.
200. Zhang DY, Guo QH, An DW, Li Y, Wang JG. A comparative meta-analysis of prospective observational studies on masked hypertension and masked uncontrolled hypertension defined by ambulatory and home blood pressure. *J Hypertens.* 2019;37(9):1775–1785.
201. Fagard RH, Celis H, Thijs L, et al. Daytime and nighttime blood pressure as predictors of death and cause-specific cardiovascular events in hypertension. *Hypertension.* 2008;51(1):55–61. https://doi.org/10.1161/HYPERTENSIONAHA.107.100727.
202. Salles GF, Reboldi G, Fagard RH, et al. Prognostic effect of the nocturnal blood pressure fall in hypertensive patients: the ambulatory blood pressure collaboration in patients with hypertension (ABC-H) meta-analysis. *Hypertension.* 2016;67(4):693–700.
203. Kasai T, Bradley TD, Friedman O, Logan AG. Effect of intensified diuretic therapy on overnight rostral fluid shift and obstructive sleep apnoea in patients with uncontrolled hypertension. *J Hypertens.* 2014;32(3):673–680. https://doi.org/10.1097/HJH.0000000000000047.
204. Boggia J, Li Y, Thijs L, et al. Prognostic accuracy of day versus night ambulatory blood pressure: a cohort study. *Lancet.* 2007;370(9594):1219–1229.
205. Fagard RH, Thijs L, Staessen JA, Clement DL, De Buyzere ML, De Bacquer DA. Night-day blood pressure ratio and dipping pattern as predictors of death and cardiovascular events in hypertension. *J Hum Hypertens.* 2009;23(10):645–653. https://doi.org/10.1038/jhh.2009.9.
206. Parati G, Bilo G, Kollias A, et al. Blood pressure variability: methodological aspects, clinical relevance and practical indications for management–a European Society of Hypertension position paper. *J Hypertens.* 2023;41(4):527–544.
207. Staessen JA, Byttebier G, Buntinx F, Celis H, O'Brien ET, Fagard R. Antihypertensive treatment based on conventional or ambulatory blood pressure measurement. A randomized controlled trial. Ambulatory Blood Pressure Monitoring and Treatment of Hypertension Investigators. *JAMA.* 1997;278(13):1065–1072. https://www.ncbi.nlm.nih.gov/pubmed/9315764.
208. Verberk WJ, Kroon AA, Lenders JW, et al. Self-measurement of blood pressure at home reduces the need for antihypertensive drugs: a randomized, controlled trial. *Hypertension.* 2007;50(6):1019–1025.
209. Verberk WJ, Kessels AG, Thien T. Telecare is a valuable tool for hypertension management, a systematic review and meta-analysis. *Blood Pres Monit.* 2011;16(3):149–155.
210. Stergiou GS, Palatini P, Parati G, et al. 2021 European Society of Hypertension practice guidelines for office and out-of-office blood pressure measurement. *J Hypertens.* 2021;39(7):1293–1302.
211. Niiranen TJ, Asayama K, Thijs L, et al. Outcome-driven thresholds for home blood pressure measurement: international database of home blood pressure in relation to cardiovascular outcome. *Hypertension.* 2013;61(1):27–34. https://doi.org/10.1161/HYPERTENSIONAHA.111.00100.
212. Carey RM, Calhoun DA, Bakris GL, et al. Resistant hypertension: detection, evaluation, and management: a scientific statement from the American heart association. *Hypertension.* 2018;72(5):e53–e90. https://doi.org/10.1161/HYP.0000000000000084.
213. Santos M, Shah AM. Alterations in cardiac structure and function in hypertension. *Curr Hypertens Rep.* 2014;16(5):428.
214. Laragh JH. Renin profiling for diagnosis, risk assessment, and treatment of hypertension. *Kidney Int.* 1993;44(5):1163–1175.
215. Blumenfeld JD, Laragh JH. Renin system analysis: a rational method for the diagnosis and treatment of the individual patient with hypertension. *Am J Hypertens.* 1998;11(7):894–896.
216. Egan BM, Basile JN, Rehman SU, et al. Plasma Renin test-guided drug treatment algorithm for correcting patients with treated but uncontrolled hypertension: a randomized controlled trial. *Am J Hypertens.* 2009;22(7):792–801.
217. Smith RD, Levy P, Ferrario CM. Consideration of noninvasive hemodynamic monitoring to target reduction of blood pressure levels study G. Value of noninvasive hemodynamics to achieve blood pressure control in hypertensive subjects. *Hypertension.* 2006;47(4):771–777.
218. Taler SJ, Textor SC, Augustine JE. Resistant hypertension: comparing hemodynamic management to specialist care. *Hypertension.* 2002;39(5):982–988.
219. Greco B, Chait Y, Nathanson BH, Germain MJ. A novel hypertension management algorithm guided by hemodynamic data. *Kidney Int Rep.* 2022;7(2):330–333.
220. Nugent JT, Jiang K, Funaro MC, et al. Does this child with high blood pressure have secondary hypertension?: the rational clinical examination systematic review. *JAMA.* 2023;329(12):1012–1021.
221. Omura M, Saito J, Yamaguchi K, Kakuta Y, Nishikawa T. Prospective study on the prevalence of secondary hypertension among hypertensive patients visiting a general outpatient clinic in Japan. *Hypertens Res.* 2004;27(3):193–202. https://doi.org/10.1291/hypres.27.193.
222. Anderson Jr GH, Blakeman N, Streeten DH. The effect of age on prevalence of secondary forms of hypertension in 4429 consecutively referred patients. *J Hypertens.* 1994;12(5):609–615. https://doi.org/10.1097/00004872-199405000-00015.
223. Peixoto AJ, Orias M, Desir GV. Does kidney disease cause hypertension? *Curr Hypertens Rep.* 2013;15(2):89–94.
224. Brown JM, Siddiqui M, Calhoun DA, et al. The unrecognized prevalence of primary aldosteronism: a cross-sectional study. *Ann Intern Med.* 2020;173(1):10–20.
225. Vaidya A, Brown JM, Carey RM, Siddiqui M, Williams GH. The unrecognized prevalence of primary aldosteronism. *Ann Intern Med.* 2020;173(8):683. https://doi.org/10.7326/L20-1097.
226. Markou A, Pappa T, Kaltsas G, et al. Evidence of primary aldosteronism in a predominantly female cohort of normotensive individuals: a very high odds ratio for progression into

arterial hypertension. *J Clin Endocrinol Metab*. 2013;98(4):1409–1416. https://doi.org/10.1210/jc.2012-3353.
227. Funder JW, Carey RM, Mantero F, et al. The management of primary aldosteronism: case detection, diagnosis, and treatment: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2016;101(5):1889–1916.
228. Huang WC, Lin YH, Wu VC, et al. Who should be screened for primary aldosteronism? A comprehensive review of current evidence. *J Clin Hypertens*. 2022;24(9):1194–1203.
229. Rossi GP, Bernini G, Caliumi C, et al. A prospective study of the prevalence of primary aldosteronism in 1,125 hypertensive patients. *J Am Coll Cardiol*. 2006;48(11):2293–2300.
230. Jaffe G, Gray Z, Krishnan G, et al. Screening rates for primary aldosteronism in resistant hypertension: a cohort study. *Hypertension*. 2020;75(3):650–659.
231. Cohen JB, Cohen DL, Herman DS, Leppert JT, Byrd JB, Bhalla V. Testing for primary aldosteronism and mineralocorticoid receptor antagonist use among U.S. Veterans: a retrospective cohort study. *Ann Intern Med*. 2021;174(3):289–297.
232. Monticone S, D'Ascenzo F, Moretti C, et al. Cardiovascular events and target organ damage in primary aldosteronism compared with essential hypertension: a systematic review and meta-analysis. *Lancet Diabetes Endocrinol*. 2018;6(1):41–50. https://doi.org/10.1016/S2213-8587(17)30319-4.
233. Brown JM, Robinson-Cohen C, Luque-Fernandez MA, et al. The spectrum of subclinical primary aldosteronism and incident hypertension: a cohort study. *Ann Intern Med*. 2017;167(9):630–641. https://doi.org/10.7326/M17-0882.
234. Wu X, Senanayake R, Goodchild E, et al. [(11)C]metomidate PET-CT versus adrenal vein sampling for diagnosing surgically curable primary aldosteronism: a prospective, within-patient trial. *Nat Med*. 2023;29(1):190–202.
235. Parthasarathy HK, Menard J, White WB, et al. A double-blind, randomized study comparing the antihypertensive effect of eplerenone and spironolactone in patients with hypertension and evidence of primary aldosteronism. *J Hypertens*. 2011;29(5):980–990.
236. Jing Y, Liao K, Li R, et al. Cardiovascular events and all-cause mortality in surgically or medically treated primary aldosteronism: a Meta-analysis. *J Renin-Angiotensin-Aldosterone Syst JRAAS*. 2021;22(1):14703203211003781.
237. Laffin LJ, Rodman D, Luther JM, et al. Aldosterone synthase inhibition with lorundrostat for uncontrolled hypertension: the target-HTN randomized clinical trial. *JAMA*. 2023;330(12):1140–1150.
238. Fernandes-Rosa FL, Boulkroun S, Fedlaoui B, et al. New advances in endocrine hypertension: from genes to biomarkers. *Kidney Int*. 2023;103(3):485–500.
239. Nieman LK, Biller BM, Findling JW, et al. The diagnosis of cushing's syndrome: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2008;93(5):1526–1540.
240. Fleseriu M, Auchus R, Bancos I, et al. Consensus on diagnosis and management of Cushing's disease: a guideline update. *Lancet Diabetes Endocrinol*. 2021;9(12):847–875.
241. Turin CG, Crenshaw MM, Fishbein L. Pheochromocytoma and paraganglioma: germline genetics and hereditary syndromes. *Endocr Oncol*. 2022;2(1):R65–R77.
242. Cohen DF, L. Pheochromocytoma and paraganglioma. In: Bakris G, Sorrentino MJ, eds. *Hypertension : A Companion to Braunwald's Heart Disease*. 3rd ed. Philadelphia, PA: Elsevier; 2018:136–143.
243. Cohen DL, Fraker D, Townsend RR. Lack of symptoms in patients with histologic evidence of pheochromocytoma: a diagnostic challenge. *Ann N Y Acad Sci*. 2006;1073:47–51.
244. Fishbein L. Pheochromocytoma and paraganglioma: genetics, diagnosis, and treatment. *Hematol Oncol Clin North Am*. 2016;30(1):135–150.
245. Eisenhofer G, Lenders JW, Timmers H, et al. Measurements of plasma methoxytyramine, normetanephrine, and metanephrine as discriminators of different hereditary forms of pheochromocytoma. *Clin Chem*. 2011;57(3):411–420.
246. Amar L, Pacak K, Steichen O, et al. International consensus on initial screening and follow-up of asymptomatic SDHx mutation carriers. *Nat Rev Endocrinol*. 2021;17(7):435–444.
247. Lenders JW, Duh QY, Eisenhofer G, et al. Pheochromocytoma and paraganglioma: an endocrine society clinical practice guideline. *J Clin Endocrinol Metab*. 2014;99(6):1915–1942.
248. Bruynzeel H, Feelders RA, Groenland TH, et al. Risk factors for hemodynamic instability during surgery for pheochromocytoma. *J Clin Endocrinol Metab*. 2010;95(2):678–685.
249. Fishbein L, Orlowski R, Cohen D. Pheochromocytoma/Paraganglioma: review of perioperative management of blood pressure and update on genetic mutations associated with pheochromocytoma. *J Clin Hypertens*. 2013;15(6):428–434.
250. Townsend MS, Fulgoni 3rd VL, Stern JS, Adu-Afarwuah S, McCarron DA. Low mineral intake is associated with high systolic blood pressure in the Third and Fourth National Health and Nutrition Examination Surveys: could we all be right? *Am J Hypertens*. 2005;18(2 Pt 1):261–269.
251. Hager A. Hypertension in aortic coarctation. *Minerva Cardioangiol*. 2009;57(6):733–742.
252. Aronow WS, Fleg JL, Pepine CJ, et al. ACCF/AHA 2011 expert consensus document on hypertension in the elderly: a report of the American College of cardiology foundation task force on clinical expert consensus documents developed in collaboration with the American Academy of neurology, American geriatrics society, American society for preventive cardiology, American society of hypertension, American society of nephrology, association of Black cardiologists, and European society of hypertension. *J Am Coll Cardiol*. 2011;57(20):2037–2114.
253. Canniffe C, Ou P, Walsh K, Bonnet D, Celermajer D. Hypertension after repair of aortic coarctation–a systematic review. *Int J Cardiol*. 2013;167(6):2456–2461.
254. Melmed S, Casanueva FF, Klibanski A, et al. A consensus on the diagnosis and treatment of acromegaly complications. *Pituitary*. 2013;16(3):294–302. https://doi.org/10.1007/s11102-012-0420-x.
255. Sardella C, Urbani C, Lombardi M, et al. The beneficial effect of acromegaly control on blood pressure values in normotensive patients. *Clin Endocrinol*. 2014;81(4):573–581.
256. Pepin JL, Borel AL, Tamisier R, Baguet JP, Levy P, Dauvilliers Y. Hypertension and sleep: overview of a tight relationship. *Sleep Med Rev*. 2014;18(6):509–519.
257. Gangwisch JE, Heymsfield SB, Boden-Albala B, et al. Short sleep duration as a risk factor for hypertension: analyses of the first National Health and Nutrition Examination Survey. *Hypertension*. 2006;47(5):833–839.
258. Sayk F, Teckentrup C, Becker C, et al. Effects of selective slow-wave sleep deprivation on nocturnal blood pressure dipping and daytime blood pressure regulation. *Am J Physiol Regul Integr Comp Physiol*. 2010;298(1):R191–R197. https://doi.org/10.1152/ajpregu.00368.2009.
259. Itani O, Jike M, Watanabe N, Kaneita Y. Short sleep duration and health outcomes: a systematic review, meta-analysis, and meta-regression. *Sleep Med*. 2017;32:246–256.
260. Dematteis M, Julien C, Guillermet C, et al. Intermittent hypoxia induces early functional cardiovascular remodeling in mice. *Am J Respir Crit Care Med*. 2008;177(2):227–235.
261. Gottlieb DJ, Punjabi NM, Mehra R, et al. CPAP versus oxygen in obstructive sleep apnea. *N Engl J Med*. 2014;370(24):2276–2285.
262. Labarca G, Schmidt A, Dreyse J, et al. Efficacy of continuous positive airway pressure (CPAP) in patients with obstructive sleep apnea (OSA) and resistant hypertension (RH): systematic review and meta-analysis. *Sleep Med Rev*. 2021;58:101446.
263. Altay S, Firat S, Peker Y, The Turcosact C. A narrative review of the association of obstructive sleep apnea with hypertension: how to treat both when they coexist? *J Clin Med*. 2023;12(12).
264. Peixoto AJ. Acute severe hypertension. *N Engl J Med*. 2019;381(19):1843–1852.
265. Vlcek M, Bur A, Woisetschlager C, Herkner H, Laggner AN, Hirschl MM. Association between hypertensive urgencies and subsequent cardiovascular events in patients with hypertension. *J Hypertens*. 2008;26(4):657–662. https://doi.org/10.1097/HJH.0b013e3282f4e8b6.
266. Garovic VD, Dechend R, Easterling T, et al. Hypertension in pregnancy: diagnosis, blood pressure goals, and pharmacotherapy: a scientific statement from the American heart association. *Hypertension*. 2022;79(2):e21–e41.
267. Minhas R, Young D, Naseem R, et al. Association of antepartum blood pressure levels and angiogenic profile among women with chronic hypertension. *Pregnancy Hypertens*. 2018;14:110–114.

268. Pan Y, Cai W, Cheng Q, Dong W, An T, Yan J. Association between anxiety and hypertension: a systematic review and meta-analysis of epidemiological studies. *Neuropsychiatr Dis Treat*. 2015;11:1121–1130.
269. Yau KK, Loke AY. Effects of diaphragmatic deep breathing exercises on prehypertensive or hypertensive adults: a literature review. *Complement Ther Clin Pract*. 2021;43:101315.
270. Hanna J, Ghazi L, Yamamoto Y, et al. Excessive blood pressure response to clonidine in hospitalized patients with asymptomatic severe hypertension. *Am J Hypertens*. 2022;35(5):433–440.
271. Emdin CA, Rahimi K, Neal B, Callender T, Perkovic V, Patel A. Blood pressure lowering in type 2 diabetes: a systematic review and meta-analysis. *JAMA*. 2015;313(6):603–615.
272. Thomopoulos C, Parati G, Zanchetti A. Effects of blood pressure lowering on outcome incidence in hypertension: 7. Effects of more vs. less intensive blood pressure lowering and different achieved blood pressure levels–updated overview and meta-analyses of randomized trials. *J Hypertens*. 2016;34(4):613–622.
273. Ettehad D, Emdin CA, Kiran A, et al. Blood pressure lowering for prevention of cardiovascular disease and death: a systematic review and meta-analysis. *Lancet*. 2016;387(10022):957–967.
274. Jamerson K, Weber MA, Bakris GL, et al. Benazepril plus amlodipine or hydrochlorothiazide for hypertension in high-risk patients. *N Engl J Med*. 2008;359(23):2417–2428.
275. Bakris GL, Sarafidis PA, Weir MR, et al. Renal outcomes with different fixed-dose combination therapies in patients with hypertension at high risk for cardiovascular events (ACCOMPLISH): a prespecified secondary analysis of a randomised controlled trial. *Lancet*. 2010;375(9721):1173–1181.
276. Gradman AH, Basile JN, Carter BL, Bakris GL, American Society of Hypertension Writing G. Combination therapy in hypertension. *J Clin Hypertens*. 2011;13(3):146–154. https://doi.org/10.1111/j.1751-7176.2010.00397.x.
277. Weber MA, Schiffrin EL, White WB, et al. Clinical practice guidelines for the management of hypertension in the community a statement by the American Society of Hypertension and the International Society of Hypertension. *J Hypertens*. 2014;32(1):3–15.
278. Jung O, Gechter JL, Wunder C, et al. Resistant hypertension? Assessment of adherence by toxicological urine analysis. *J Hypertens*. 2013;31(4):766–774.
279. Tomaszewski M, White C, Patel P, et al. High rates of non-adherence to antihypertensive treatment revealed by high-performance liquid chromatography-tandem mass spectrometry (HP LC-MS/MS) urine analysis. *Heart*. 2014;100(11):855–861. https://doi.org/10.1136/heartjnl-2013-305063.
280. Chen R, Suchard MA, Krumholz HM, et al. Comparative first-line effectiveness and safety of ACE (Angiotensin-Converting enzyme) inhibitors and angiotensin receptor blockers: a multinational cohort study. *Hypertension*. 2021;78(3):591–603.
281. Ruggenenti P, Perna A, Loriga G, et al. Blood-pressure control for renoprotection in patients with non-diabetic chronic renal disease (REIN-2): multicentre, randomised controlled trial. *Lancet*. 2005;365(9463):939–946.
282. Appel LJ, Wright Jr JT, Greene T, et al. Intensive blood-pressure control in hypertensive chronic kidney disease. *N Engl J Med*. 2010;363(10):918–929.
283. Cheung AK, Rahman M, Reboussin DM, et al. Effects of intensive BP control in CKD. *J Am Soc Nephrol*. 2017;28(9):2812–2823.
284. Klahr S, Levey AS, Beck GJ, et al. The effects of dietary protein restriction and blood-pressure control on the progression of chronic renal disease. Modification of Diet in Renal Disease Study Group. *N Engl J Med*. 1994;330(13):877–884.
285. Kidney disease: improving global outcomes blood pressure work G. KDIGO 2021 clinical practice guideline for the management of blood pressure in chronic kidney disease. *Kidney Int*. 2021;99(3S):S1–S87.
286. Pajewski NM, Beddhu S, Bress AP, et al. The legacy effect of intensive versus standard BP control on the incidence of needing dialysis or kidney transplantation. *J Am Soc Nephrol*. 2024;35(12):1737–1745.
287. Flack JM, Sica DA, Bakris G, et al. Management of high blood pressure in Blacks: an update of the International Society on Hypertension in Blacks consensus statement. *Hypertension*. 2010;56(5):780–800.
288. Pogue V, Rahman M, Lipkowitz M, et al. Disparate estimates of hypertension control from ambulatory and clinic blood pressure measurements in hypertensive kidney disease. *Hypertension*. 2009;53(1):20–27.
289. Crespo JJ, Pineiro L, Otero A, et al. Administration-time-dependent effects of hypertension treatment on ambulatory blood pressure in patients with chronic kidney disease. *Chronobiol Int*. 2013;30(1–2):159–175.
290. Maschio G, Alberti D, Janin G, et al. Effect of the angiotensin-converting-enzyme inhibitor benazepril on the progression of chronic renal insufficiency. The angiotensin-converting-enzyme inhibition in progressive renal insufficiency study group. *N Engl J Med*. 1996;334(15):939–945. https://doi.org/10.1056/NEJM199604113341502.
291. Hou FF, Zhang X, Zhang GH, et al. Efficacy and safety of benazepril for advanced chronic renal insufficiency. *N Engl J Med*. 2006;354(2):131–140.
292. Hansson L. The Hypertension Optimal Treatment study and the importance of lowering blood pressure. *J Hypertens Suppl*. 1999;17(1):S9–S13.
293. Mann JF, Anderson C, Gao P, et al. Dual inhibition of the renin-angiotensin system in high-risk diabetes and risk for stroke and other outcomes: results of the ONTARGET trial. *J Hypertens*. 2013;31(2):414–421.
294. Group AS, Cushman WC, Evans GW, et al. Effects of intensive blood-pressure control in type 2 diabetes mellitus. *N Engl J Med*. 2010;362(17):1575–1585.
295. Cooper-DeHoff RM, Gong Y, Handberg EM, et al. Tight blood pressure control and cardiovascular outcomes among hypertensive patients with diabetes and coronary artery disease. *JAMA*. 2010;304(1):61–68.
296. Zhang W, Zhang S, Deng Y, et al. Trial of intensive blood-pressure control in older patients with hypertension. *N Engl J Med*. 2021;385(14):1268–1279.
297. de Boer I, Bangalore S, Benetos A, et al. Diabetes and hypertension: a position statement by the American diabetes association. *Diabetes Care*. 2017;40(9):1273–1284.
298. American Diabetes Association Professional Practice C. 10. Cardiovascular disease and risk management: standards of care in diabetes-2024. *Diabetes Care*. 2024;47(suppl 1):S179–S218.
299. Bakris GL, Molitch M. Microalbuminuria as a risk predictor in diabetes: the continuing saga. *Diabetes Care*. 2014;37(3):867–875.
300. Mauer M, Zinman B, Gardiner R, et al. Renal and retinal effects of enalapril and losartan in type 1 diabetes. *N Engl J Med*. 2009;361(1):40–51.
301. Deedwania P, Shea J, Chen W, Brener L. Effects of add-on nebivolol on blood pressure and glucose parameters in hypertensive patients with prediabetes. *J Clin Hypertens*. 2013;15(4):270–278.
302. Barzilay JI, Davis BR, Pressel SL, et al. Long-term effects of incident diabetes mellitus on cardiovascular outcomes in people treated for hypertension: the ALLHAT Diabetes Extension Study. *Circ Cardiovasc Qual Outcomes*. 2012;5(2):153–162.
303. Lewis EJ, Hunsicker LG, Bain RP, Rohde RD. The effect of angiotensin-converting-enzyme inhibition on diabetic nephropathy. The Collaborative Study Group. *N Engl J Med*. 1993;329(20):1456–1462.
304. Brenner BM, Cooper ME, de Zeeuw D, et al. Effects of losartan on renal and cardiovascular outcomes in patients with type 2 diabetes and nephropathy. *N Engl J Med*. 2001;345(12):861–869.
305. Lewis EJ, Hunsicker LG, Clarke WR, et al. Renoprotective effect of the angiotensin-receptor antagonist irbesartan in patients with nephropathy due to type 2 diabetes. *N Engl J Med*. 2001;345(12):851–860.
306. Fried LF, Emanuele N, Zhang JH, et al. Combined angiotensin inhibition for the treatment of diabetic nephropathy. *N Engl J Med*. 2013;369(20):1892–1903.
307. Parving HH, Brenner BM, McMurray JJ, et al. Cardiorenal end points in a trial of aliskiren for type 2 diabetes. *N Engl J Med*. 2012;367(23):2204–2213. https://doi.org/10.1056/NEJMoa1208799.
308. Bakris G, Burgess E, Weir M, Davidai G, Koval S, Investigators AS. Telmisartan is more effective than losartan in reducing proteinuria in patients with diabetic nephropathy. *Kidney Int*. 2008;74(3):364–369.

309. Bakris GL, Toto RD, McCullough PA, et al. Effects of different ACE inhibitor combinations on albuminuria: results of the GUARD study. *Kidney Int.* 2008;73(11):1303–1309.
310. Zoungas S, Chalmers J, Neal B, et al. Follow-up of blood-pressure lowering and glucose control in type 2 diabetes. *N Engl J Med.* 2014;371(15):1392–1406.
311. O'Malley K, McCormack P, O'Brien ET. Isolated systolic hypertension: data from the European working party on high blood pressure in the elderly. *J Hypertens Suppl.* 1988;6(1):S105–S108.
312. Messerli FH, Mancia G, Conti CR, Pepine CJ. Guidelines on the management of stable angina pectoris: executive summary: the task force on the management of stable angina pectoris of the European society of cardiology. *Eur Heart J.* 2006;27(23):2902–2903. https://doi.org/10.1093/eurheartj/ehl308; author reply 2903.
313. Oliva RV, Bakris GL. Management of hypertension in the elderly population. *J Gerontol A Biol Sci Med Sci.* 2012;67(12):1343–1351. https://doi.org/10.1093/gerona/gls148.
314. Kovesdy CP, Bleyer AJ, Molnar MZ, et al. Blood pressure and mortality in U.S. veterans with chronic kidney disease: a cohort study. *Ann Intern Med.* 2013;159(4):233–242.
315. Beddhu S, Chertow GM, Cheung AK, et al. Influence of baseline diastolic blood pressure on effects of intensive compared with standard blood pressure control. *Circulation.* 2018;137(2):134–143.
316. Young JH, Klag MJ, Muntner P, Whyte JL, Pahor M, Coresh J. Blood pressure and decline in kidney function: findings from the Systolic Hypertension in the Elderly Program (SHEP). *J Am Soc Nephrol.* 2002;13(11):2776–2782. https://doi.org/10.1097/01.asn.0000031805.09178.37.
317. Curb JD, Pressel SL, Cutler JA, et al. Effect of diuretic-based antihypertensive treatment on cardiovascular disease risk in older diabetic patients with isolated systolic hypertension. Systolic Hypertension in the Elderly Program Cooperative Research Group. *JAMA.* 1996;276(23):1886–1892. https://www.ncbi.nlm.nih.gov/pubmed/8968014.
318. Rahman M, Pressel S, Davis BR, et al. Renal outcomes in high-risk hypertensive patients treated with an angiotensin-converting enzyme inhibitor or a calcium channel blocker vs a diuretic: a report from the Antihypertensive and Lipid-Lowering Treatment to Prevent Heart Attack Trial (ALLHAT). *Arch Intern Med.* 2005;165(8):936–946.
319. Whelton PK, Appel LJ, Espeland MA, et al. Sodium reduction and weight loss in the treatment of hypertension in older persons: a randomized controlled trial of nonpharmacologic interventions in the elderly (TONE). TONE Collaborative Research Group. *JAMA.* 1998;279(11):839–846.
320. Neal B, Wu Y, Feng X, et al. Effect of salt substitution on cardiovascular events and death. *N Engl J Med.* 2021;385(12):1067–1077.
321. Johnson AG, Nguyen TV, Owe-Young R, Williamson DJ, Day RO. Potential mechanisms by which nonsteroidal anti-inflammatory drugs elevate blood pressure: the role of endothelin-1. *J Hum Hypertens.* 1996;10(4):257–261.
322. Izhar M, Alausa T, Folker A, Hung E, Bakris GL. Effects of COX inhibition on blood pressure and kidney function in ACE inhibitor-treated blacks and hispanics. *Hypertension.* 2004;43(3):573–577.
323. Ishiguro C, Fujita T, Omori T, Fujii Y, Mayama T, Sato T. Assessing the effects of non-steroidal anti-inflammatory drugs on antihypertensive drug therapy using post-marketing surveillance database. *J Epidemiol.* 2008;18(3):119–124. https://doi.org/10.2188/jea.je2007413.
324. Birkenhager WH. Diuretics and blood pressure reduction: physiologic aspects. *J Hypertens Suppl.* 1990;8(2):S3–S7.
325. Sica DA, Carter B, Cushman W, Hamm L. Thiazide and loop diuretics. *J Clin Hypertens.* 2011;13(9):639–643.
326. Frishman WH. Comparative pharmacokinetic and clinical profiles of angiotensin-converting enzyme inhibitors and calcium antagonists in systemic hypertension. *Am J Cardiol.* 1992;69(10):17C–25C. https://doi.org/10.1016/0002-9149(92)90277-6.
327. Chen N, Zhou M, Yang M, et al. Calcium channel blockers versus other classes of drugs for hypertension. *Cochrane Database Syst Rev.* 2010;(8):CD003654.
328. Pappoe LS, Winkelmayer WC. ACE inhibitor and angiotensin II type 1 receptor antagonist therapies in elderly patients with diabetes mellitus: are they underutilized? *Drugs Aging.* 2010;27(2):87–94.
329. Xie X, Liu Y, Perkovic V, et al. Renin-angiotensin system inhibitors and kidney and cardiovascular outcomes in patients with CKD: a bayesian network meta-analysis of randomized clinical trials. *Am J Kidney Dis.* 2016;67(5):728–741.
330. Moser M, Menard J. Clinical significance of the metabolic effects of antihypertensive drugs. *J Hum Hypertens.* 1993;7(suppl 1):S50–S55.
331. Moen MD, Wagstaff AJ. Nebivolol: a review of its use in the management of hypertension and chronic heart failure. *Drugs (Basel).* 2006;66(10):1389–1409; discussion 1410.
332. Ladage D, Reidenbach C, Rieckeheer E, Graf C, Schwinger RH, Brixius K. Nebivolol lowers blood pressure and increases weight loss in patients with hypertension and diabetes in regard to age. *J Cardiovasc Pharmacol.* 2010;56(3):275–281.
333. Susic D, Varagic J, Ahn J, Matavelli L, Frohlich ED. Long-term mineralocorticoid receptor blockade reduces fibrosis and improves cardiac performance and coronary hemodynamics in elderly SHR. *Am J Physiol Heart Circ Physiol.* 2007;292(1):H175–H179.
334. Currie G, Taylor AH, Fujita T, et al. Effect of mineralocorticoid receptor antagonists on proteinuria and progression of chronic kidney disease: a systematic review and meta-analysis. *BMC Nephrol.* 2016;17(1):127.
335. Sarafidis PA, Persu A, Agarwal R, et al. Hypertension in dialysis patients: a consensus document by the European renal and cardiovascular medicine (EURECA-m) working group of the European renal association-European dialysis and transplant association (ERA-EDTA) and the hypertension and the kidney working group of the European society of hypertension (ESH). *Nephrol Dial Transplant.* 2017;32(4):620–640. official publication of the European Dialysis and Transplant Association - European Renal Association.
336. Agarwal R, Sinha AD, Pappas MK, Abraham TN, Tegegne GG. Hypertension in hemodialysis patients treated with atenolol or lisinopril: a randomized controlled trial. official publication of the European Dialysis and Transplant Association - European Renal Association *Nephrol Dial Transplant.* 2014;29(3):672–681.
337. Nakao K, Makino H, Morita S, et al. Beta-blocker prescription and outcomes in hemodialysis patients from the Japan dialysis outcomes and practice patterns study. *Nephron Clin Pract.* 2009;113(3):c132–c139.
338. Port FK, Hulbert-Shearon TE, Wolfe RA, et al. Predialysis blood pressure and mortality risk in a national sample of maintenance hemodialysis patients. *Am J Kidney Dis.* 1999;33(3):507–517.
339. Agarwal R, Sinha AD. Cardiovascular protection with antihypertensive drugs in dialysis patients: systematic review and meta-analysis. *Hypertension.* 2009;53(5):860–866.
340. Miskulin DC, Gassman J, Schrader R, et al. BP in dialysis: results of a pilot study. *J Am Soc Nephrol.* 2018;29(1):307–316.
341. Agarwal R, Peixoto AJ, Santos SF, Zoccali C. Out-of-office blood pressure monitoring in chronic kidney disease. *Blood Pres Monit.* 2009;14(1):2–11.
342. Khan YH, Sarriff A, Adnan AS, Khan AH, Mallhi TH. Diuretics prescribing in chronic kidney disease patients: physician assessment versus bioimpedence spectroscopy. *Clin Exp Nephrol.* 2017;21(3):488–496. https://doi.org/10.1007/s10157-016-1303-7.
343. Taler SJ, Agarwal R, Bakris GL, et al. KDOQI US commentary on the 2012 KDIGO clinical practice guideline for management of blood pressure in CKD. *Am J Kidney Dis.* 2013;62(2):201–213.
344. Upadhyay A, Uhlig K. Is the lower blood pressure target for patients with chronic kidney disease supported by evidence? *Curr Opin Cardiol.* 2012;27(4):370–373.
345. Lazich I, Bakris GL. Prediction and management of hyperkalemia across the spectrum of chronic kidney disease. *Semin Nephrol.* 2014;34(3):333–339.
346. Bakris GL, Pitt B, Weir MR, et al. Effect of patiromer on serum potassium level in patients with hyperkalemia and diabetic kidney disease: the AMETHYST-DN randomized clinical trial. *JAMA.* 2015;314(2):151–161.
347. Packham DK, Rasmussen HS, Lavin PT, et al. Sodium zirconium cyclosilicate in hyperkalemia. *N Engl J Med.* 2015;372(3):222–231.
348. Agarwal R, Rossignol P, Budden J, et al. Patiromer and spironolactone in resistant hypertension and advanced CKD: analysis of the randomized AMBER trial. *Kidney360.* 2021;2(3):425–434.
349. Ali W, Bakris G. Evolution of patiromer use: a review. *Curr Cardiol Rep.* 2020;22(9):94.

350. Paolillo S, Basile C, Dell'Aversana S, et al. Novel potassium binders to optimize RAASi therapy in heart failure: a systematic review and meta-analysis. *Eur J Intern Med.* 2024;119:109–117.
351. Boudville N, Ward S, Benaroia M, House AA. Increased sodium intake correlates with greater use of antihypertensive agents by subjects with chronic kidney disease. *Am J Hypertens.* 2005;18(10):1300–1305.
352. Manfredini F, Mallamaci F, D'Arrigo G, et al. Exercise in patients on dialysis: a multicenter, randomized clinical trial. *J Am Soc Nephrol.* 2017;28(4):1259–1268.
353. Chobanian AV, Bakris GL, Black HR, et al. Seventh report of the Joint national committee on prevention, detection, evaluation, and treatment of high blood pressure. *Hypertension.* 2003;42(6):1206–1252.
354. Feldman RD, Zou GY, Vandervoort MK, Wong CJ, Nelson SA, Feagan BG. A simplified approach to the treatment of uncomplicated hypertension: a cluster randomized, controlled trial. *Hypertension.* 2009;53(4):646–653.
355. Bakris GL, Weir MR, Study of H. The Efficacy of Lotrel in Diabetes I. Achieving goal blood pressure in patients with type 2 diabetes: conventional versus fixed-dose combination approaches. *J Clin Hypertens.* 2003;5(3):202–209.
356. Bakris GL, Williams M, Dworkin L, et al. Preserving renal function in adults with hypertension and diabetes: a consensus approach. *Am J Kidney Dis.* 2000;36(3):646–661. National Kidney Foundation Hypertension and Diabetes Executive Committees Working Group.
357. Ferdinand KC. A compendium of antihypertensive therapy. *J Clin Hypertens.* 2011;13(9):636–638.
358. Strippoli GF, Craig M, Craig JC. Antihypertensive agents for preventing diabetic kidney disease. *Cochrane Database Syst Rev.* 2005;(4):CD004136.
359. Daien V, Duny Y, Ribstein J, et al. Treatment of hypertension with renin-angiotensin system inhibitors and renal dysfunction: a systematic review and meta-analysis. *Am J Hypertens.* 2012;25(1):126–132.
360. Mangrum AJ, Bakris GL. Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers in chronic renal disease: safety issues. *Semin Nephrol.* 2004;24(2):168–175.
361. Caldeira D, David C, Sampaio C. Tolerability of angiotensin-receptor blockers in patients with intolerance to angiotensin-converting enzyme inhibitors: a systematic review and meta-analysis. *Am J Cardiovasc Drugs.* 2012;12(4):263–277.
362. Bakris GL, Weir MR. Angiotensin-converting enzyme inhibitor-associated elevations in serum creatinine: is this a cause for concern? *Arch Intern Med.* 2000;160(5):685–693.
363. Hirsch S, Hirsch J, Bhatt U, Rovin BH. Tolerating increases in the serum creatinine following aggressive treatment of chronic kidney disease, hypertension and proteinuria: pre-renal success. *Am J Nephrol.* 2012;36(5):430–437.
364. Ruggenenti P, Remuzzi G. Dealing with renin-angiotensin inhibitors, don't mind serum creatinine. *Am J Nephrol.* 2012;36(5):427–429.
365. Bakris GL, Oparil S, Purkayastha D, Yadao AM, Alessi T, Sowers JR. Randomized study of antihypertensive efficacy and safety of combination aliskiren/valsartan vs valsartan monotherapy in hypertensive participants with type 2 diabetes mellitus. *J Clin Hypertens.* 2013;15(2):92–100.
366. Bolignano D, Palmer SC, Navaneethan SD, Strippoli GF. Aldosterone antagonists for preventing the progression of chronic kidney disease. *Cochrane Database Syst Rev.* 2014;(4):CD007004.
367. Georgianos PI, Vaios V, Eleftheriadis T, Zebekakis P, Liakopoulos V. Mineralocorticoid antagonists in ESRD: an overview of their efficacy and safety. *Curr Vasc Pharmacol.* 2017;15(6):599–606.
368. Sinnott SJ, Tomlinson LA, Root AA, et al. Comparative effectiveness of fourth-line anti-hypertensive agents in resistant hypertension: a systematic review and meta-analysis. *Eur J Prev Cardiol.* 2017;24(3):228–238.
369. Carter BE,M. Diuretics and hypertension. In: Bakris GS, M J, eds. *Hypertension : A Companion to Braunwald's Heart Disease.* 3rd ed. Philadelphia, PA: Elsevier; 2018:211–221.
370. Agarwal R, Sinha AD, Tu W. Chlorthalidone for hypertension in advanced CKD. Reply. *N Engl J Med.* 2022;386(14):1384.
371. Ishani A, Cushman WC, Leatherman SM, et al. Chlorthalidone vs. Hydrochlorothiazide for hypertension-cardiovascular events. *N Engl J Med.* 2022;387(26):2401–2410.
372. Ernst ME, Carter BL, Goerdt CJ, et al. Comparative antihypertensive effects of hydrochlorothiazide and chlorthalidone on ambulatory and office blood pressure. *Hypertension.* 2006;47(3):352–358.
373. Khosla N, Chua DY, Elliott WJ, Bakris GL. Are chlorthalidone and hydrochlorothiazide equivalent blood-pressure-lowering medications? *J Clin Hypertens.* 2005;7(6):354–356.
374. Bakris GL, Sica D, White WB, et al. Antihypertensive efficacy of hydrochlorothiazide vs chlorthalidone combined with azilsartan medoxomil. *Am J Med.* 2012;125(12):1229 e1–1229 e10.
375. Bakris GL, Weir MR, Secic M, Campbell B, Weis-McNulty A. Differential effects of calcium antagonist subclasses on markers of nephropathy progression. *Kidney Int.* 2004;65(6):1991–2002.
376. Boero R, Rollino C, Massara C, et al. Verapamil versus amlodipine in proteinuric non-diabetic nephropathies treated with trandolapril (VVANNTT study): design of a prospective randomized multicenter trial. *J Nephrol.* 2001;14(1):15–18.
377. Smith AC, Toto R, Bakris GL. Differential effects of calcium channel blockers on size selectivity of proteinuria in diabetic glomerulopathy. *Kidney Int.* 1998;54(3):889–896.
378. Agodoa LY, Appel L, Bakris GL, et al. Effect of ramipril vs amlodipine on renal outcomes in hypertensive nephrosclerosis: a randomized controlled trial. *JAMA.* 2001;285(21):2719–2728.
379. Weir MR, Rosenberger C, Fink JC. Pilot study to evaluate a water displacement technique to compare effects of diuretics and ACE inhibitors to alleviate lower extremity edema due to dihydropyridine calcium antagonists. *Am J Hypertens.* 2001;14(9 Pt 1):963–968.
380. Saseen JJ, Carter BL, Brown TE, Elliott WJ, Black HR. Comparison of nifedipine alone and with diltiazem or verapamil in hypertension. *Hypertension.* 1996;28(1):109–114.
381. Huang BT, Huang FY, Zuo ZL, et al. Meta-analysis of relation between oral beta-blocker therapy and outcomes in patients with acute myocardial infarction who underwent percutaneous coronary intervention. *Am J Cardiol.* 2015;115(11):1529–1538.
382. Chatterjee S, Udell JA, Sardar P, Lichstein E, Ryan JJ. Comparable benefit of beta-blocker therapy in heart failure across regions of the world: meta-analysis of randomized clinical trials. *Can J Cardiol.* 2014;30(8):898–903.
383. Cice G, Ferrara L, D'Andrea A, et al. Carvedilol increases two-year survivalin dialysis patients with dilated cardiomyopathy: a prospective, placebo-controlled trial. *J Am Coll Cardiol.* 2003;41(9):1438–1444.
384. Bakris GL, Fonseca V, Katholi RE, et al. Metabolic effects of carvedilol vs metoprolol in patients with type 2 diabetes mellitus and hypertension: a randomized controlled trial. *JAMA.* 2004;292(18):2227–2236.
385. Bakris GL, Hart P, Ritz E. Beta blockers in the management of chronic kidney disease. *Kidney Int.* 2006;70(11):1905–1913.
386. Tham TC, Guy S, McDermott BJ, Shanks RG, Riddell JG. The dose dependency of the alpha- and beta-adrenoceptor antagonist activity of carvedilol in man. *Br J Clin Pharmacol.* 1995;40(1):19–23.
387. Hansson L, Himmelmann A. Carvedilol in the treatment of hypertension–a review of the clinical data base. *Scand Cardiovasc J Suppl.* 1998;47:67–80.
388. Carlsen JE, Kardel T, Lund JO, McNair A, Trap-Jensen J. Acute hemodynamic effects of pinacidil and hydralazine in essential hypertension. *Clin Pharmacol Ther.* 1985;37(3):253–259.
389. Cohn JN, McInnes GT, Shepherd AM. Direct-acting vasodilators. *J Clin Hypertens.* 2011;13(9):690–692.
390. Weder AB, Erickson S. Treatment of hypertension in the inpatient setting: use of intravenous labetalol and hydralazine. *J Clin Hypertens.* 2010;12(1):29–33.
391. Teran E, Briones F, Nicolalde V, Vivero S, Chedraui P, Perez-Lopez FR. Effect of intravenous hydralazine infusion on maternal plasma nitric oxide levels in gestations complicated with severe preeclampsia: a pilot study. *Arch Gynecol Obstet.* 2012;285(4):1183–1187.
392. Franciosa JA. Fixed combination isosorbide dinitrate-hydralazine for nitric-oxide-enhancing therapy in heart failure. *Expert Opin Pharmacother.* 2006;7(18):2521–2531.
393. Sarafidis PA, Georgianos P, Bakris GL. Resistant hypertension–its identification and epidemiology. *Nat Rev Nephrol.* 2013;9(1):51–58. https://doi.org/10.1038/nrneph.2012.260.
394. Bhatt DL, Kandzari DE, O'Neill WW, et al. A controlled trial of renal denervation for resistant hypertension. *N Engl J Med.* 2014;370(15):1393–1401.

395. Azizi M, Sapoval M, Gosse P, et al. Optimum and stepped care standardised antihypertensive treatment with or without renal denervation for resistant hypertension (DENERHTN): a multicentre, open-label, randomised controlled trial. *Lancet*. 2015;385(9981):1957–1965.
396. Williams B, MacDonald TM, Morant S, et al. Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2): a randomised, double-blind, crossover trial. *Lancet*. 2015;386(10008):2059–2068.
397. Zhao D, Liu H, Dong P, Zhao J. A meta-analysis of add-on use of spironolactone in patients with resistant hypertension. *Int J Cardiol*. 2017;233:113–117.
398. Rosa J, Widimsky P, Tousek P, et al. Randomized comparison of renal denervation versus intensified pharmacotherapy including spironolactone in true-resistant hypertension: six-month results from the Prague-15 study. *Hypertension*. 2015;65(2):407–413. https://doi.org/10.1161/HYPERTENSIONAHA.114.04019.
399. Carey RM, Calhoun DA, Bakris GL, et al. Resistant hypertension: detection, evaluation, and management: a scientific statement from the American Heart Association. *Hypertension*. 2018;72(5):e53–e90.
400. Blazek O, Bakris GL. Novel therapies on the horizon of hypertension management. *Am J Hypertens*. 2023;36(2):73–81.
401. Rey-Garcia J, Townsend RR. Renal denervation: a review. *Am J Kidney Dis*. 2022;80(4):527–535.